

Sector × Quintile Return Book

Average monthly returns of each factor's quintile portfolios, by GICS sector. 271 factors, ordered by in-sample L–S CAGR.

HOW TO READ

■ Q1 ■ Q2 ■ Q3 ■ Q4 ■ Q5

Each cluster of five bars is one sector; bars run Q1 (highest factor score) to Q5 (lowest), dark to light.

■ Grey clusters mark sectors excluded from the factor — the Q1–Q5 spread is negative there.

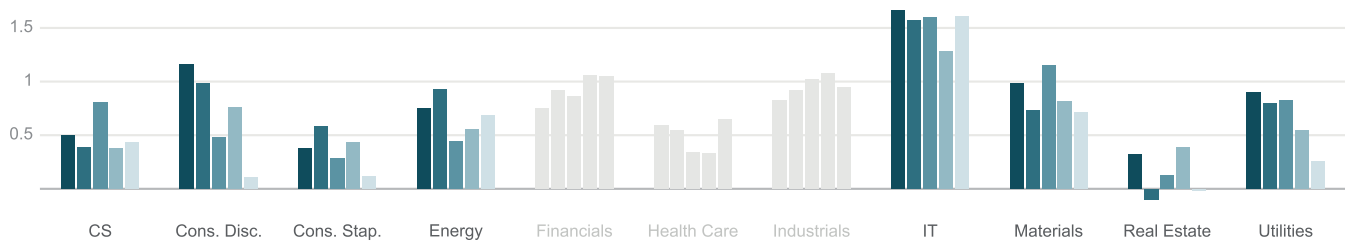
FACTOR STYLES

● Valuation	41	● Price Momentum	56
● Earnings Quality	55	● Size	1
● Analyst Expectations	24	● Historical Growth	56
● Capital Efficiency	28	● Volatility	10

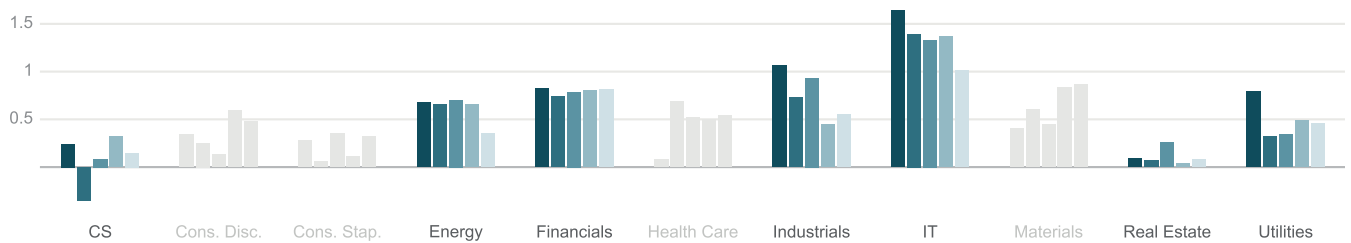
001 In current model portfolio (50 factors)

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

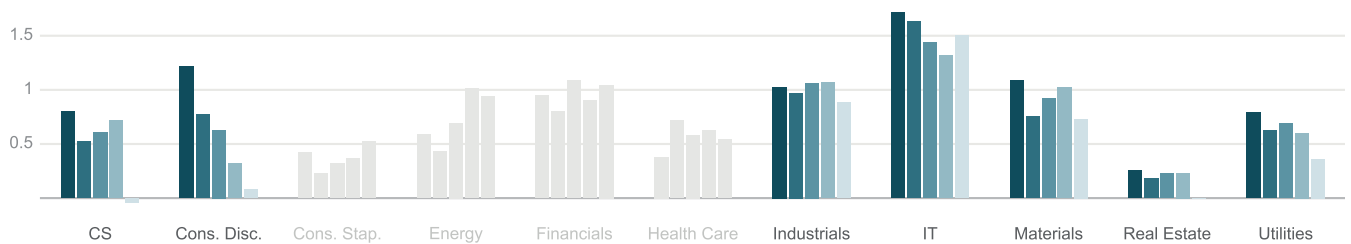
Historical Growth L-S CAGR +17.2%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

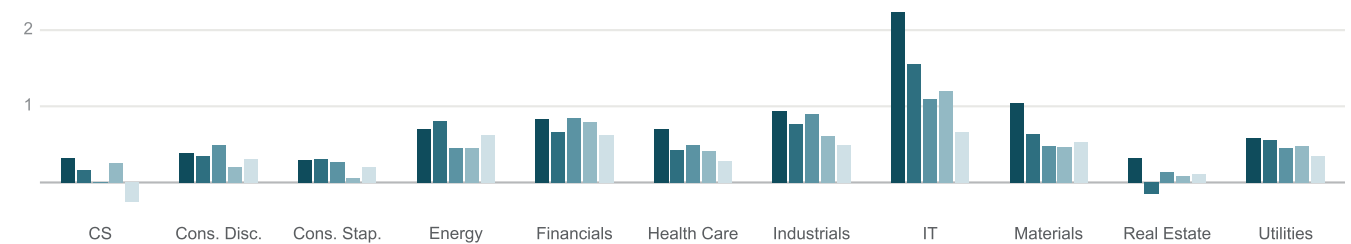
Historical Growth L-S CAGR +15.4%

**003 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

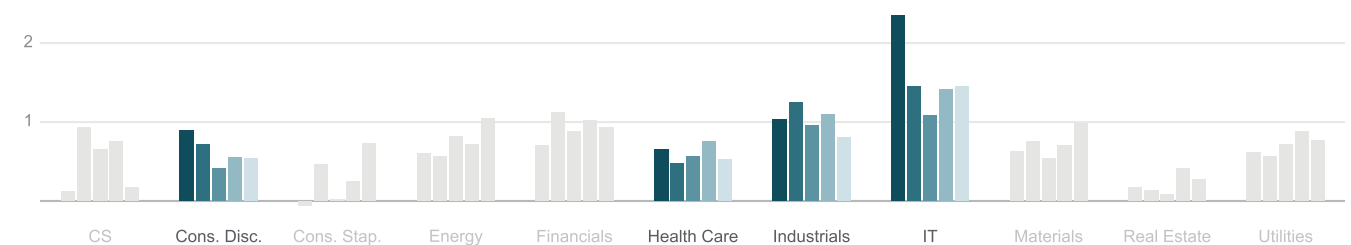
Historical Growth L-S CAGR +12.3%

**004 ● 50 Day to 200 Day Stock Price**

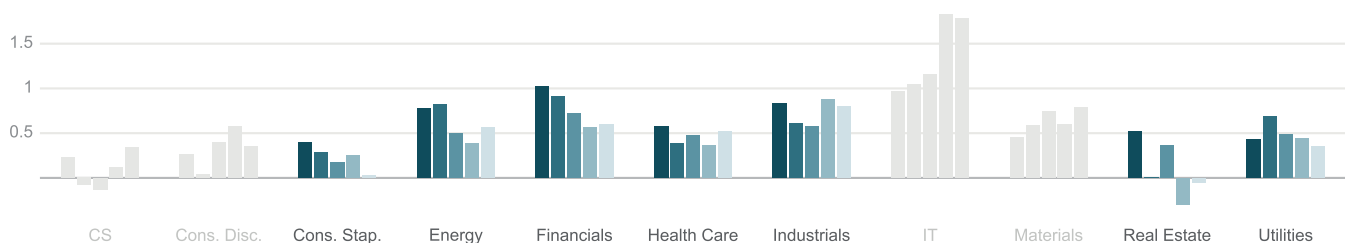
Price Momentum L-S CAGR +12.2%

**005 ● Change of Asset Turnover**

Historical Growth L-S CAGR +11.8%

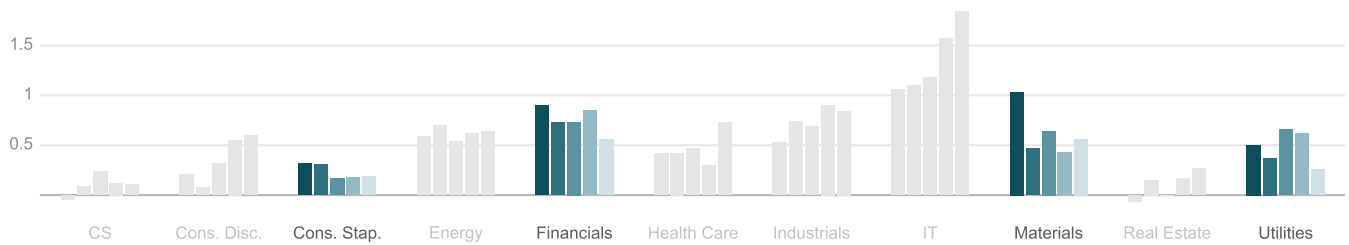
**006 ● Current Liability to Price**

Valuation L-S CAGR +11.7%

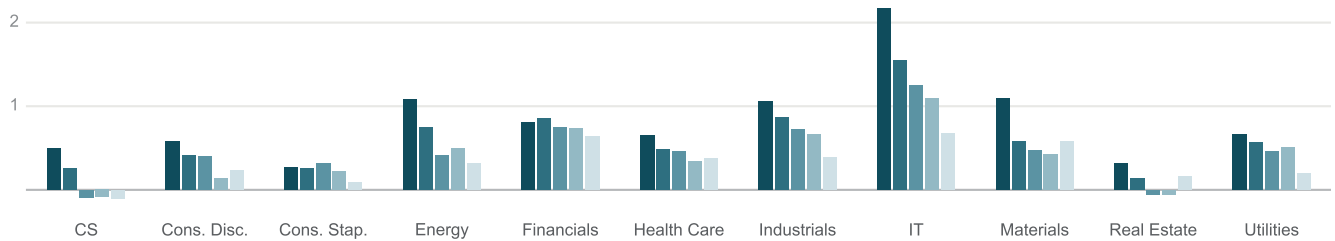


007 ● **Tobin's Q Ratio**

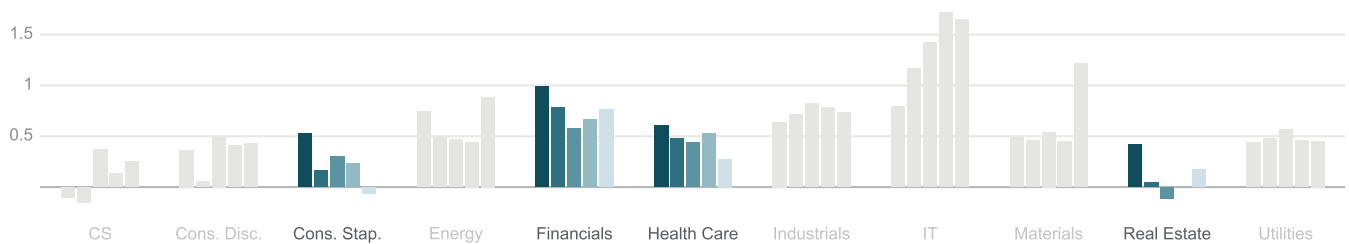
Capital Efficiency L-S CAGR +11.5%

008 ● **39W Lagged Return**

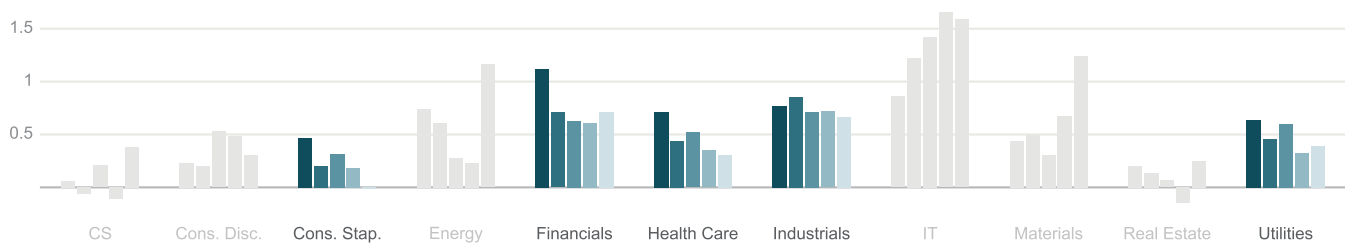
Price Momentum L-S CAGR +11.4%

009 ● **Pretax Income Per Share to Price**

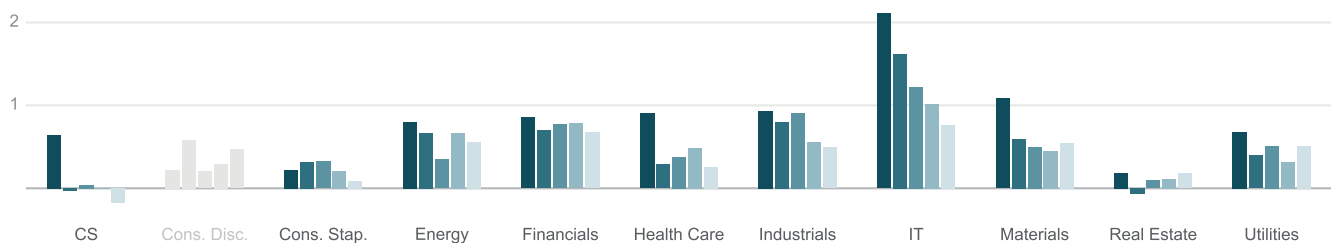
Valuation L-S CAGR +11.3%

010 ● **Operating Earnings to Price Ratio**

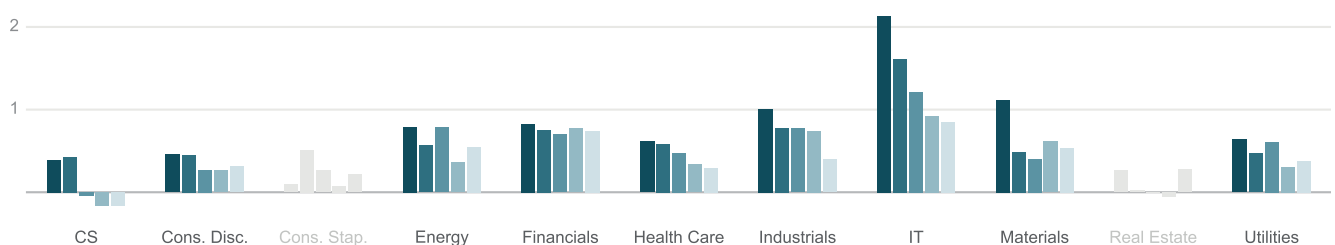
Valuation L-S CAGR +11.2%

011 ● **26W Relative Price Strength**

Price Momentum L-S CAGR +11.1%

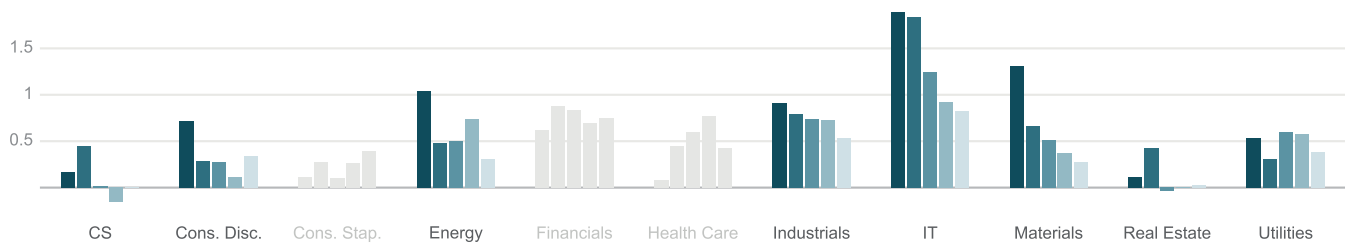
012 ● **9 Month Price Momentum**

Price Momentum L-S CAGR +11.1%



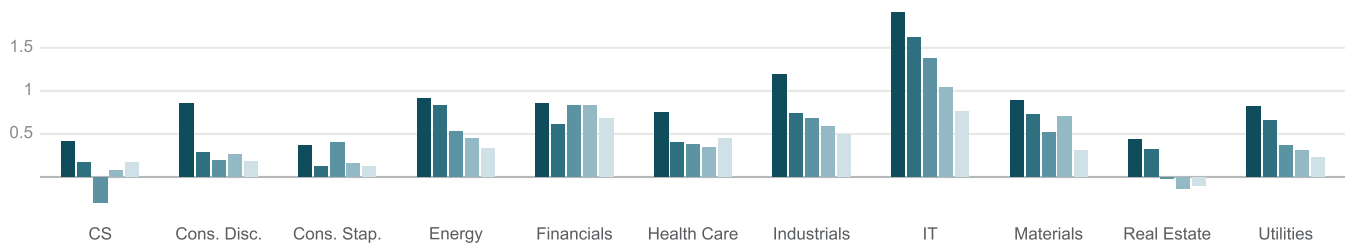
013 ● 12M Realized Price Volatility

Volatility L-S CAGR +11.0%



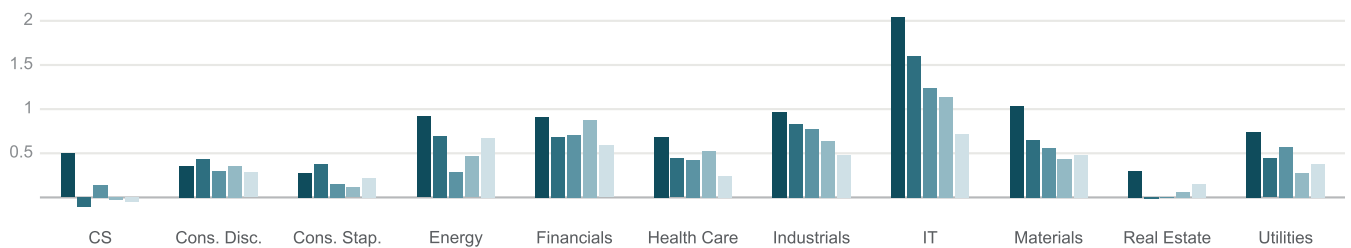
014 ● Slope of 52W Price Trend

Price Momentum L-S CAGR +11.0%



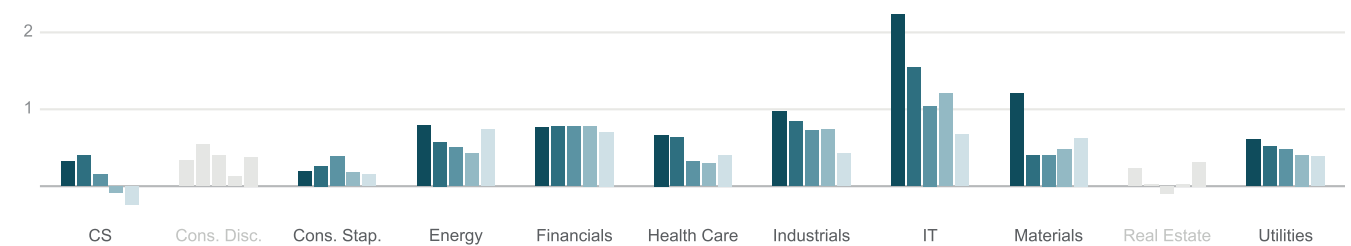
015 ● 15/36 Week Price Ratio

Price Momentum L-S CAGR +10.9%



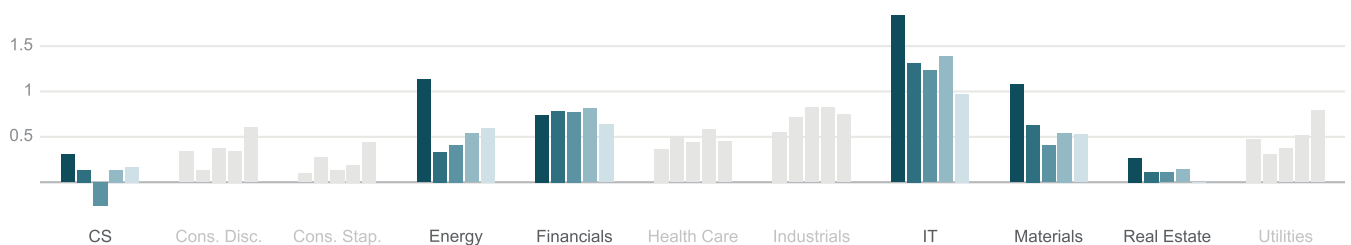
016 ● 4W to 52W Price Oscillator

Price Momentum L-S CAGR +10.8%



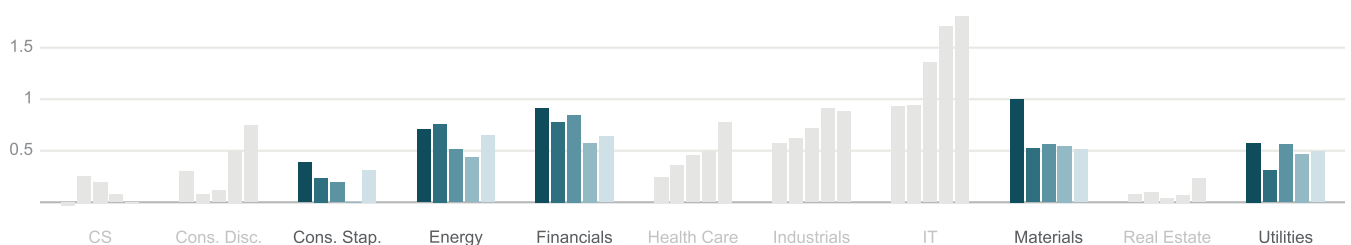
017 ● Analyst Dispersion for FY2 EPS

Analyst Expectations L-S CAGR +10.7%



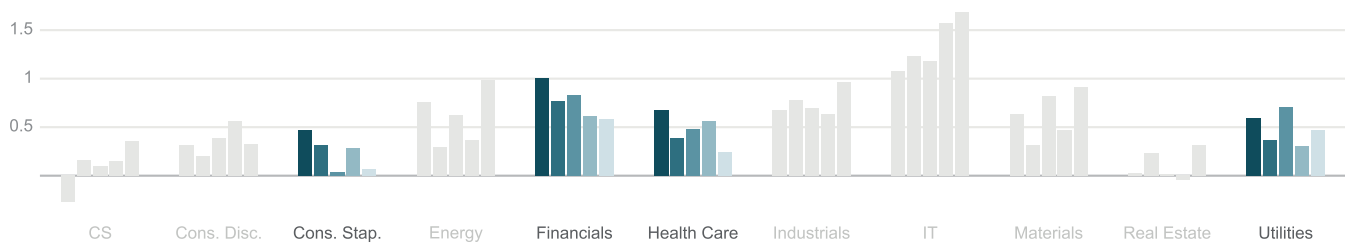
018 ● Book to Price

Valuation L-S CAGR +10.6%



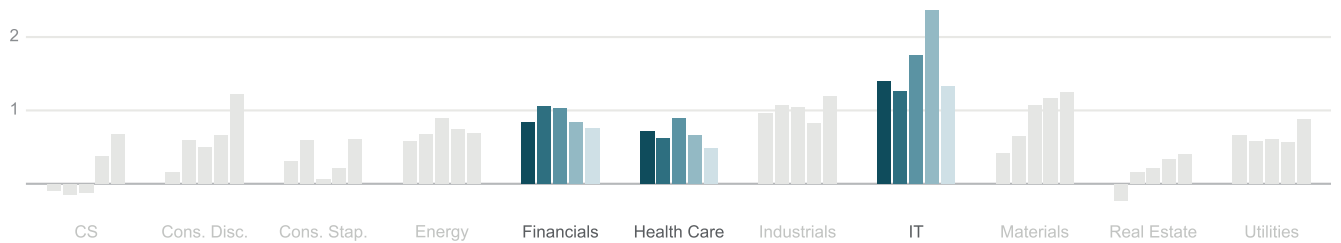
019 ● Normalized EP

Valuation L-S CAGR +10.4%



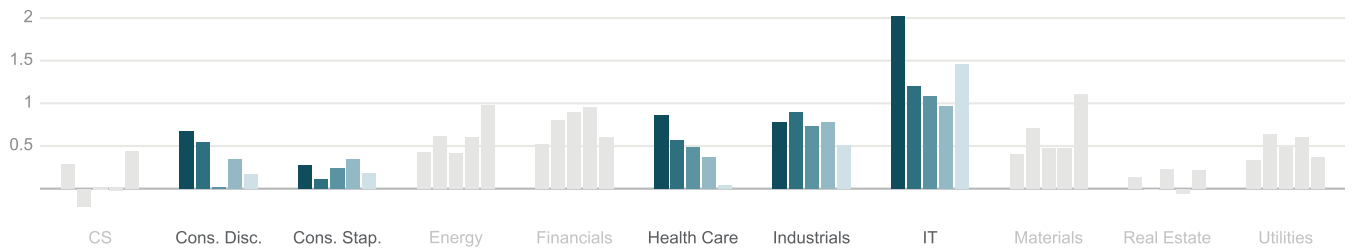
020 ● Indicated Annual Dividends to Price Ratio

Valuation L-S CAGR +10.2%



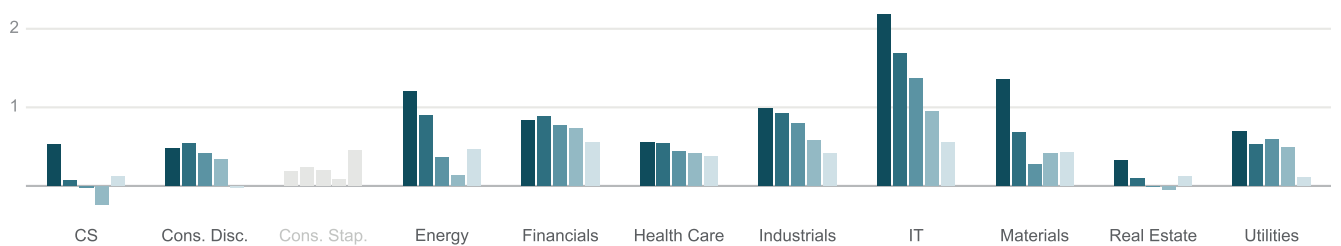
021 ● Operating Earnings to Assets Ratio

Earnings Quality L-S CAGR +10.2%



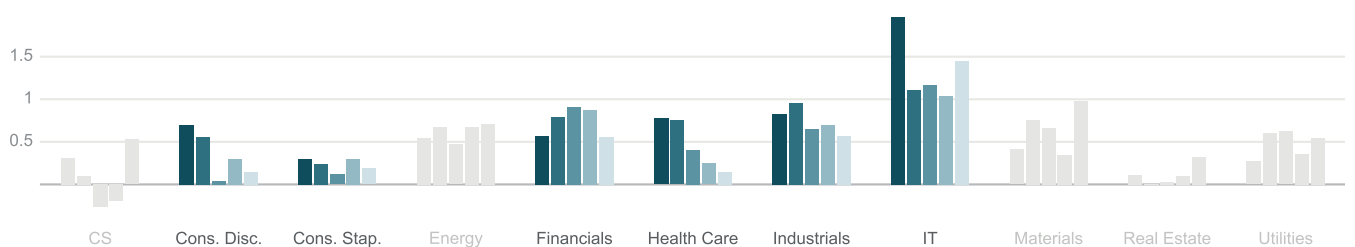
022 ● Percent Above 260-Day Low

Price Momentum L-S CAGR +10.2%



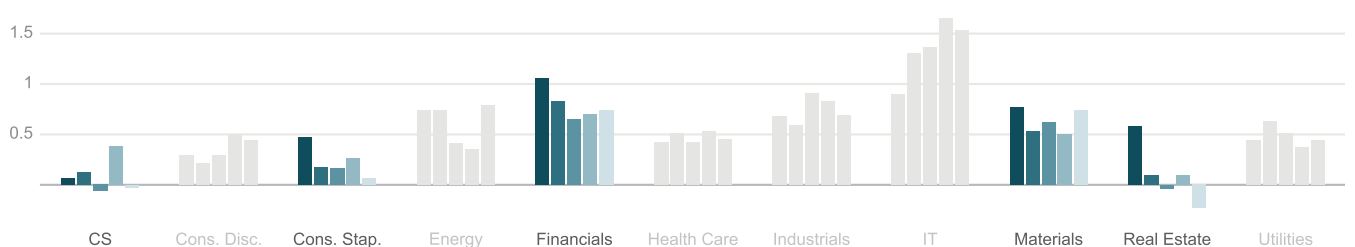
023 ● Return on Assets

Capital Efficiency L-S CAGR +10.1%



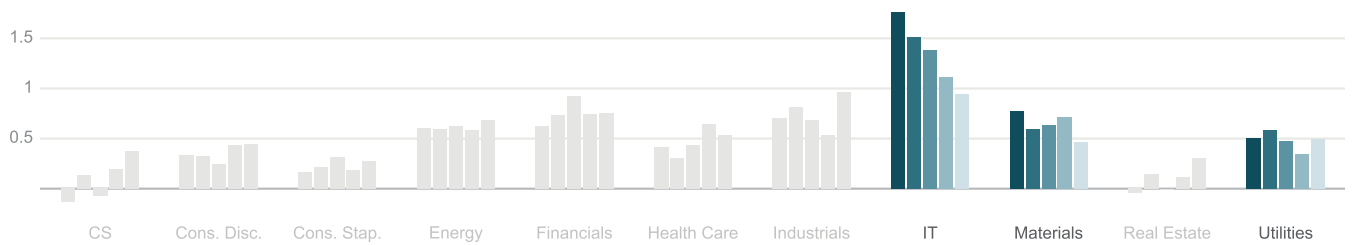
024 ● EBITDA to Price

Valuation L-S CAGR +10.0%

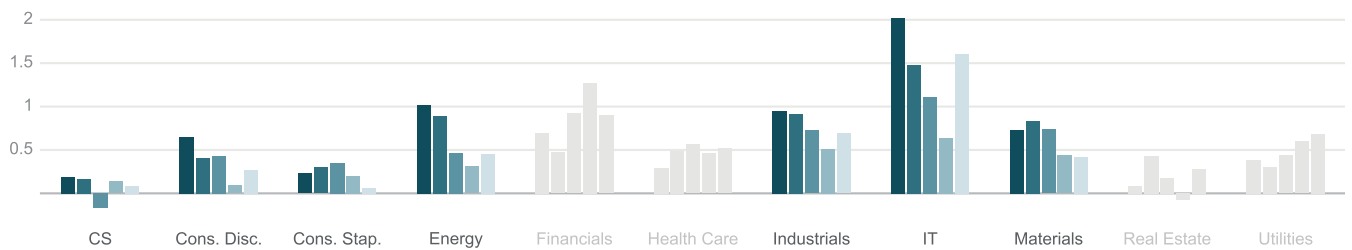


025 ● **Stability of Change in Number of Holders**

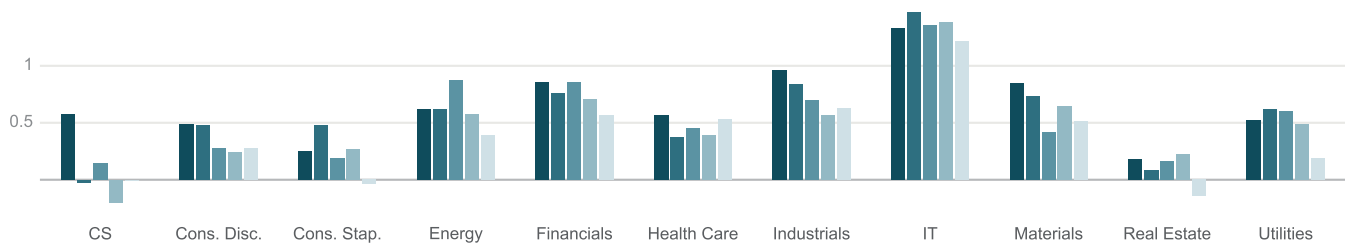
Price Momentum L-S CAGR +10.0%

026 ● **Unexpected Inventory Change**

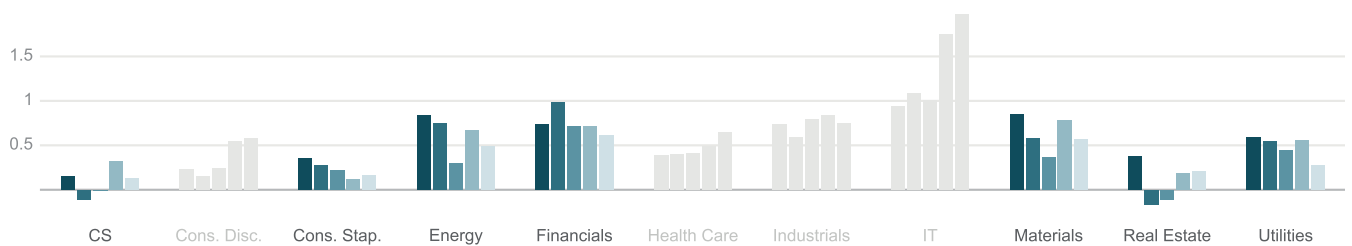
Earnings Quality L-S CAGR +9.9%

027 ● **Log of Total Last Quarter Assets**

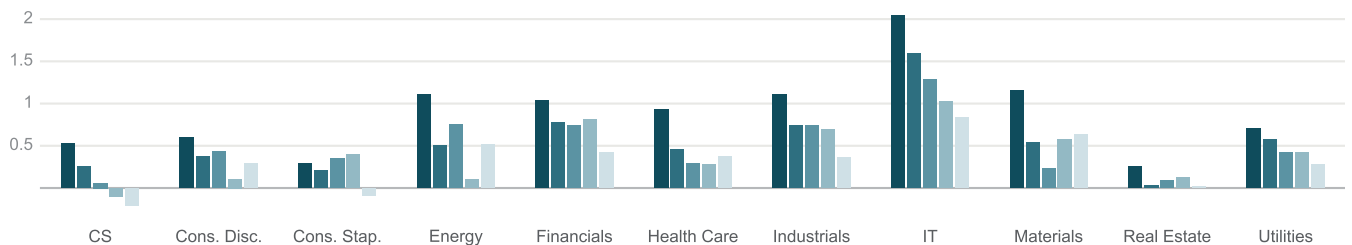
Size L-S CAGR +9.8%

028 ● **Assets to Price Ratio**

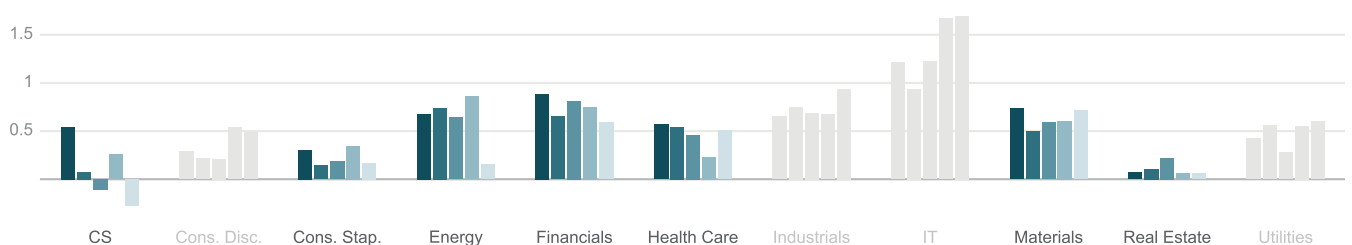
Valuation L-S CAGR +9.8%

029 ● **6M Chg in Target Price**

Analyst Expectations L-S CAGR +9.7%

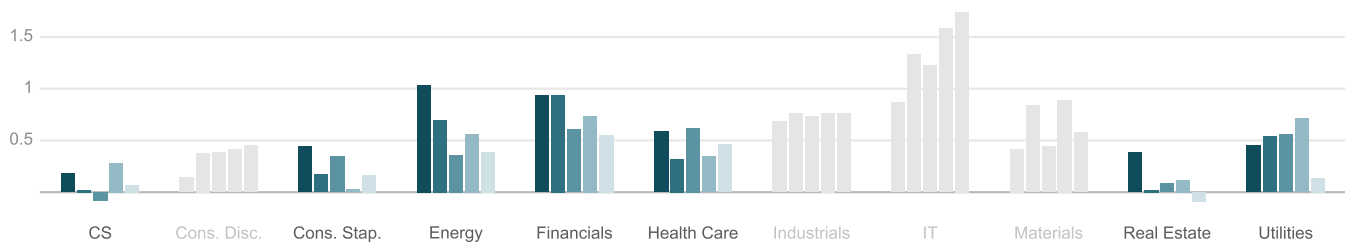
030 ● **Cash to Price**

Valuation L-S CAGR +9.7%

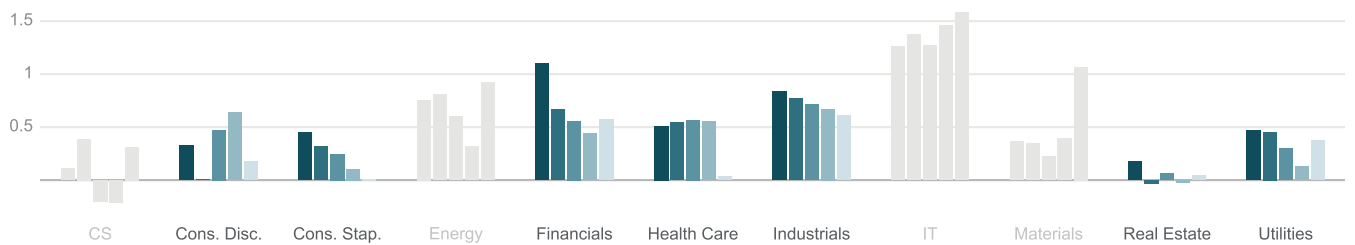


031 ● Sales to Price Ratio

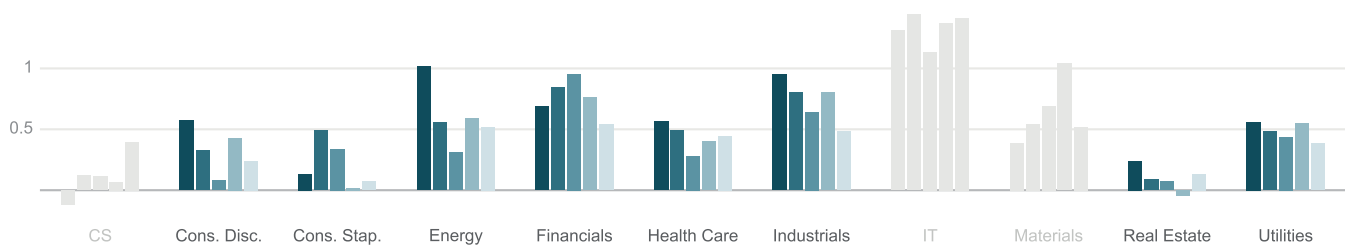
Valuation L-S CAGR +9.7%

**032 ● Growth Flow to Price**

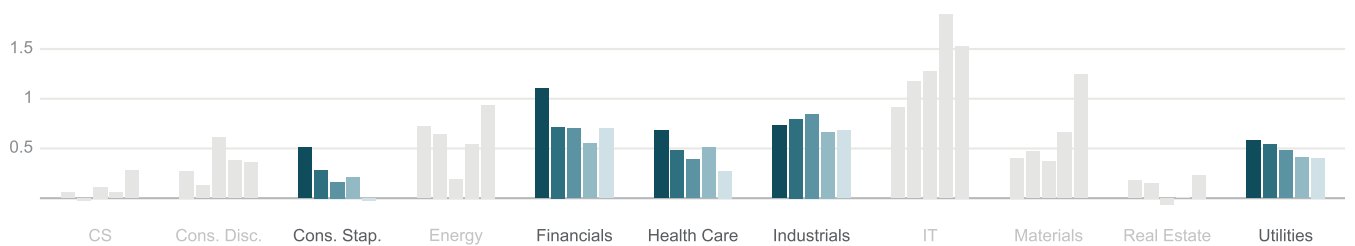
Valuation L-S CAGR +9.6%

**033 ● Book Leverage**

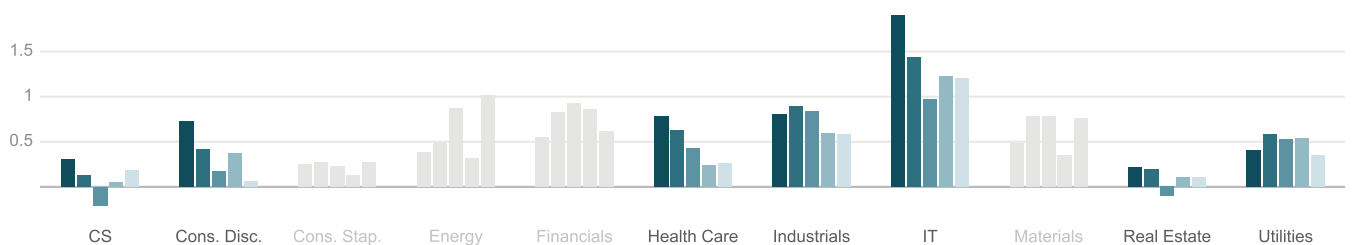
Volatility L-S CAGR +9.6%

**034 ● Earnings to Price**

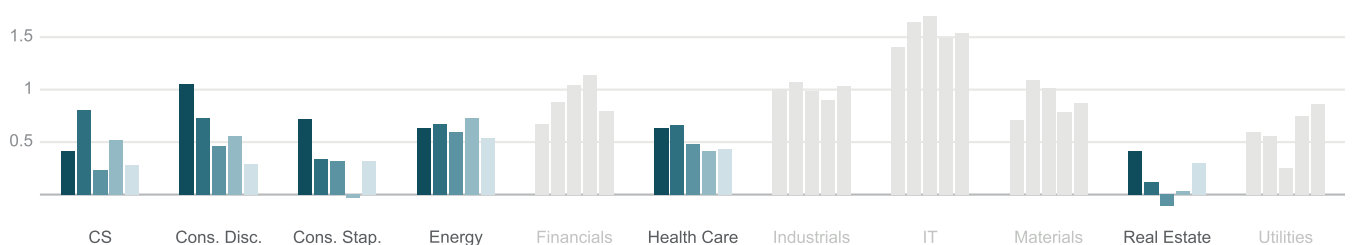
Valuation L-S CAGR +9.4%

**035 ● Cash Flow to Total Asset**

Earnings Quality L-S CAGR +9.4%

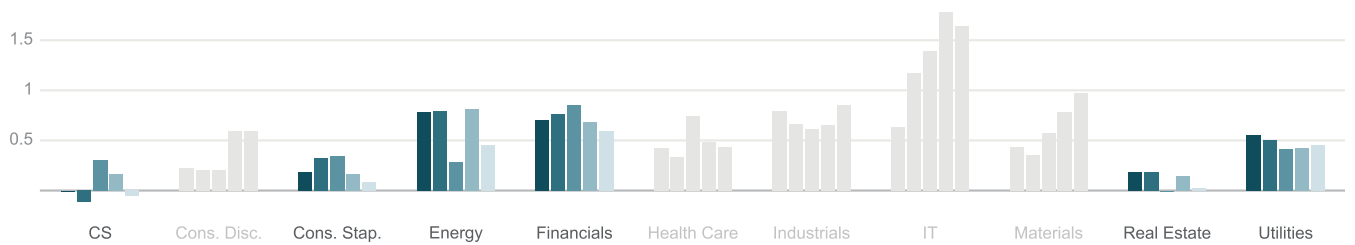
**036 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings**

Historical Growth L-S CAGR +9.2%

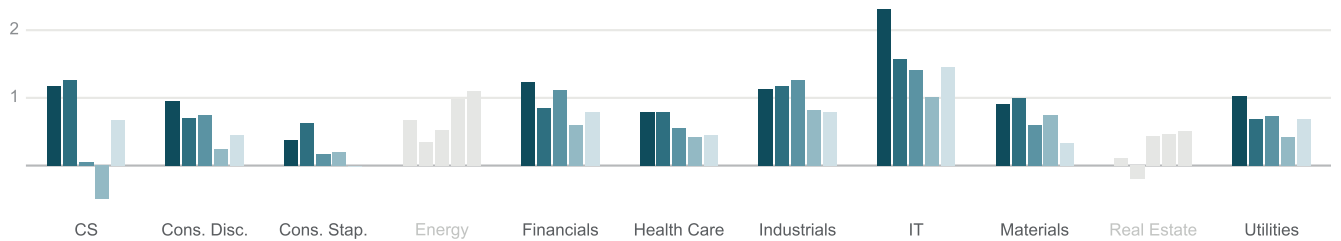


037 ROA 20 Qtr Standard Deviation

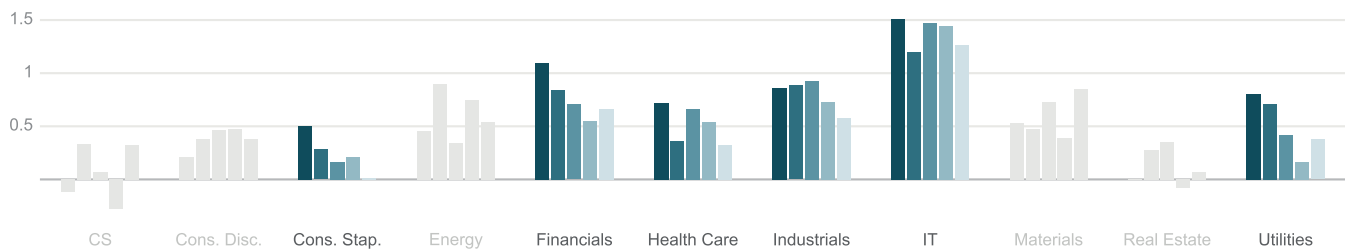
Earnings Quality L-S CAGR +9.2%

**038 Adjusted 3M Revision in FY1 EPS Estimate**

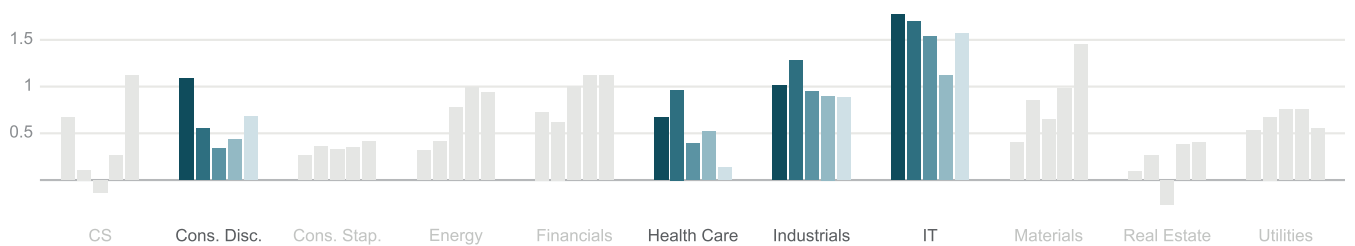
Analyst Expectations L-S CAGR +9.2%

**039 Inverse P/E Ratio Adj for Growth and Yield**

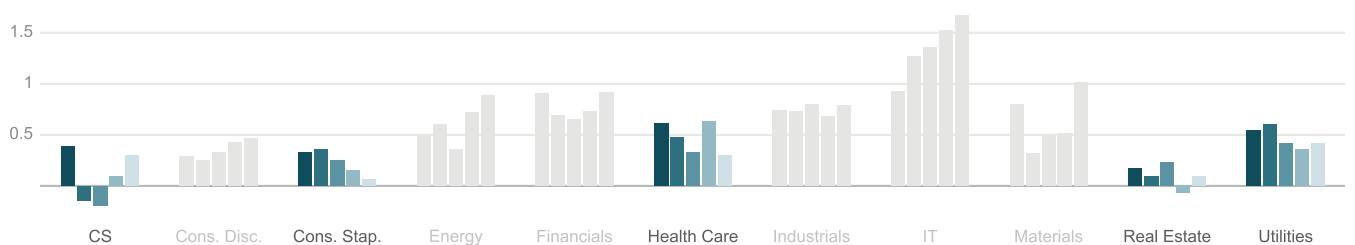
Valuation L-S CAGR +9.1%

**040 Trailing 12-Month Earnings/Trailing 12-Month Sales**

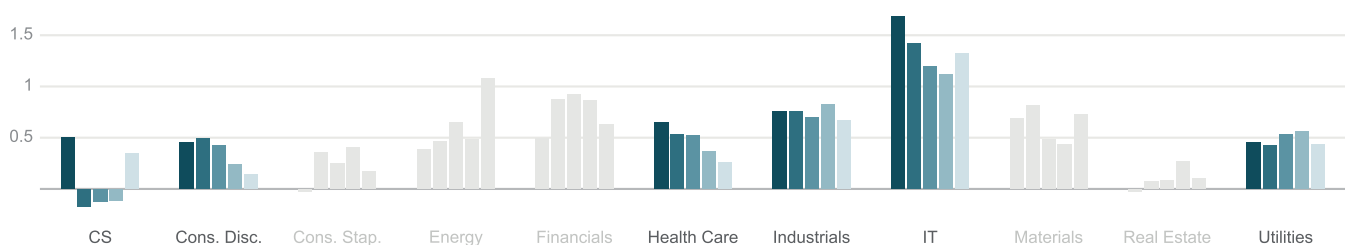
Earnings Quality L-S CAGR +9.1%

**041 Net Cash Flow to Enterprise Value**

Valuation L-S CAGR +9.1%

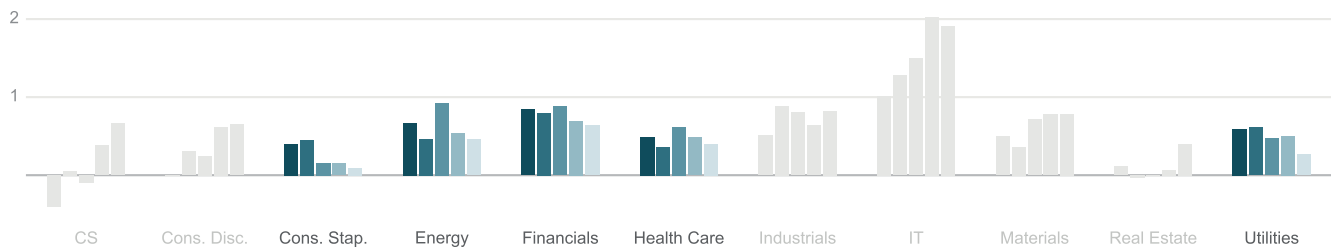
**042 Solvency Ratio**

Valuation L-S CAGR +9.1%



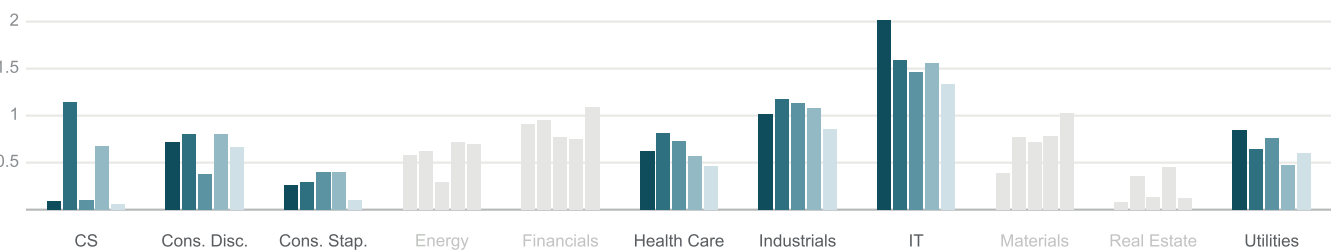
043 ● Dividends to Price Ratio

Valuation L-S CAGR +9.1%



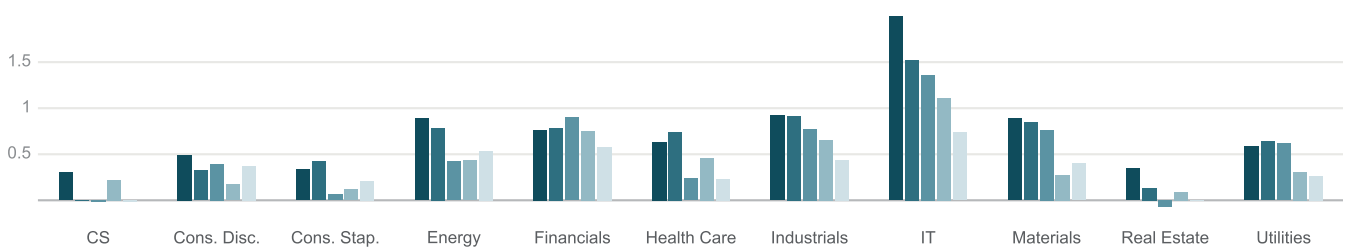
044 ● Adjusted Inverse PEG

Analyst Expectations L-S CAGR +9.1%



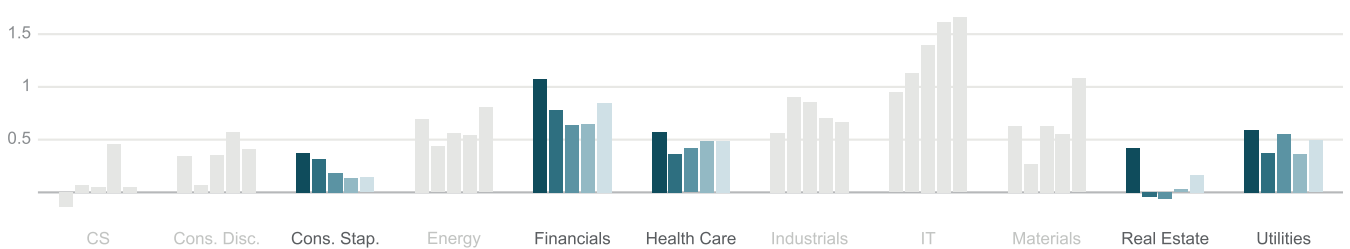
045 ● 180D Price TStat

Price Momentum L-S CAGR +8.9%



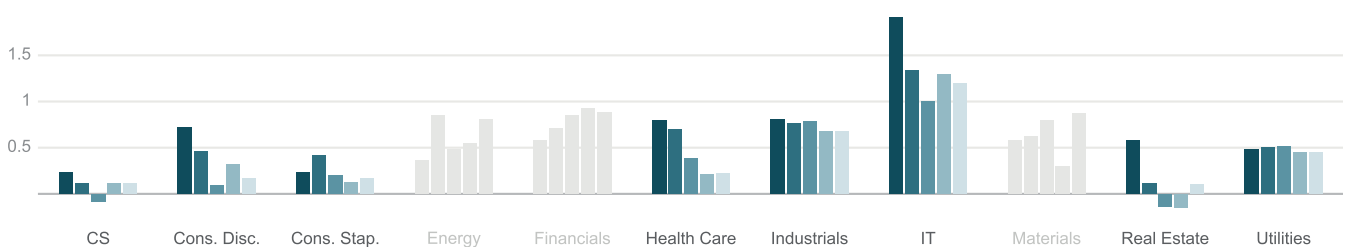
046 ● Adjusted EBIT to Price

Valuation L-S CAGR +8.9%



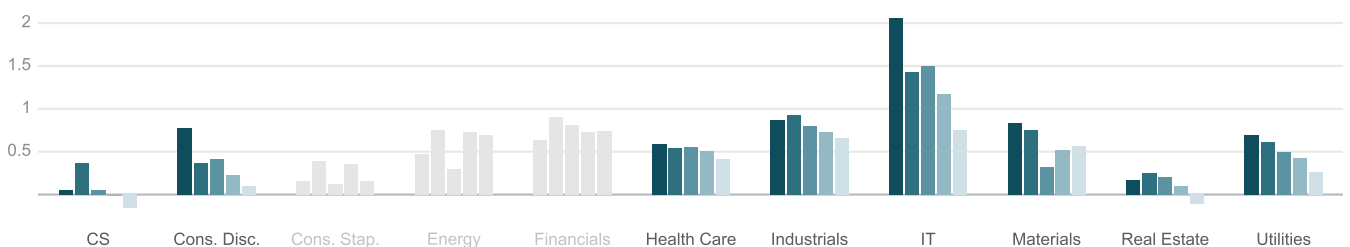
047 ● EBITDA to Assets

Capital Efficiency L-S CAGR +8.8%



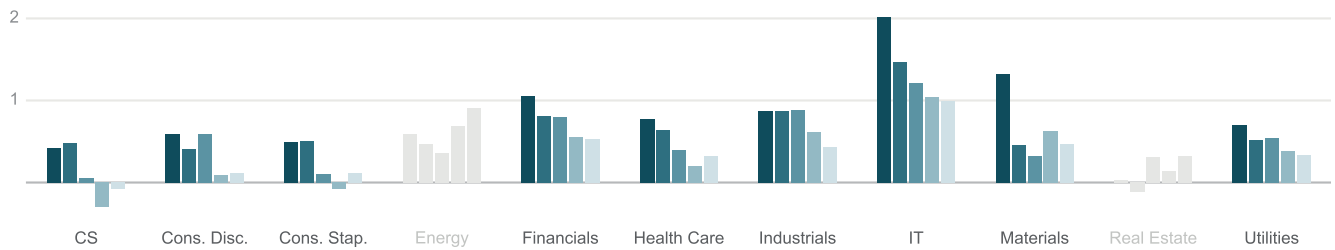
048 ● Expected LTG

Analyst Expectations L-S CAGR +8.7%

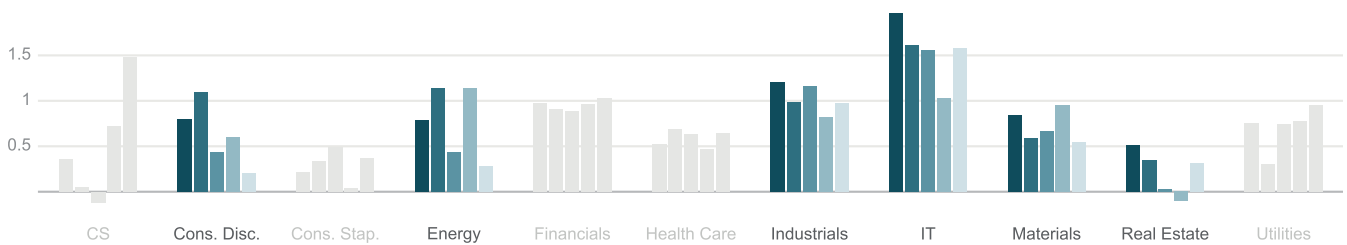


049 3M Revision in FY1 EPS Estimate

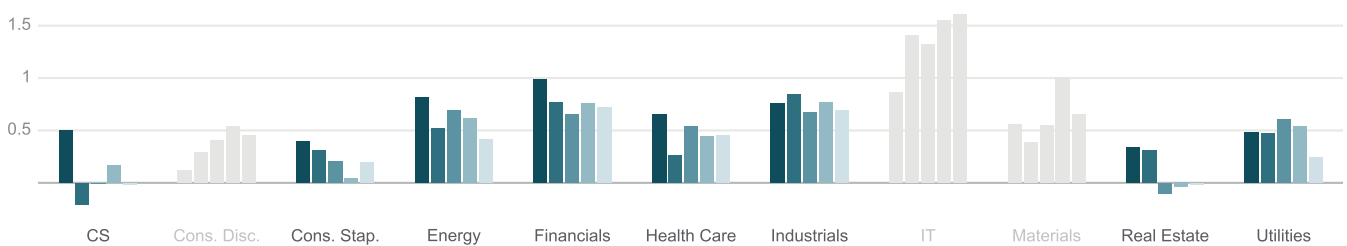
Analyst Expectations L-S CAGR +8.7%

**050 Adj 1Y Chg in Asset Adjusted FCF**

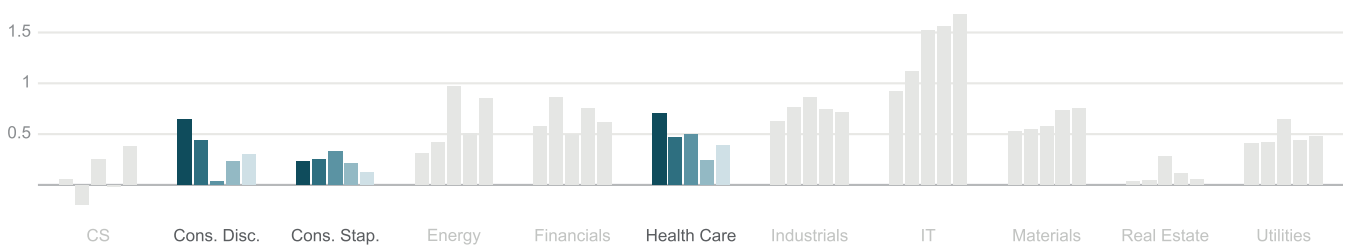
Historical Growth L-S CAGR +8.5%

**051 Sales to EV Ratio**

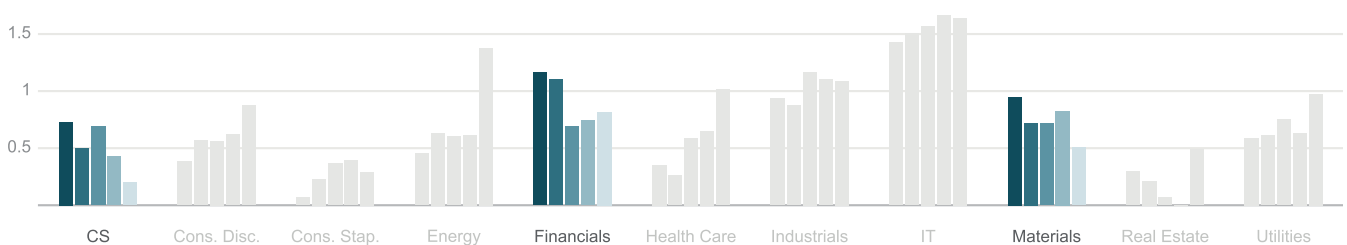
Valuation L-S CAGR +8.5%

**052 Working Capital to Sales**

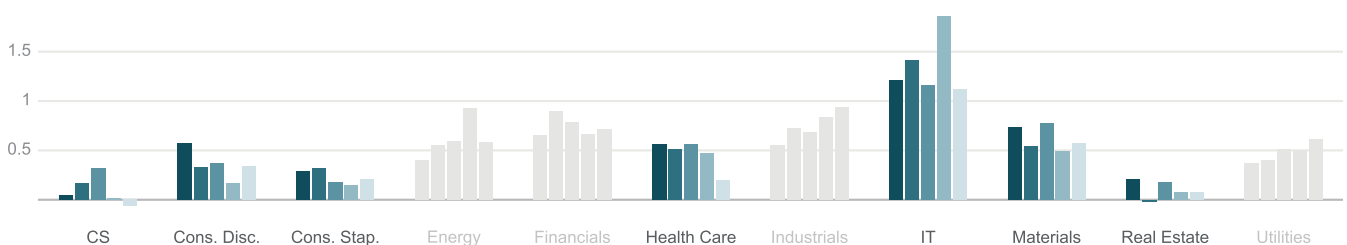
Earnings Quality L-S CAGR +8.5%

**053 Adjusted Book to Enterprise Value**

Valuation L-S CAGR +8.5%

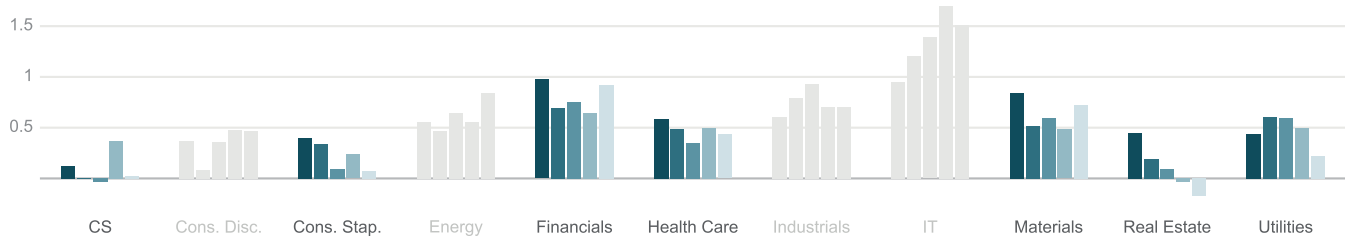
**054 Balance Sheet Accruals**

Earnings Quality L-S CAGR +8.4%

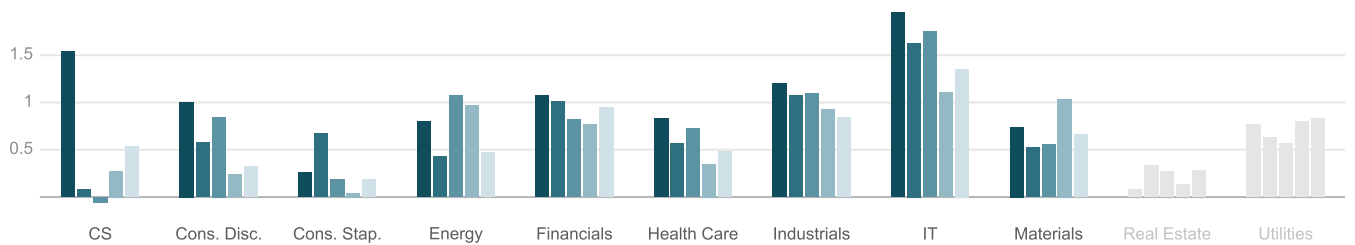


055 EBITDA to Enterprise Value

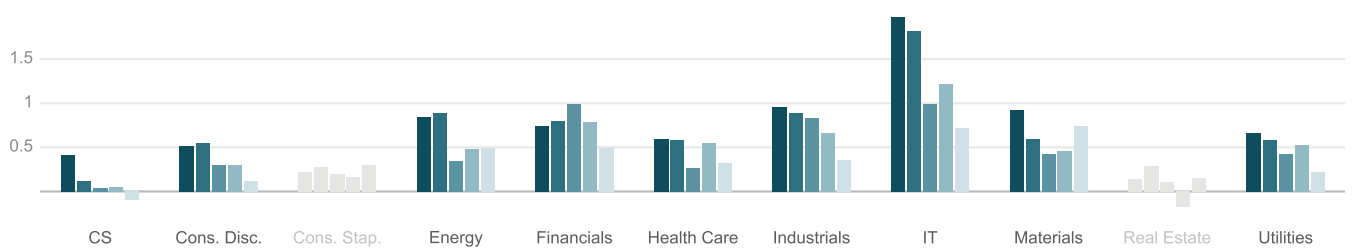
Valuation L-S CAGR +8.4%

**056 Short Interest Trading Volume**

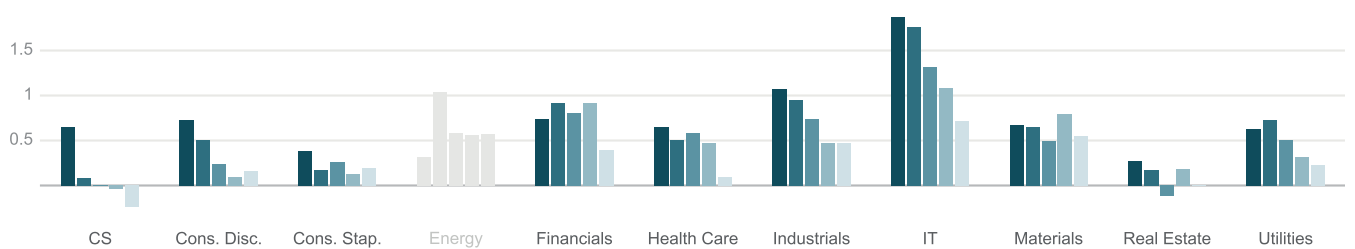
Price Momentum L-S CAGR +8.4%

**057 12M - 1M Price Momentum**

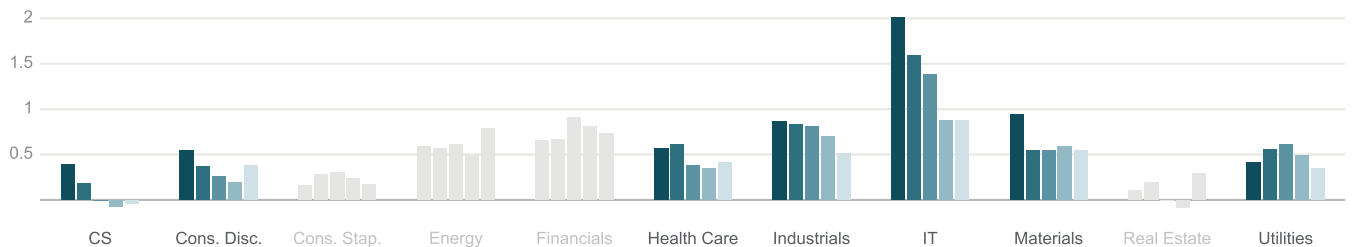
Price Momentum L-S CAGR +8.3%

**058 Amihud Illiquidity Measure**

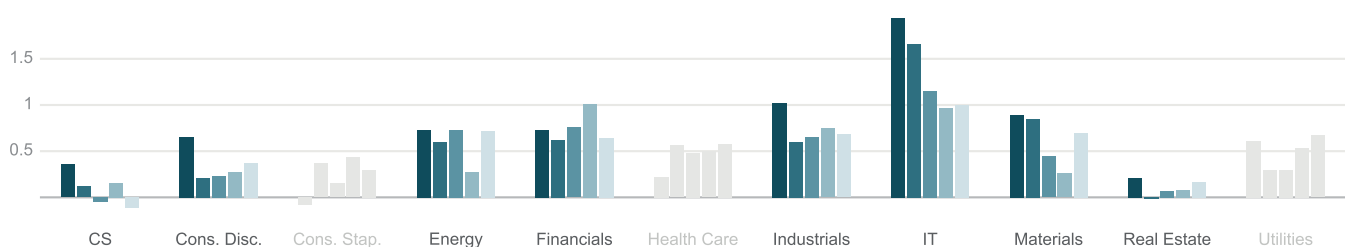
Volatility L-S CAGR +8.3%

**059 52W High Low**

Price Momentum L-S CAGR +8.3%

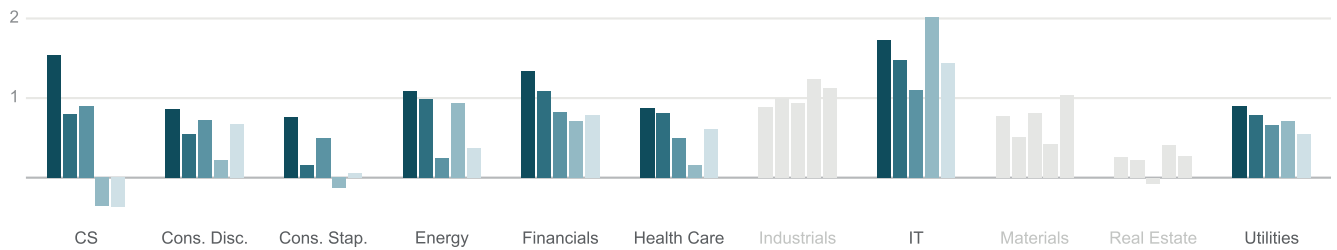
**060 52-Week Volume Price Trend with 20-Day Lag**

Price Momentum L-S CAGR +8.2%

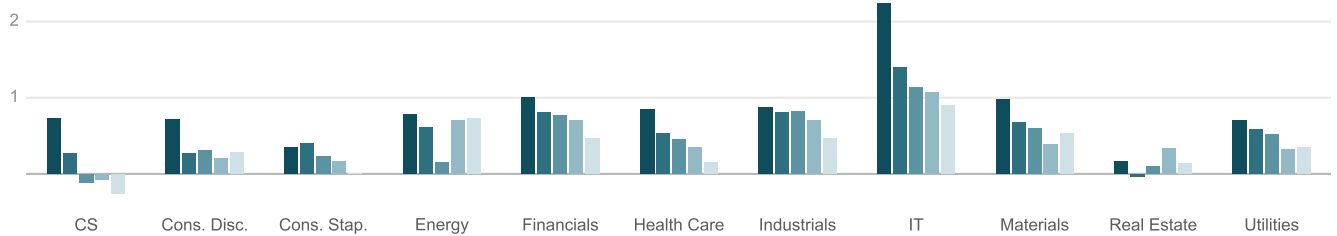


061 ● **Cash to Adjusted Enterprise Value**

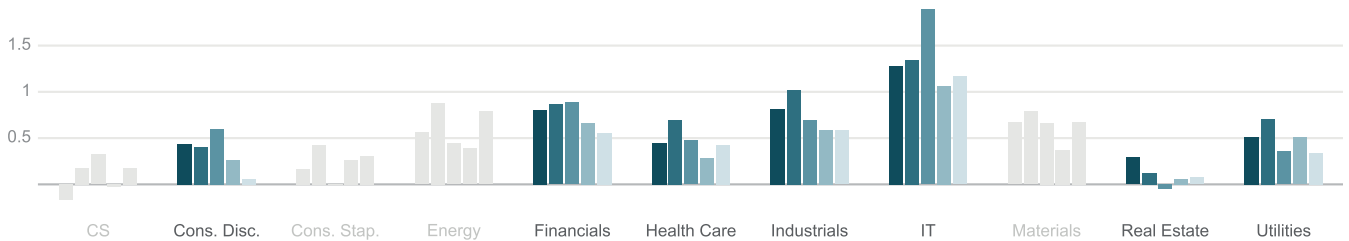
Valuation L-S CAGR +8.2%

062 ● **3M Revision in FY2 EPS Estimate**

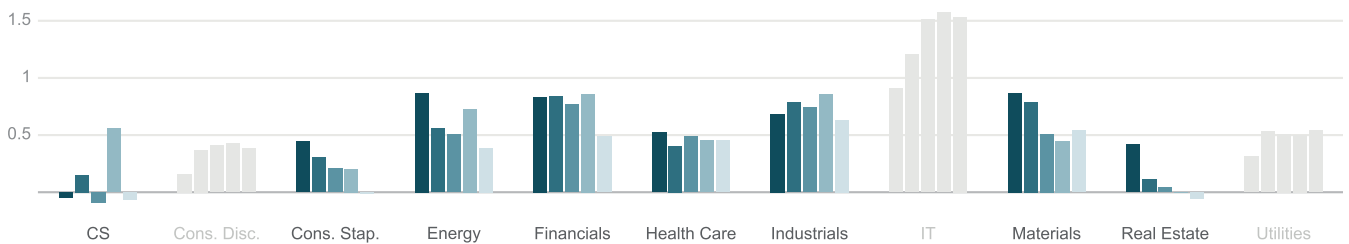
Analyst Expectations L-S CAGR +8.1%

063 ● **Institution Ownership Level - Active Manager**

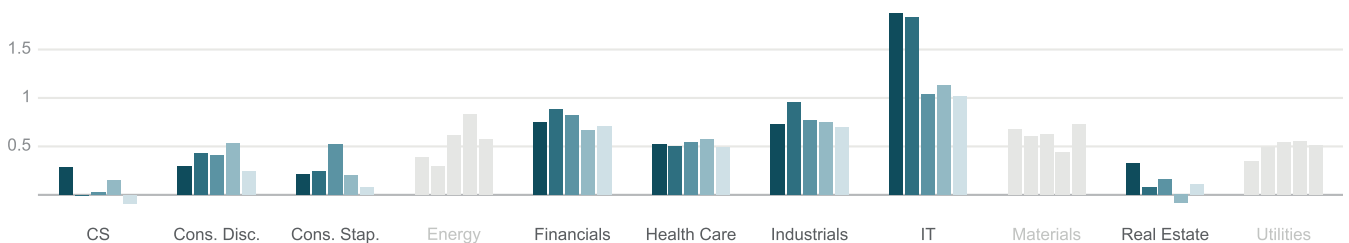
Price Momentum L-S CAGR +8.1%

064 ● **Operating Cash Flow to Price**

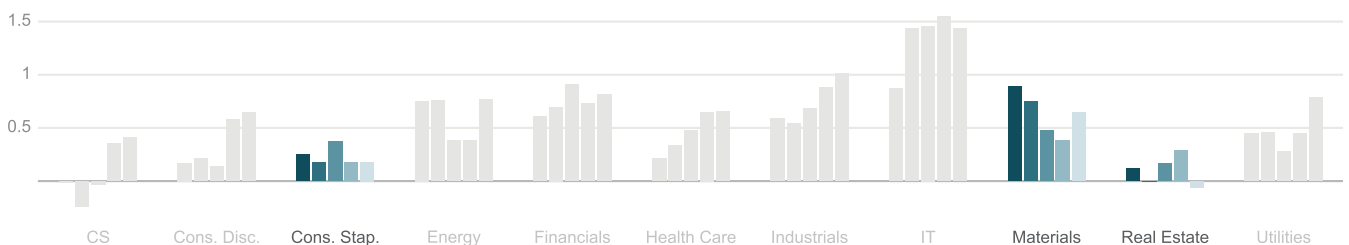
Valuation L-S CAGR +8.1%

065 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

Historical Growth L-S CAGR +8.0%

066 ● **Log of Unadjusted Stock Price**

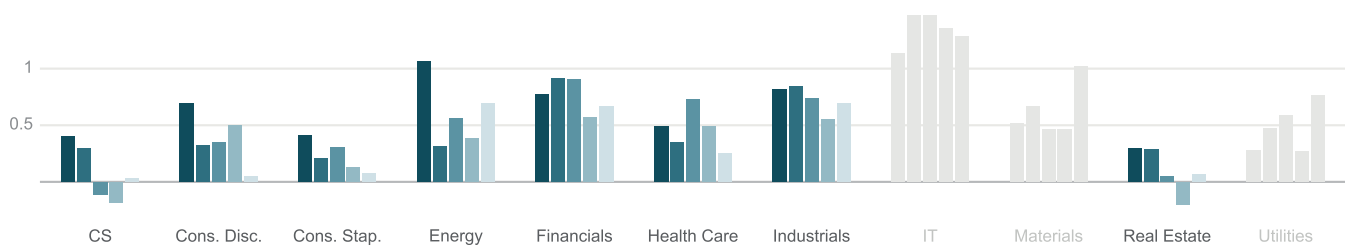
Price Momentum L-S CAGR +8.0%



067 3Y Chg in Operating Cash Flow to Price

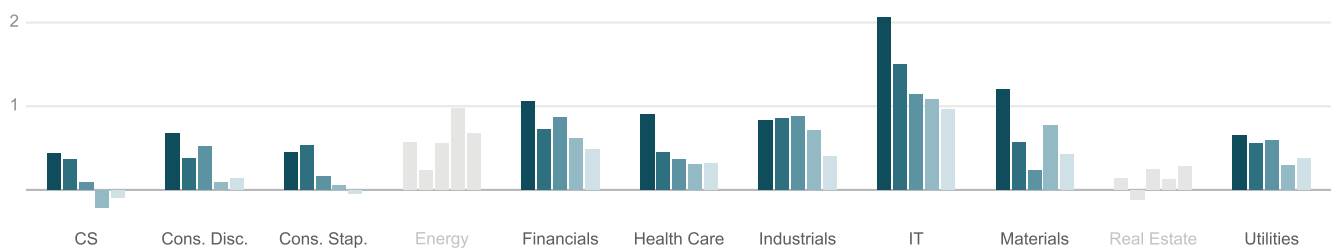
Historical Growth

L-S CAGR +8.0%

**068 Street Revision Magnitude**

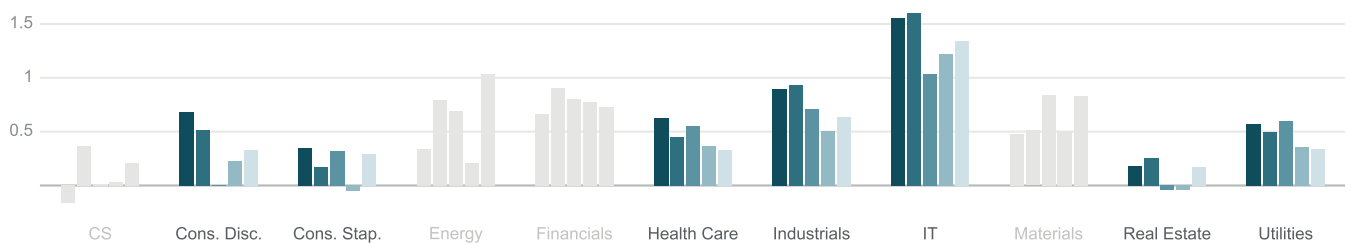
Analyst Expectations

L-S CAGR +7.9%

**069 Cash Flow to Equity**

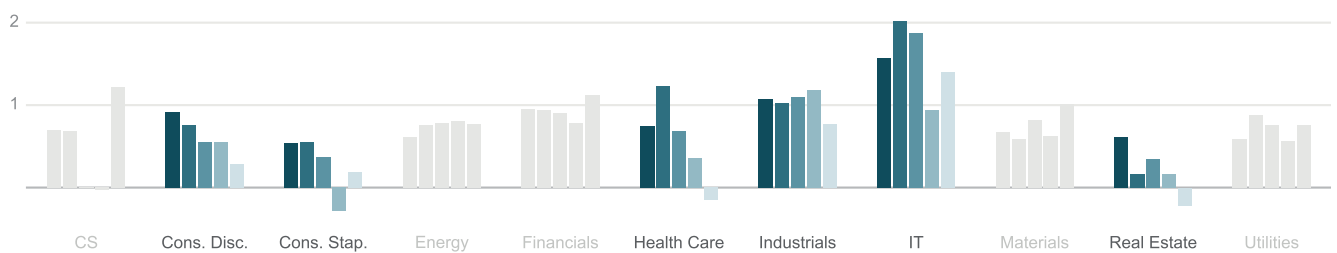
Valuation

L-S CAGR +7.9%

**070 Operating Cash Flow to Invested Capital**

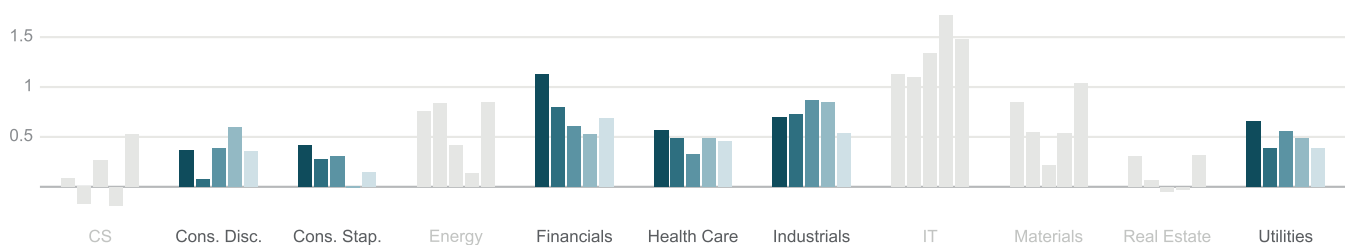
Earnings Quality

L-S CAGR +7.9%

**071 Forward Earnings to Price**

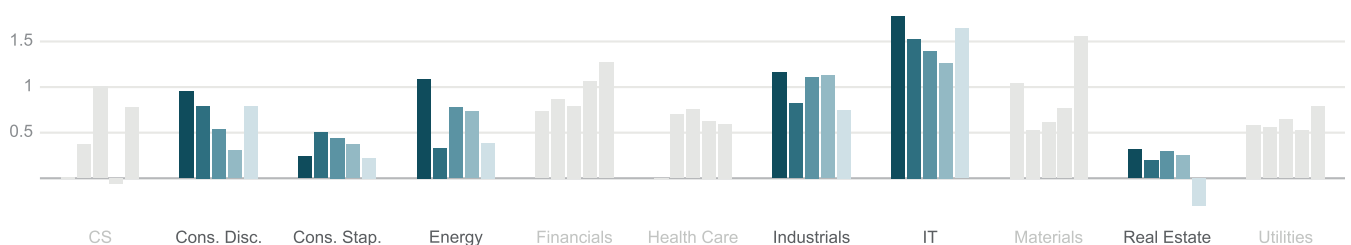
Valuation

L-S CAGR +7.9%

**072 12-Quarter Trendline in Trailing 12-Month Earnings**

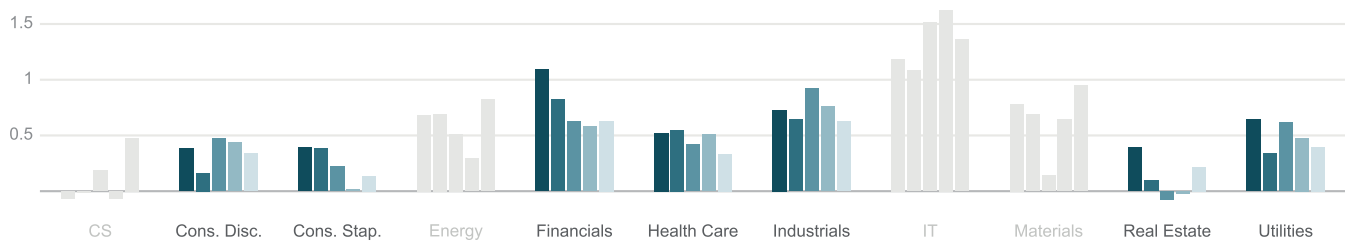
Historical Growth

L-S CAGR +7.7%



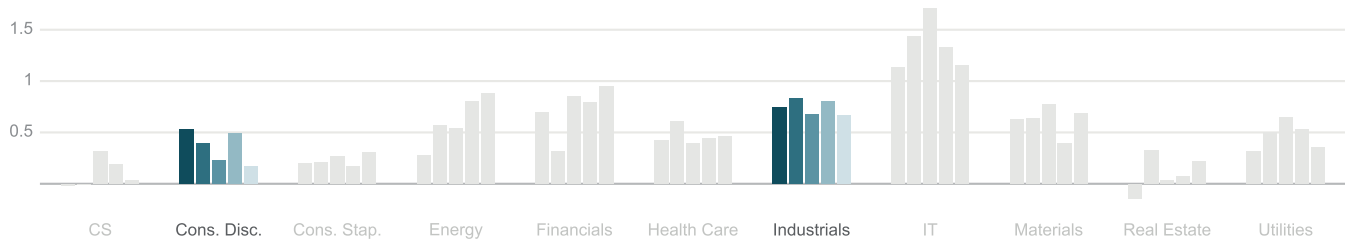
073 ● Adjusted Forward Earnings to Price

Valuation L-S CAGR +7.7%



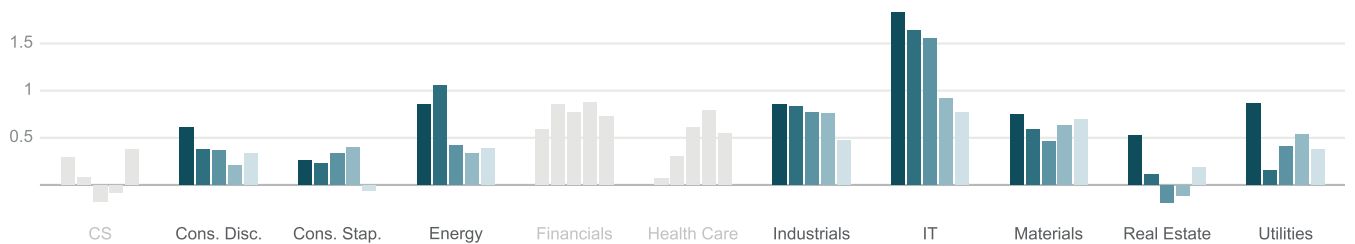
074 ● Gross Profit Margin

Earnings Quality L-S CAGR +7.7%



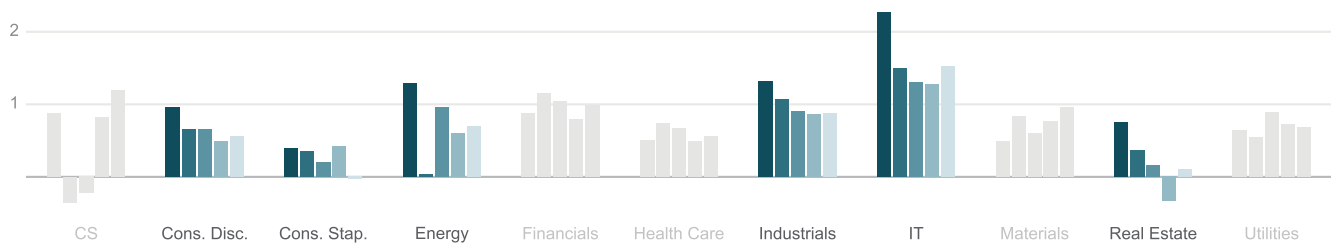
075 ● 60M CAPM Beta

Volatility L-S CAGR +7.6%



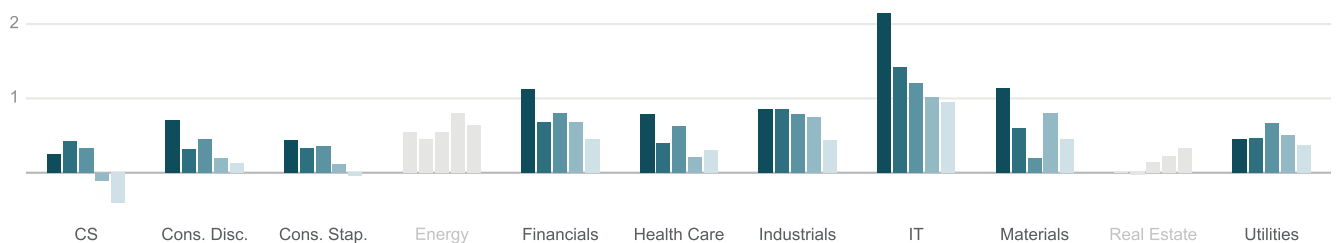
076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +7.6%



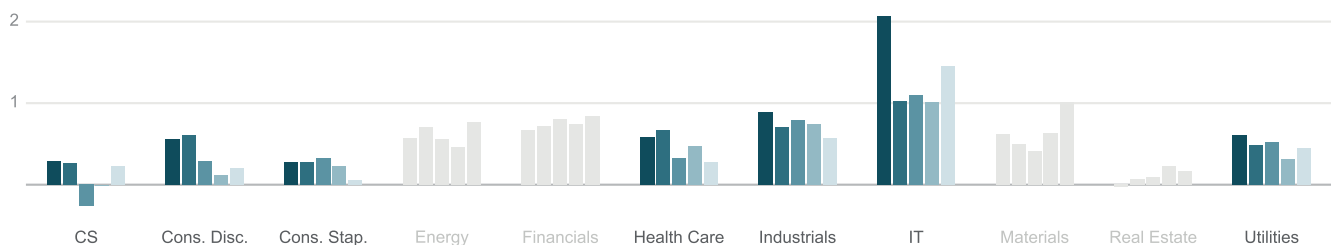
077 ● Adjusted Revision Magnitude

Analyst Expectations L-S CAGR +7.6%



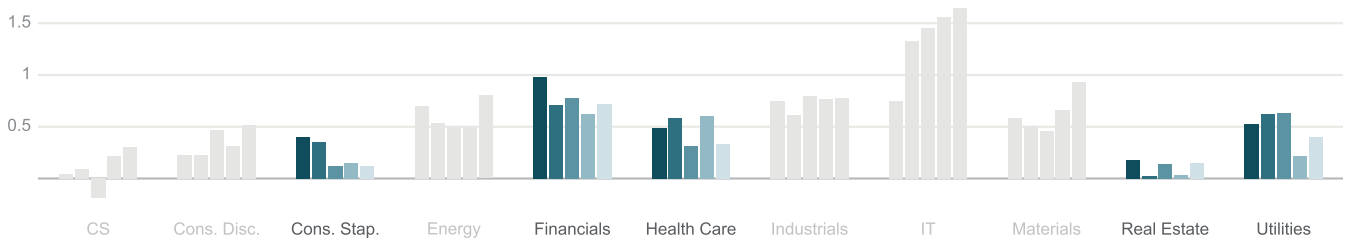
078 ● 1Y Chg in Asset Adjusted EPS

Historical Growth L-S CAGR +7.5%



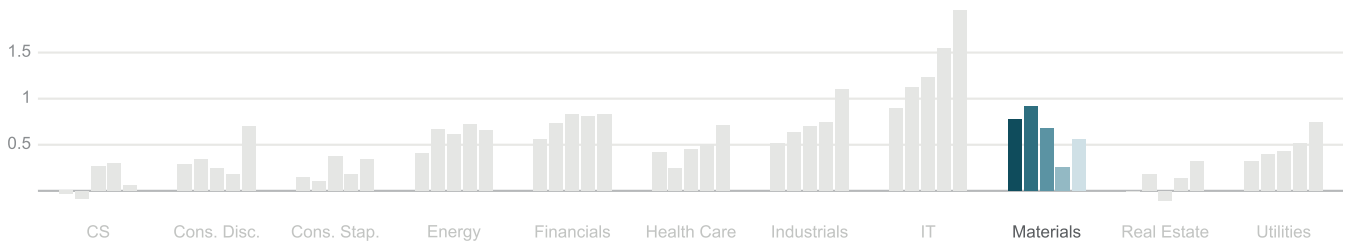
079 ● Net Cash Flow to Price

Valuation L-S CAGR +7.5%



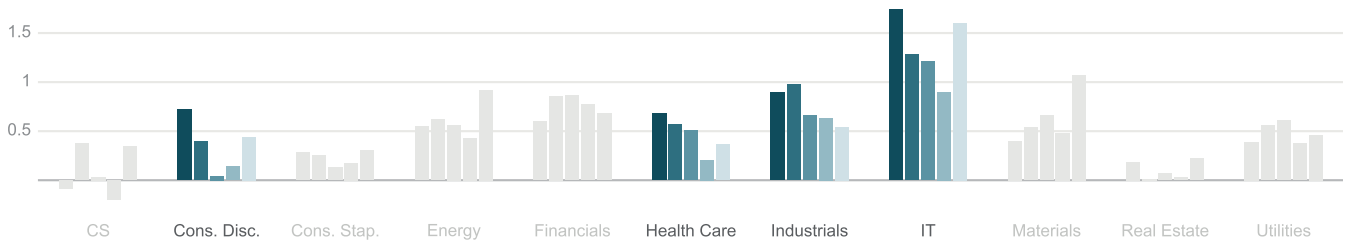
080 ● Sharpe Ratio

Price Momentum L-S CAGR +7.3%



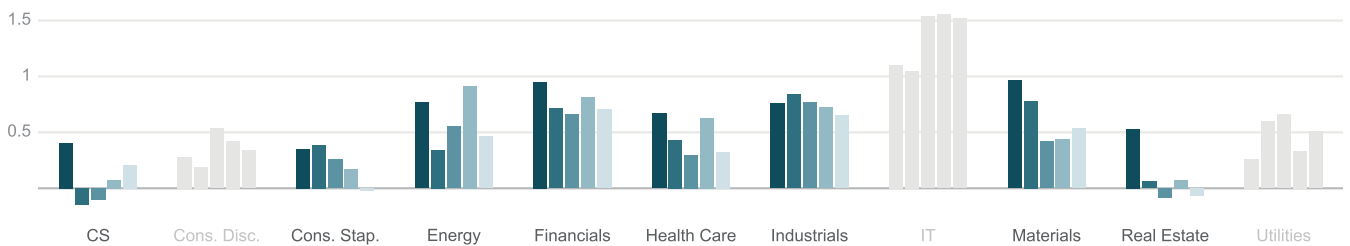
081 ● Return on Equity

Capital Efficiency L-S CAGR +7.2%



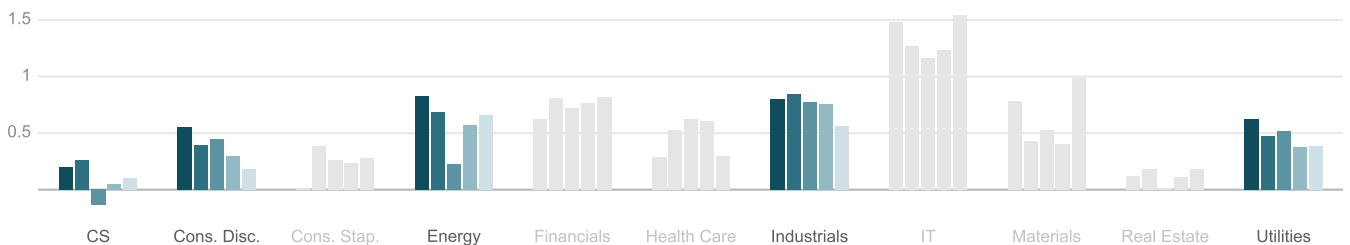
082 ● Operating Cash Flow to Enterprise Value

Valuation L-S CAGR +7.2%



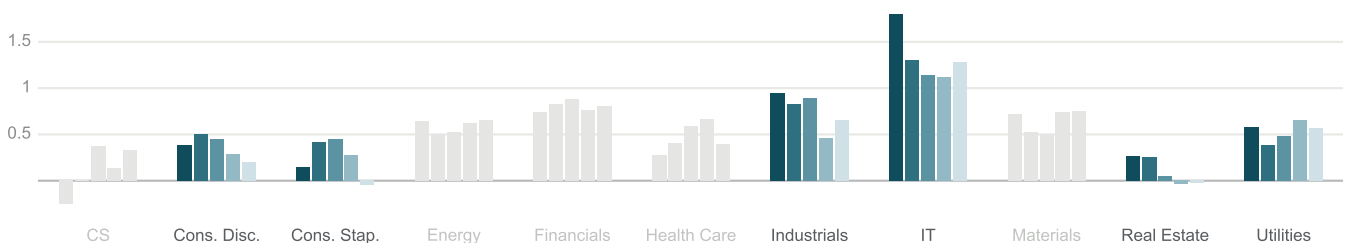
083 ● 1Y Chg in Operating Margin

Historical Growth L-S CAGR +7.2%



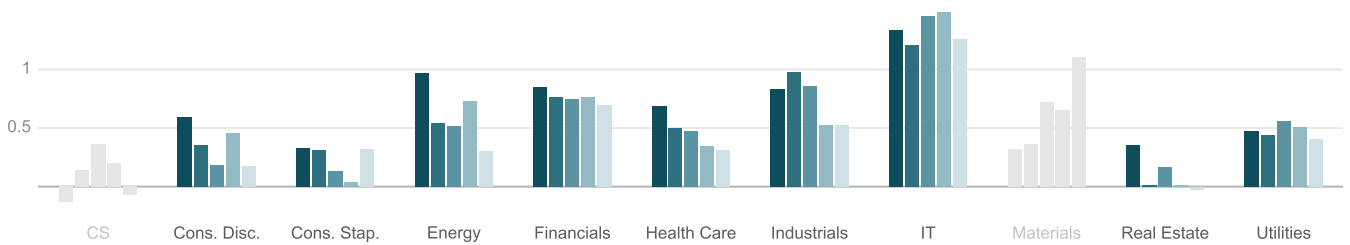
084 ● 1Y Chg in Free Cash Flow per Share

Historical Growth L-S CAGR +7.1%

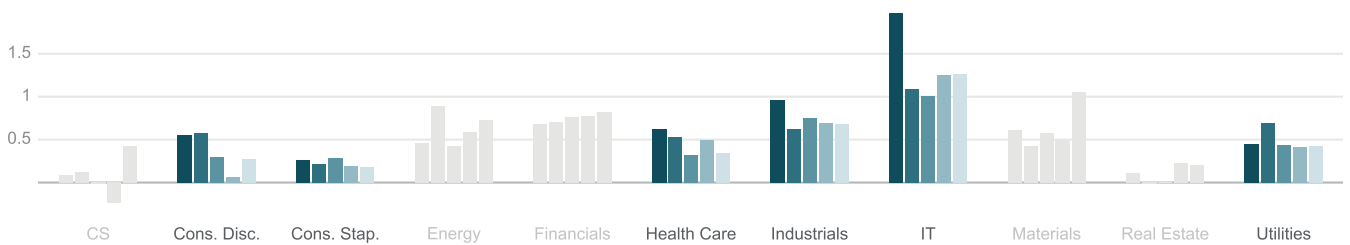


085 Equity Turnover

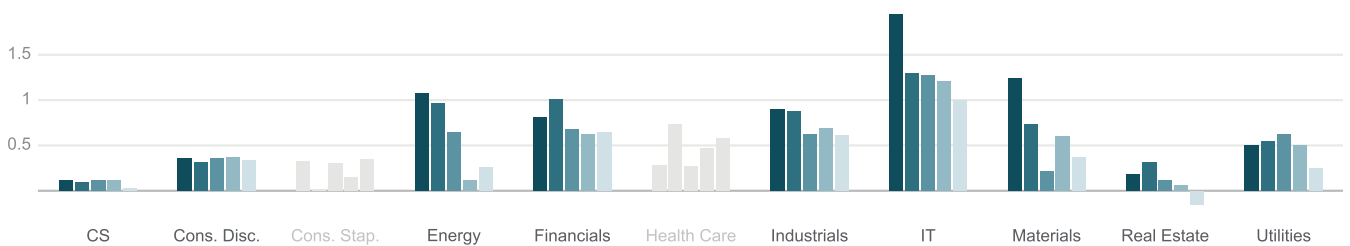
Earnings Quality L-S CAGR +7.1%

**086 1Y Chg in Asset Adjusted Cash Flow**

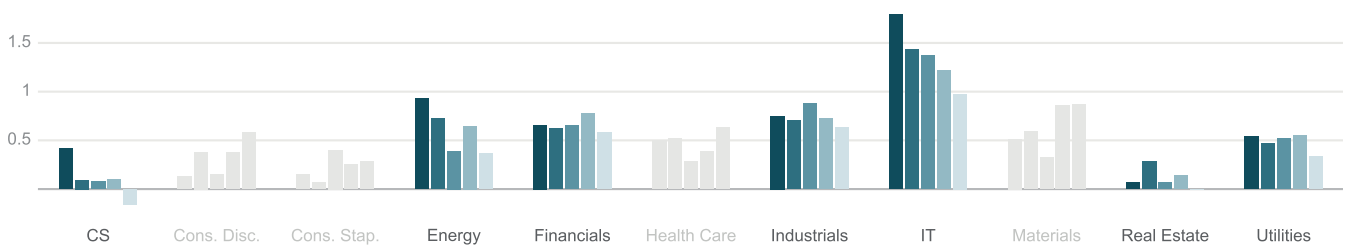
Historical Growth L-S CAGR +7.1%

**087 90 Day Coefficient of Variation**

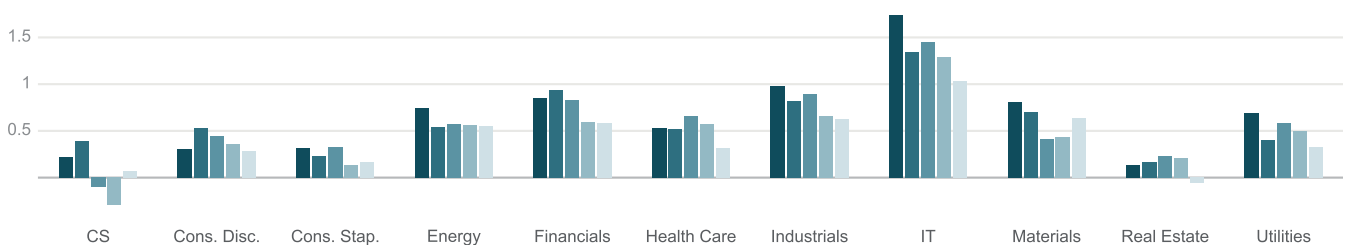
Volatility L-S CAGR +7.1%

**088 Working Capital to Assets Ratio**

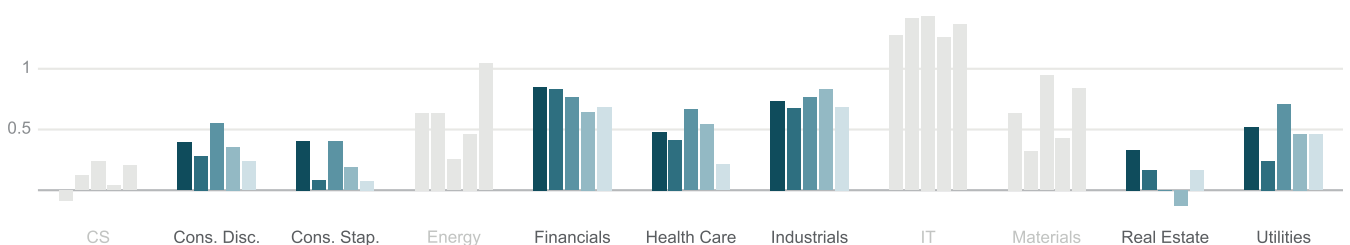
Earnings Quality L-S CAGR +7.0%

**089 Inverse PEG**

Analyst Expectations L-S CAGR +7.0%

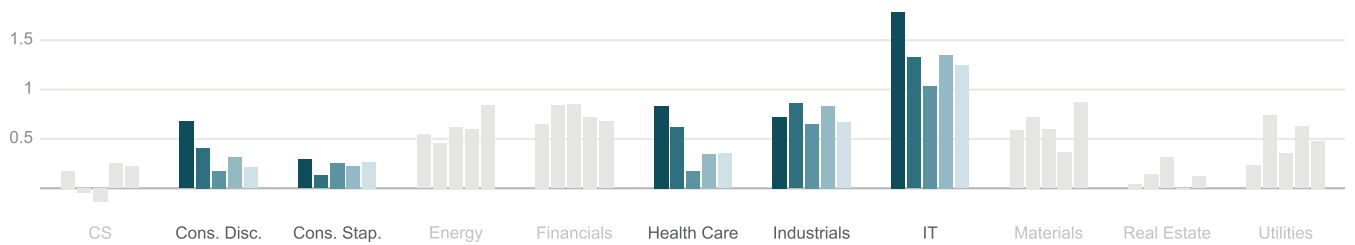
**090 1Y Chg in Sales to Price**

Historical Growth L-S CAGR +7.0%

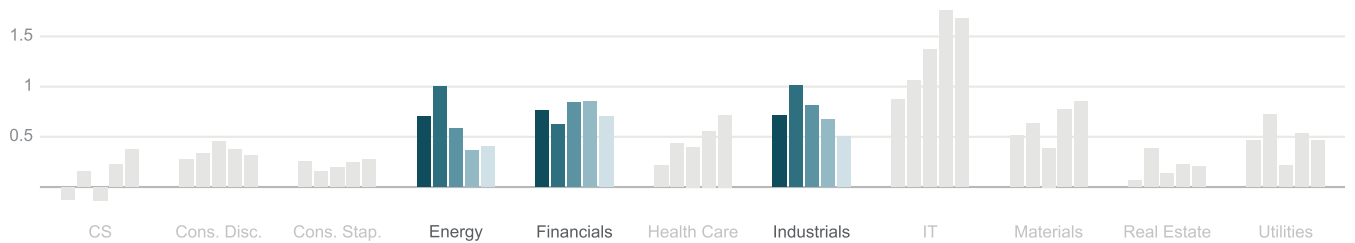


091 ● **Normalized ROA**

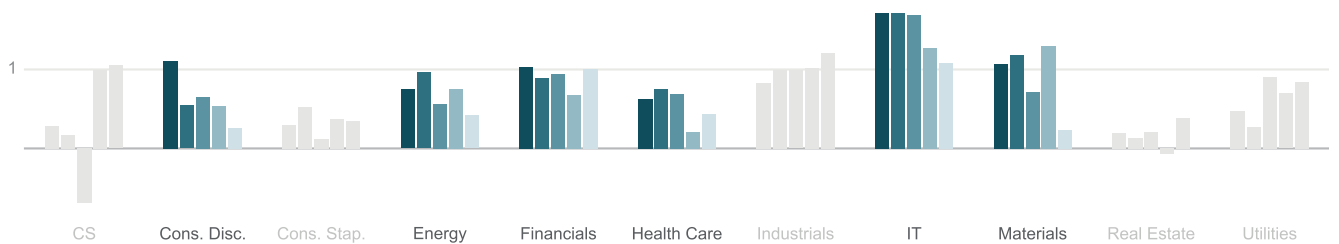
Capital Efficiency L-S CAGR +7.0%

092 ● **Capital Expenditure to Total Assets**

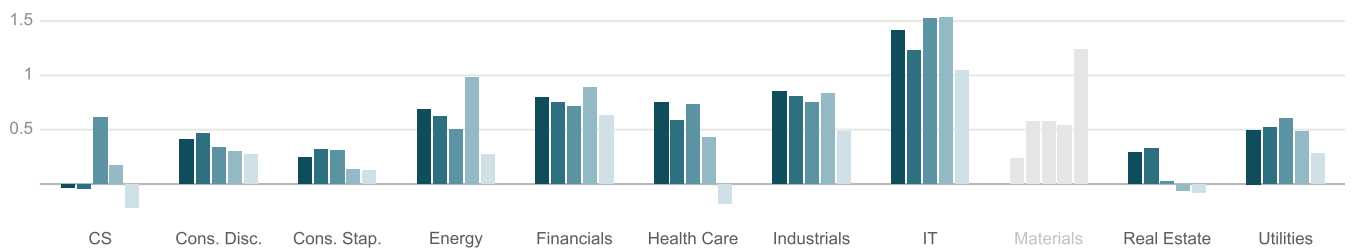
Capital Efficiency L-S CAGR +6.9%

093 ● **Consecutive Quarters of Declines in TTM Overhead/Sales**

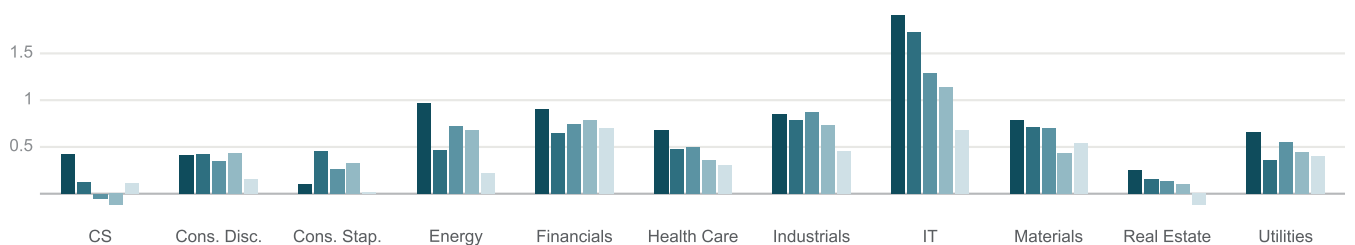
Historical Growth L-S CAGR +6.9%

094 ● **Sales to Invested Capital**

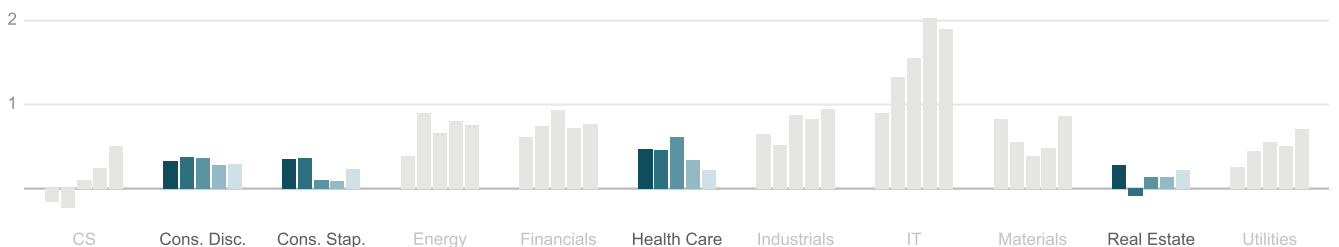
Earnings Quality L-S CAGR +6.9%

095 ● **26W Price Stat**

Price Momentum L-S CAGR +6.9%

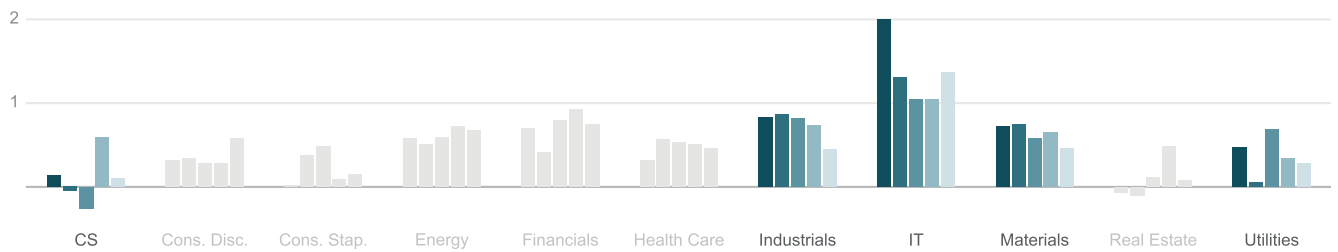
096 ● **Inventory Turnover**

Earnings Quality L-S CAGR +6.8%



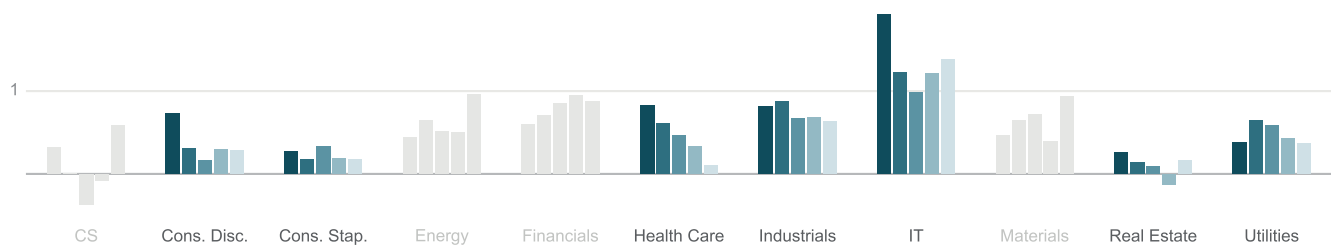
097 ● Year over Year Change in SGA to Sales

Earnings Quality L-S CAGR +6.7%



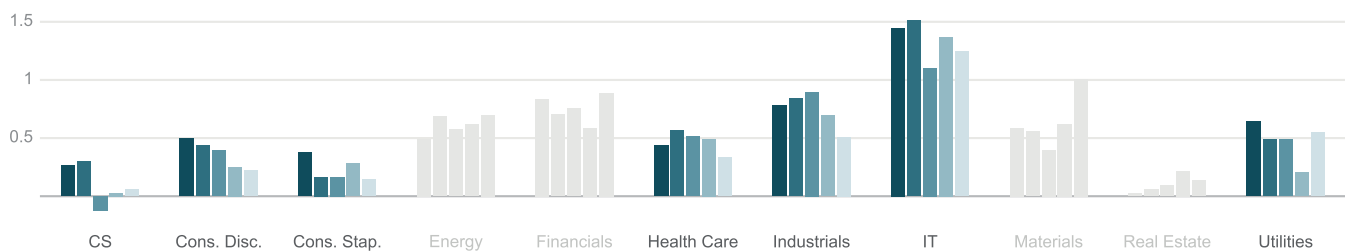
098 ● EBIT to Assets

Capital Efficiency L-S CAGR +6.7%



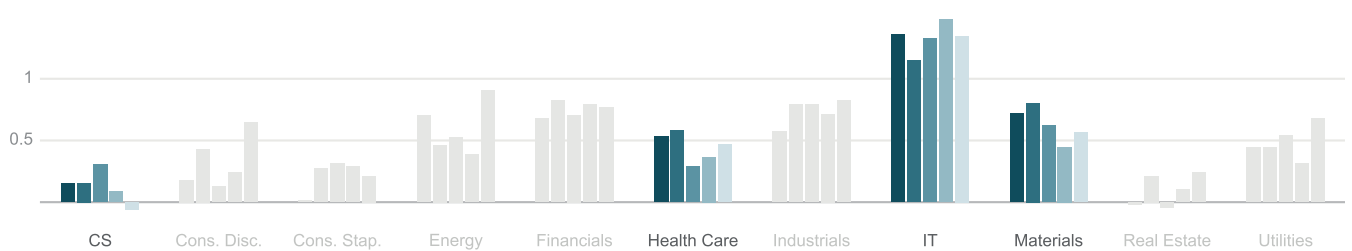
099 ● 1Y Chg in Earnings to Price

Historical Growth L-S CAGR +6.7%



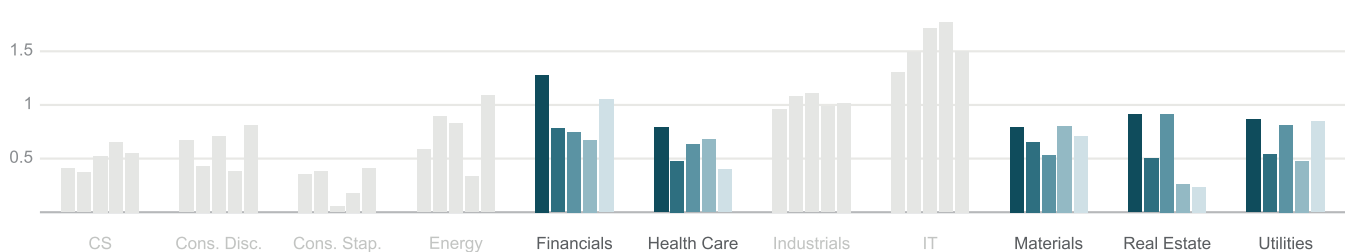
100 ● Long Term Debt to Equity Ratio

Capital Efficiency L-S CAGR +6.7%



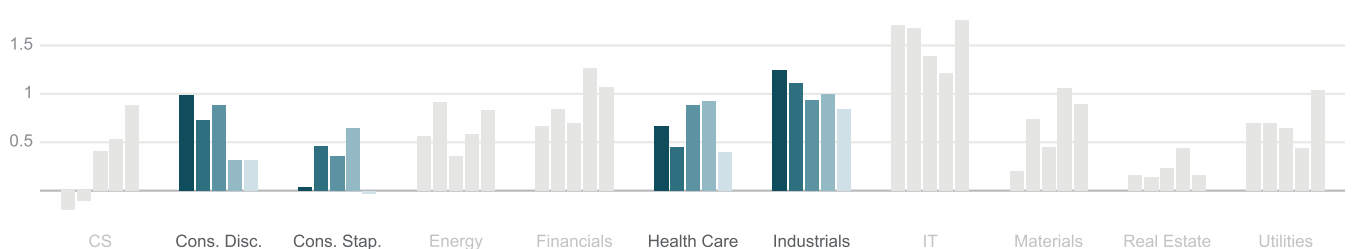
101 ● Adj. EBITDA to Enterprise Value

Valuation L-S CAGR +6.6%



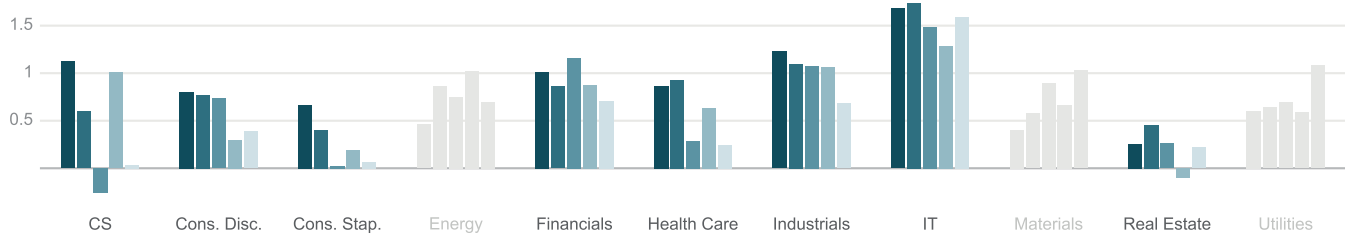
102 ● Adj 1Y Chg in Earnings per Share

Historical Growth L-S CAGR +6.6%



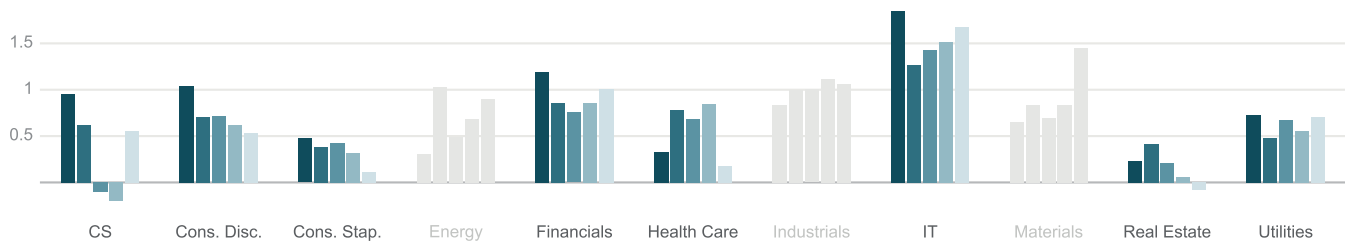
103 ● Short Interest Ratio

Price Momentum L-S CAGR +6.6%



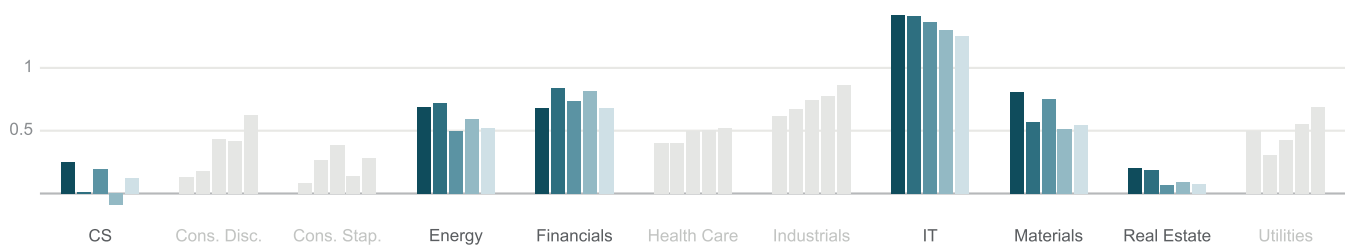
104 ● 12-Quarter Trendline in Trailing 12-Month Cash Flows

Historical Growth L-S CAGR +6.5%



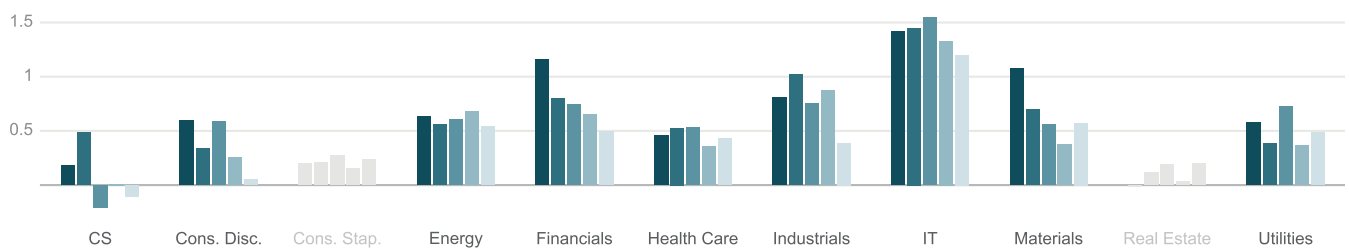
105 ● Analyst Dispersion for FY1 EPS

Analyst Expectations L-S CAGR +6.5%



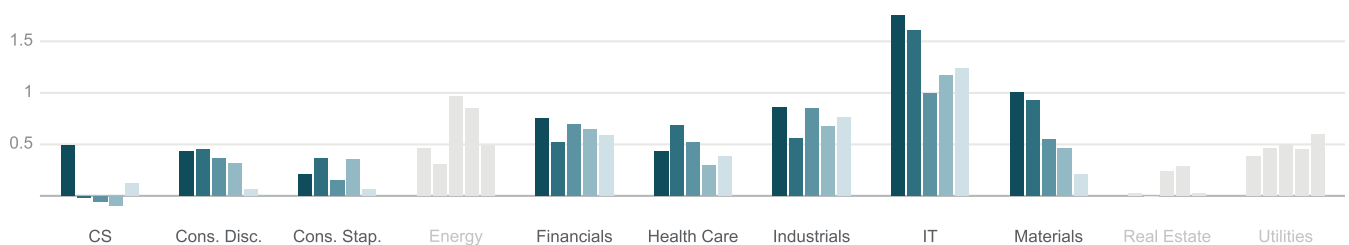
106 ● Earnings Surprise

Analyst Expectations L-S CAGR +6.4%



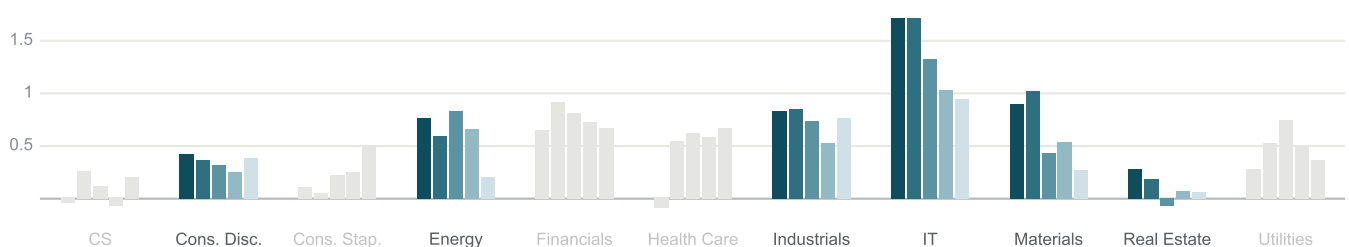
107 ● Operating Cash Flow Ratio

Capital Efficiency L-S CAGR +6.3%



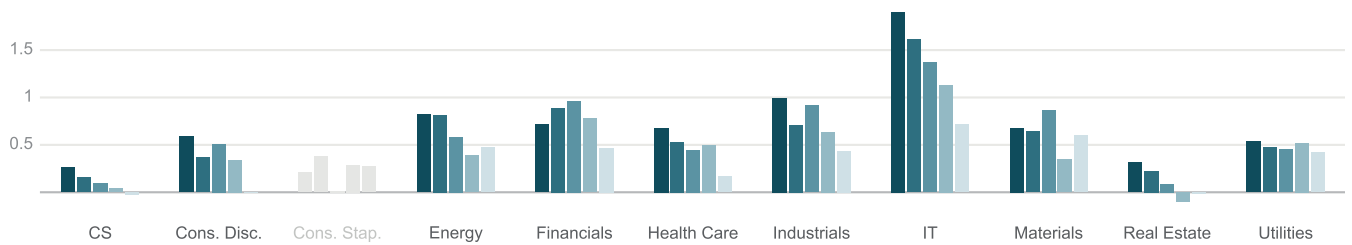
108 ● 1M Realized Price Volatility

Volatility L-S CAGR +6.2%



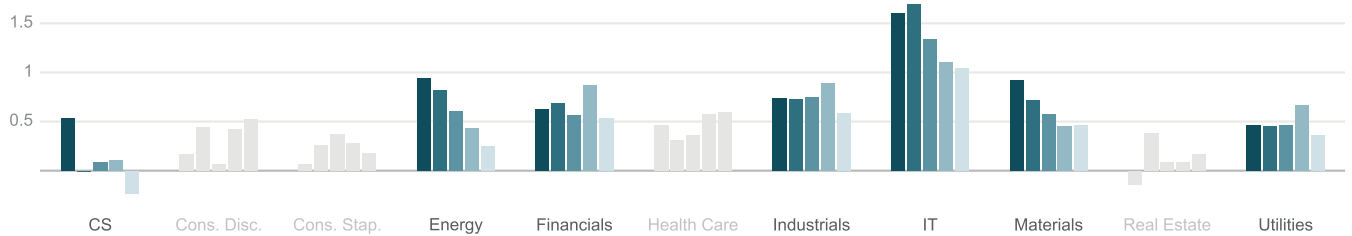
109 ● Volatility adjusted 12M return

Price Momentum L-S CAGR +6.2%



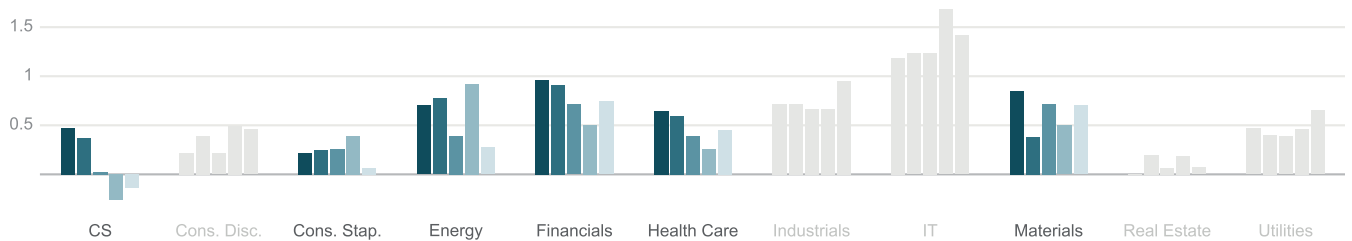
110 ● Current Ratio

Earnings Quality L-S CAGR +6.2%



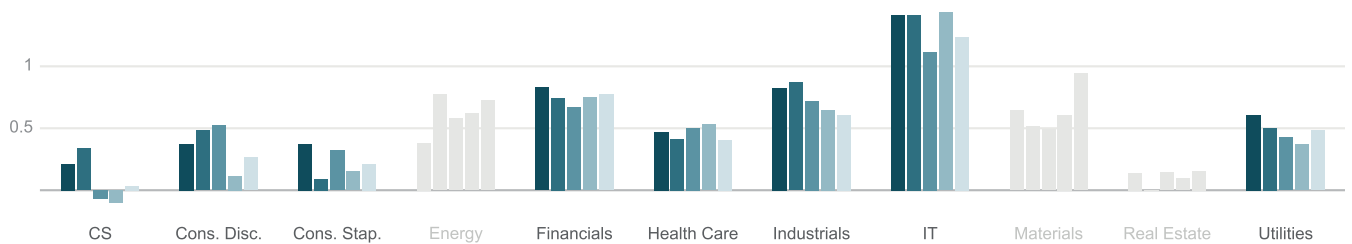
111 ● Cash to EV Ratio

Valuation L-S CAGR +6.2%



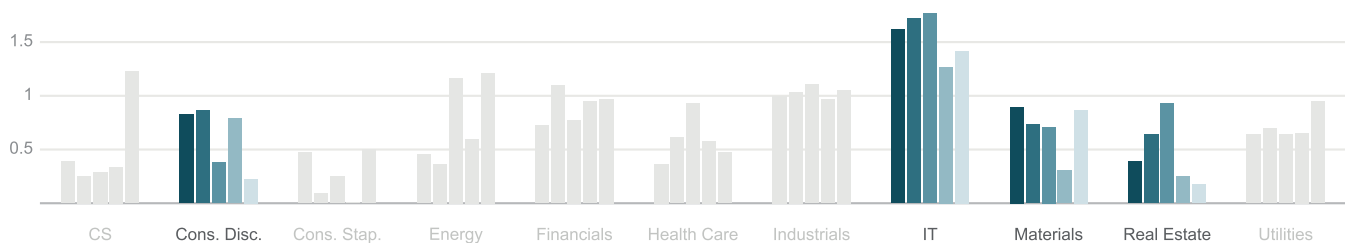
112 ● 1Y Chg in Cash Flow to Price

Historical Growth L-S CAGR +6.1%



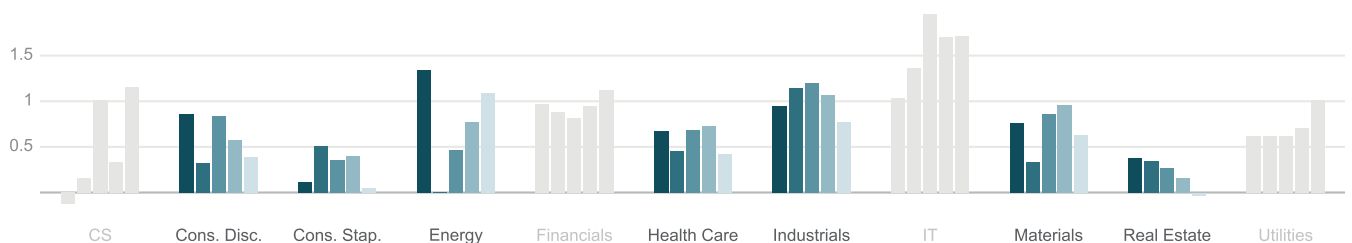
113 ● EBITDA Margin

Earnings Quality L-S CAGR +6.1%



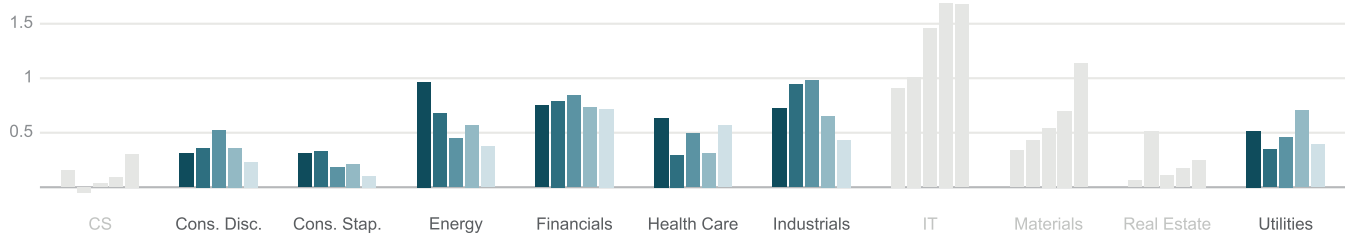
114 ● Adj Free Cash Flow to Price

Valuation L-S CAGR +6.0%

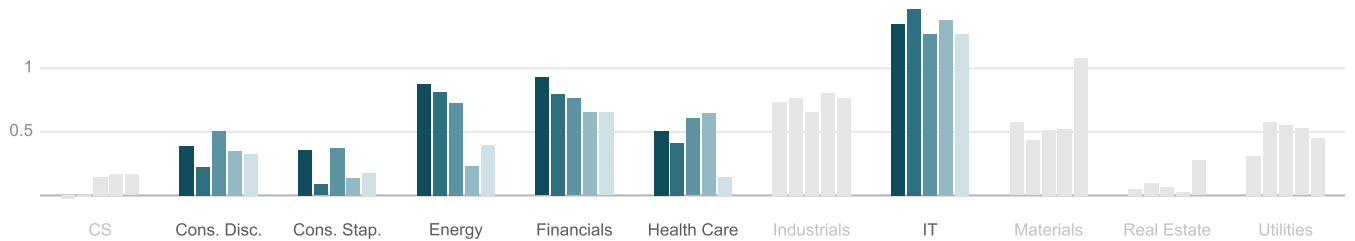


115 Capital Expenditure to Sales

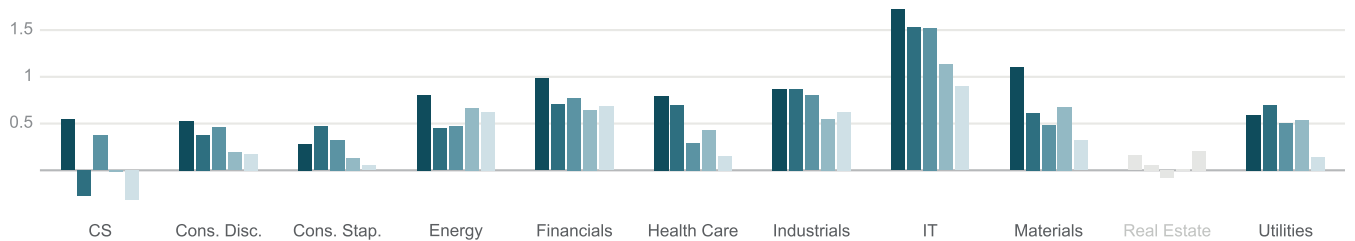
Earnings Quality L-S CAGR +6.0%

**116 Trailing 12-Month Overhead/Trailing 12-Month Sales**

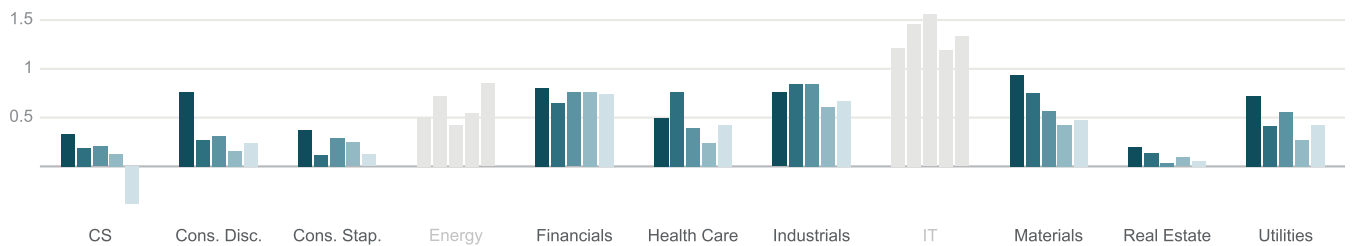
Historical Growth L-S CAGR +6.0%

**117 Adj Number of EPS FY2 Revisions**

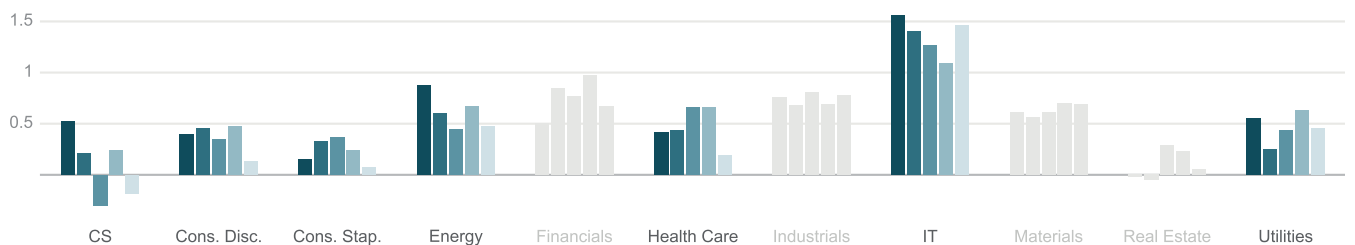
Analyst Expectations L-S CAGR +6.0%

**118 Working Capital Accruals**

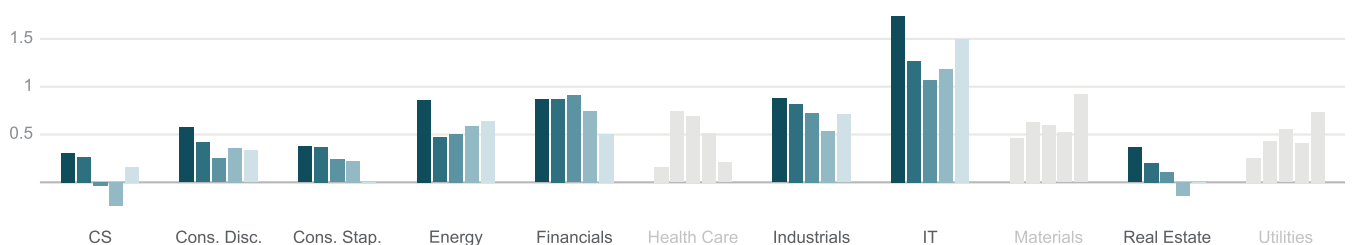
Earnings Quality L-S CAGR +6.0%

**119 Asset Growth - 1 Year**

Capital Efficiency L-S CAGR +6.0%

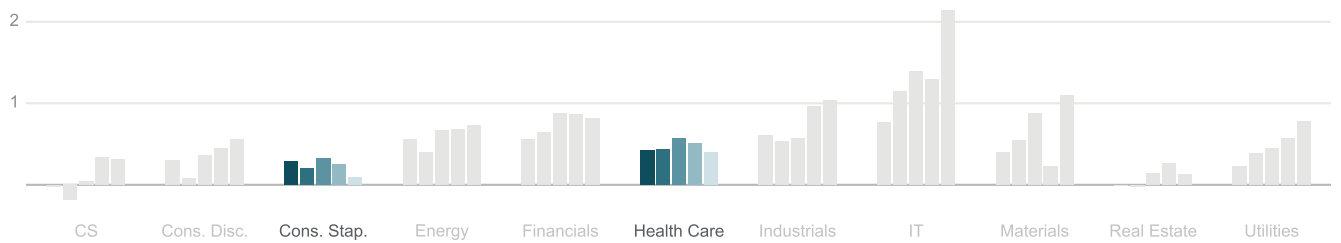
**120 3Y Chg in Operating Cash Flow per Share**

Historical Growth L-S CAGR +6.0%



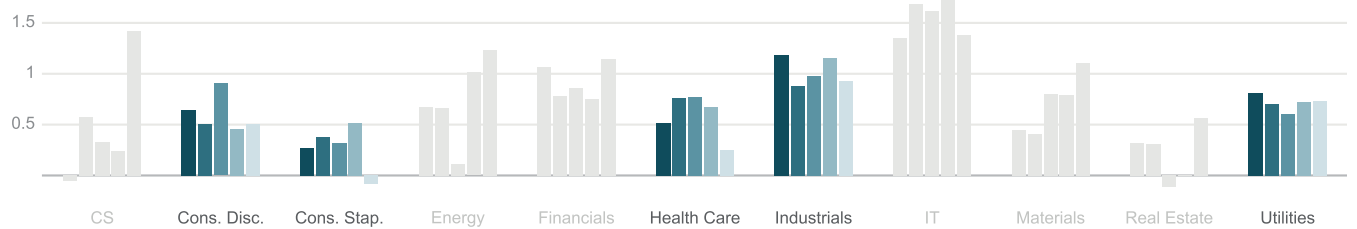
121 Adjusted 6 Month Average Share Turn Over

Price Momentum L-S CAGR +6.0%



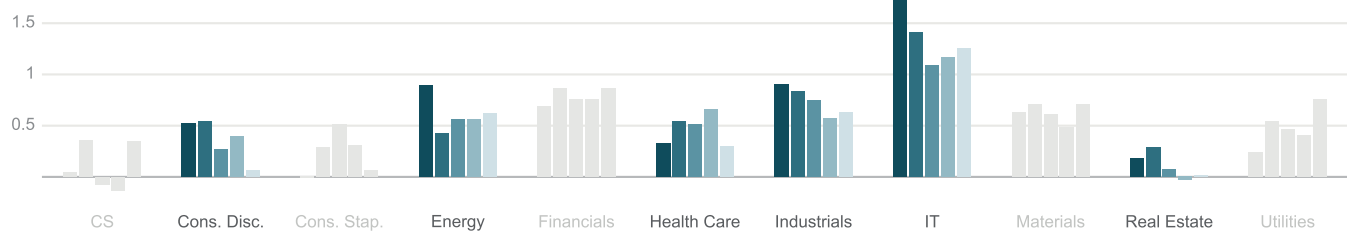
122 Adj Growth Flow to Price

Valuation L-S CAGR +6.0%



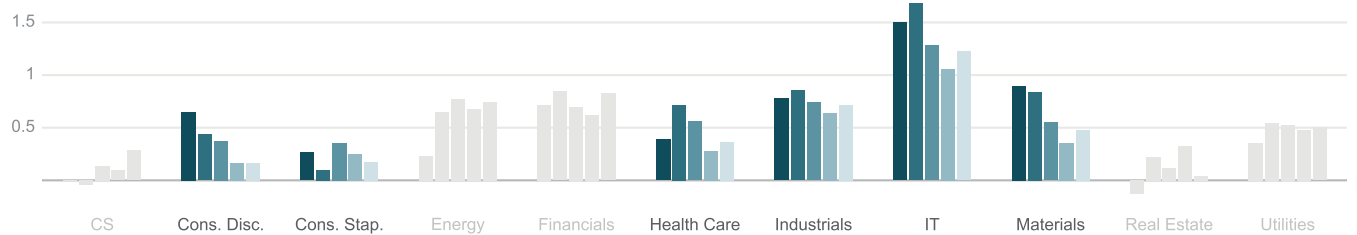
123 1Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +6.0%



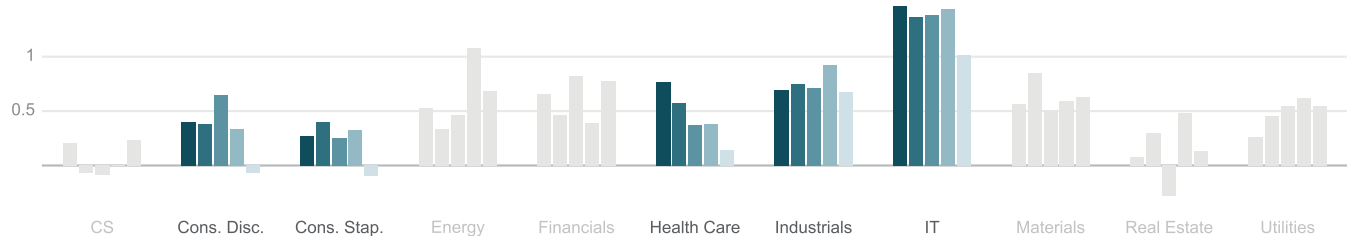
124 Operating Margin

Earnings Quality L-S CAGR +5.9%



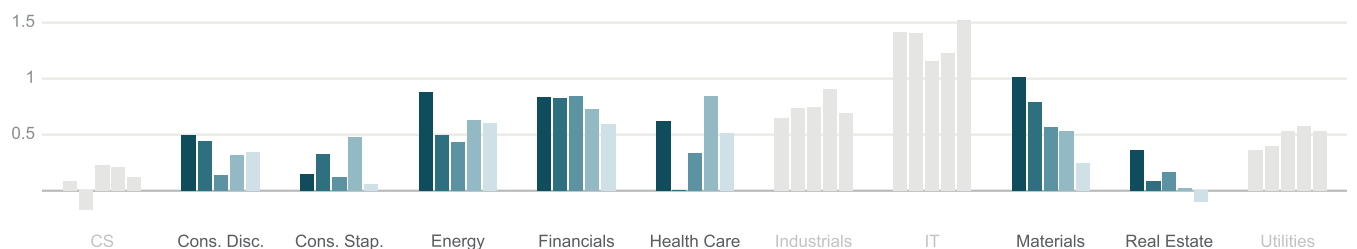
125 Adjusted Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.8%



126 Adjusted Accruals

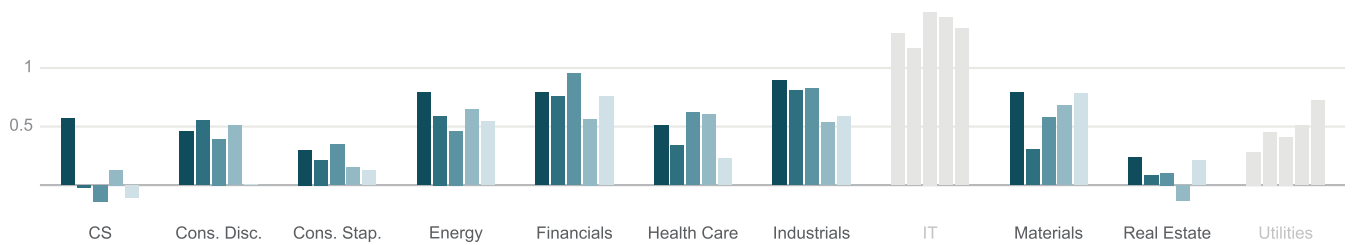
Earnings Quality L-S CAGR +5.8%



127 ● 3Y Chg in Free Cash Flow to Price

Historical Growth

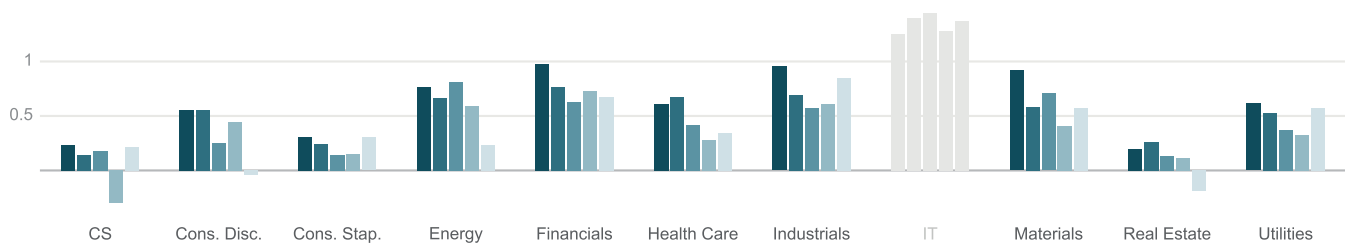
L-S CAGR +5.7%



128 ● Ownership Breadth Stability of Institution Ownership by Buyers

Price Momentum

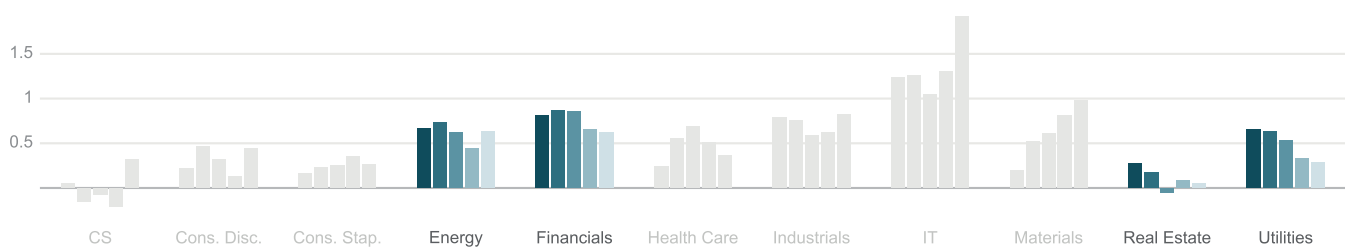
L-S CAGR +5.7%



129 ● Dividend per Share Growth

Historical Growth

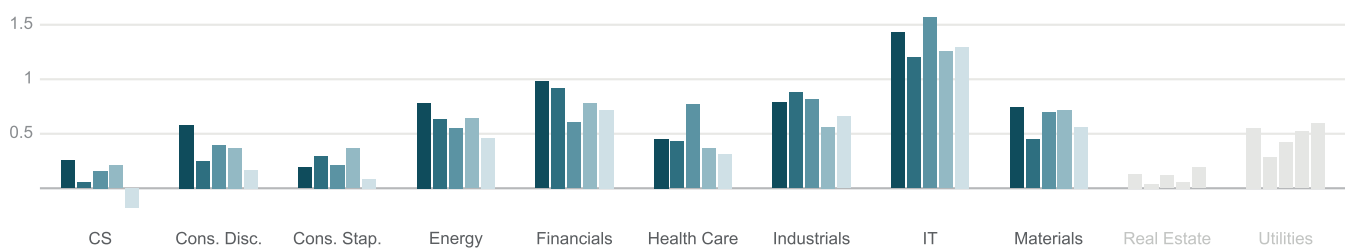
L-S CAGR +5.7%



130 ● Accrual Ratio - Balance Sheet

Earnings Quality

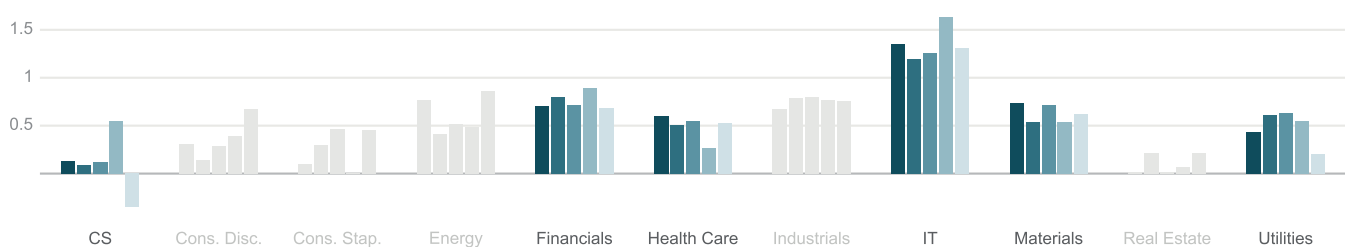
L-S CAGR +5.6%



131 ● Long Term Debt to Assets Ratio

Capital Efficiency

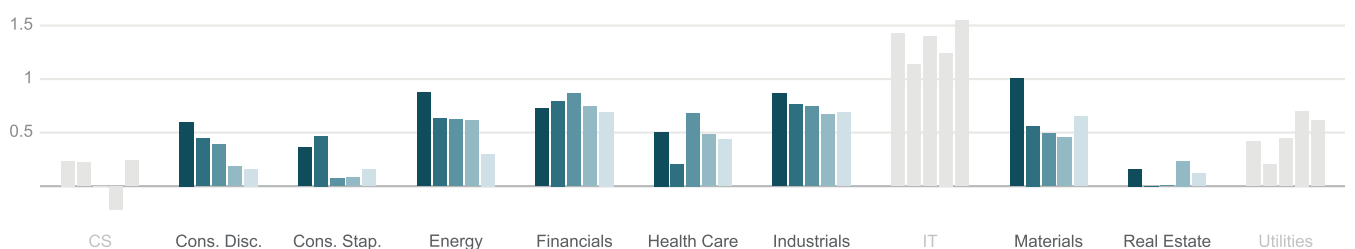
L-S CAGR +5.6%



132 ● Accrual Ratio - Cash Flows

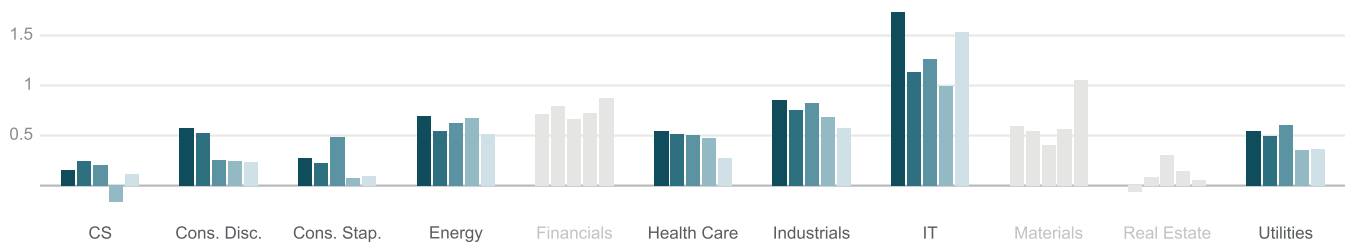
Earnings Quality

L-S CAGR +5.6%



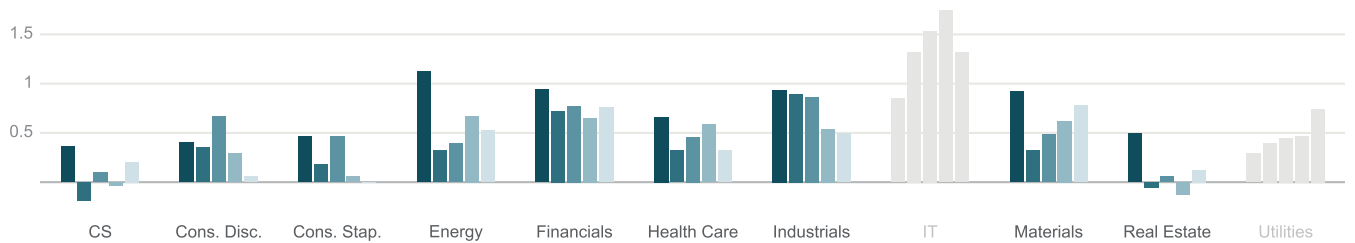
133 ● Year over Year Change of EPS to Sales

Earnings Quality L-S CAGR +5.5%



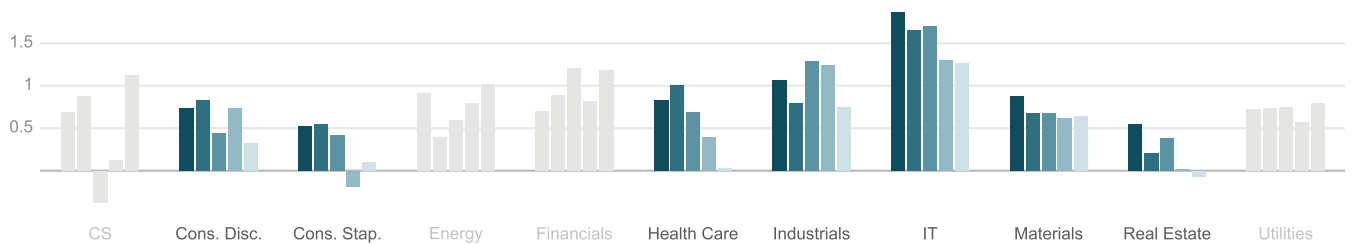
134 ● Free Cash Flow to Enterprise Value

Valuation L-S CAGR +5.5%



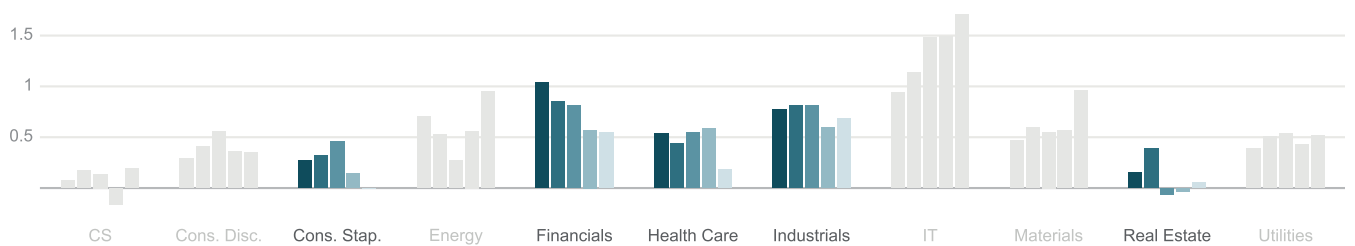
135 ● Operating Cash Flow to Asset

Earnings Quality L-S CAGR +5.5%



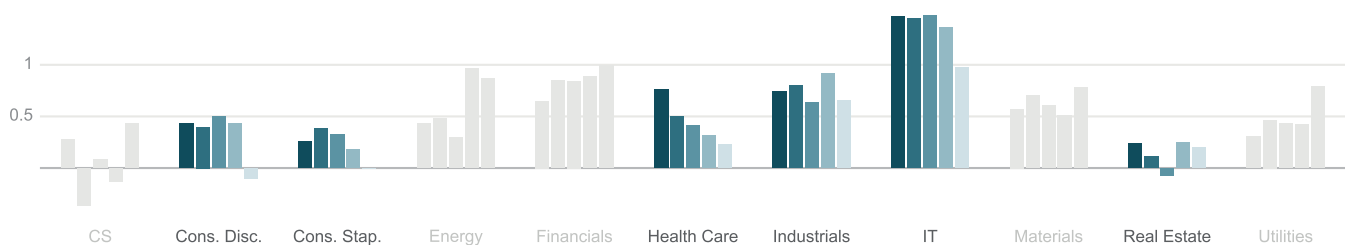
136 ● 3Y Chg in Sales to Price

Historical Growth L-S CAGR +5.5%



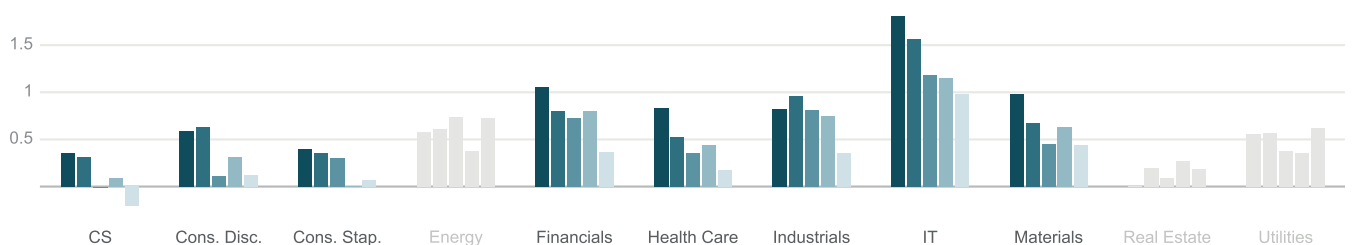
137 ● Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.4%



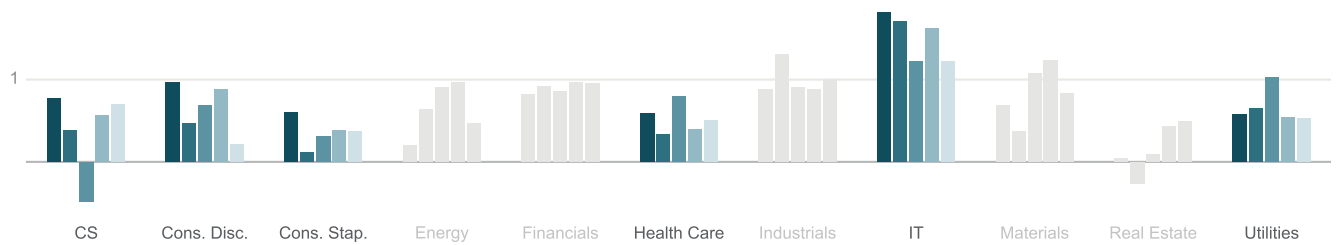
138 ● Number of EPS FY1 Revisions

Analyst Expectations L-S CAGR +5.4%



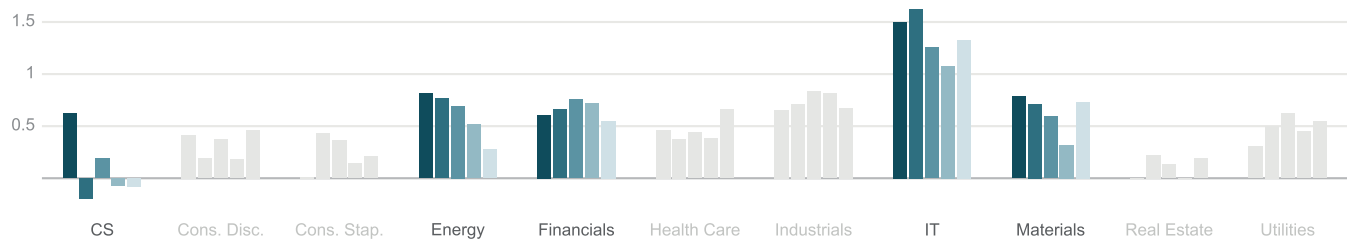
139 ● Num of Quarters of Declines in Receivables and Inventories/TTM Sales

Historical Growth L-S CAGR +5.4%



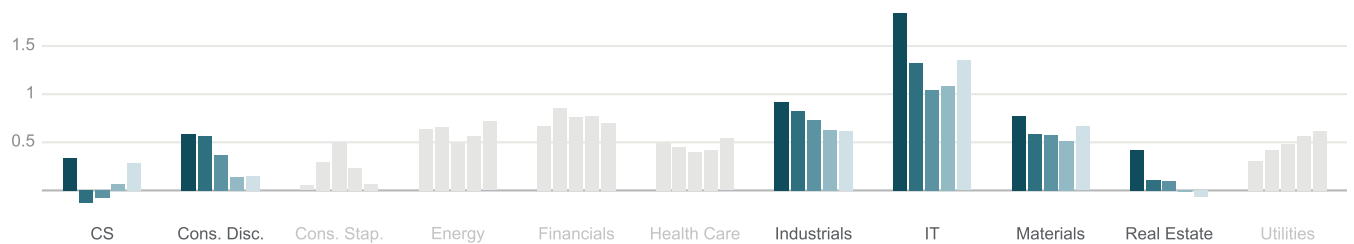
140 ● Quick Ratio

Earnings Quality L-S CAGR +5.3%



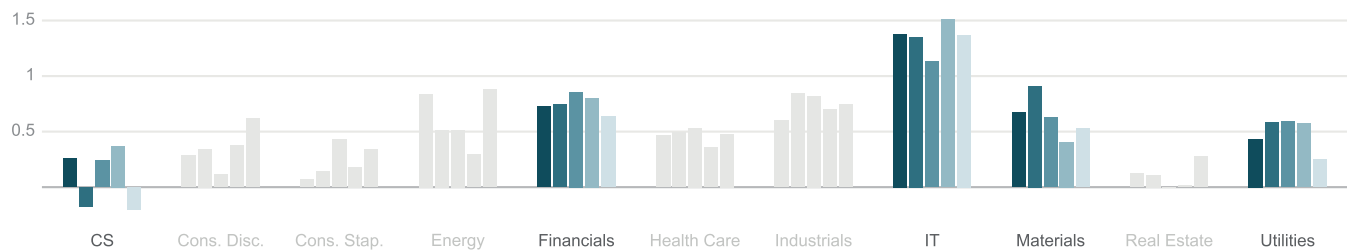
141 ● 1Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth L-S CAGR +5.3%



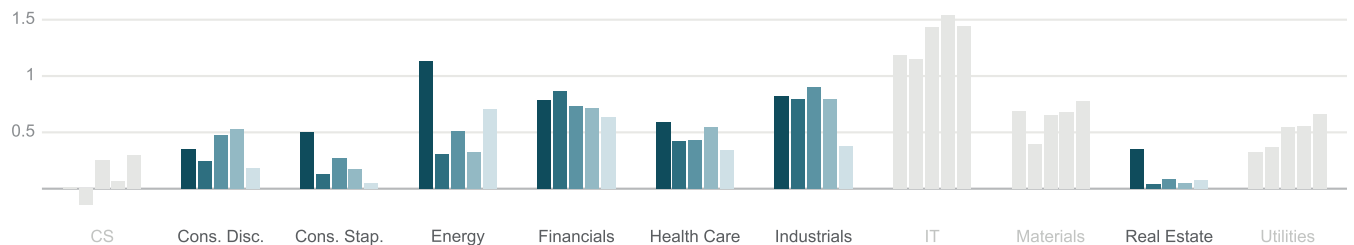
142 ● Debt to Assets Ratio

Capital Efficiency L-S CAGR +5.3%



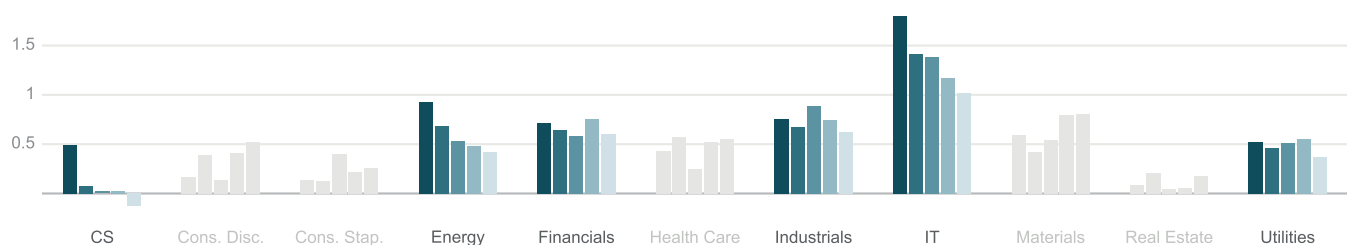
143 ● Forward Free Cash Flow to Price Ratio

Valuation L-S CAGR +5.3%



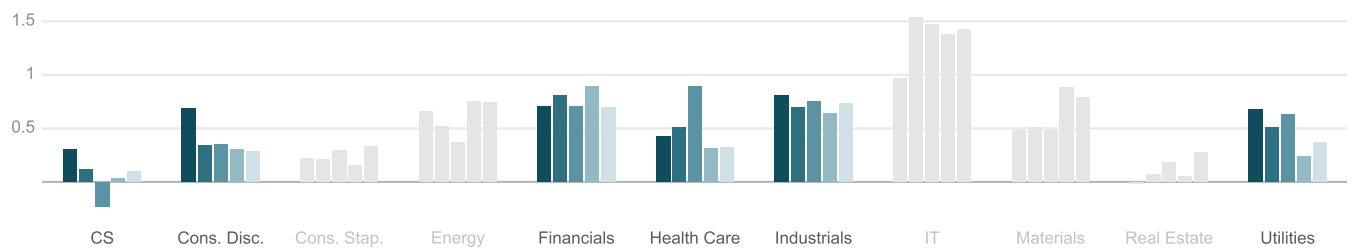
144 ● Working Capital to Total Assets

Earnings Quality L-S CAGR +5.2%



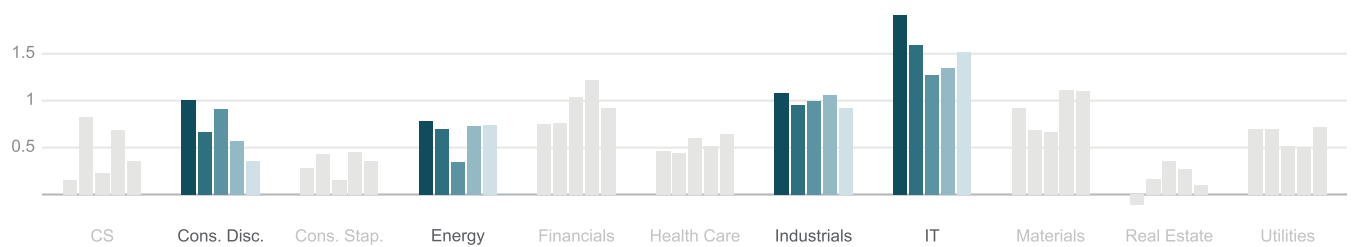
145 ● Net Income Stability

Earnings Quality L-S CAGR +5.2%



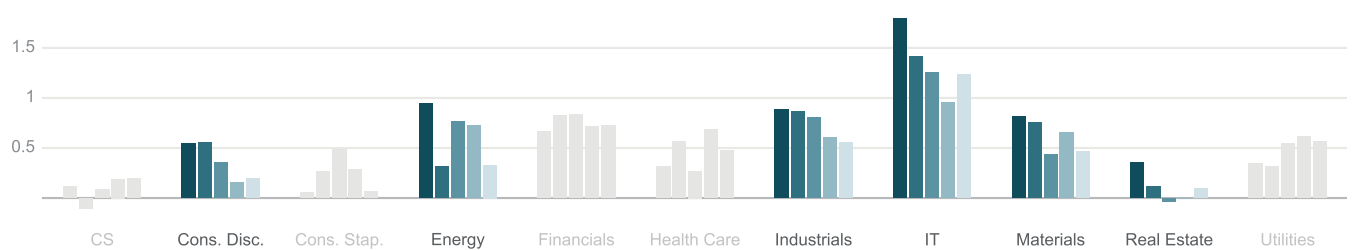
146 ● Return on Equity Stability

Historical Growth L-S CAGR +5.2%



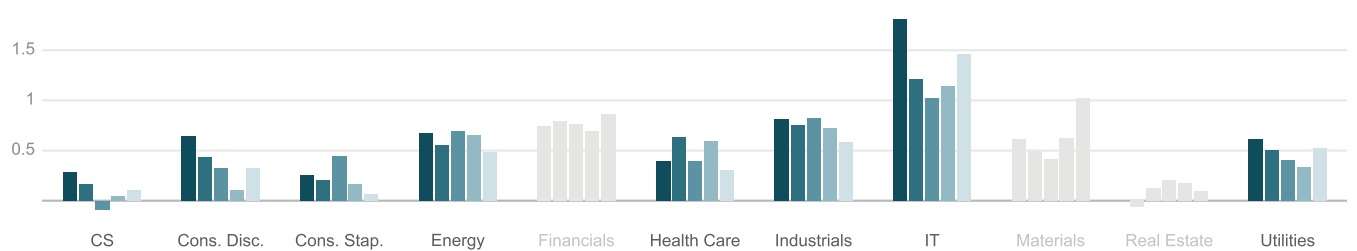
147 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +5.2%



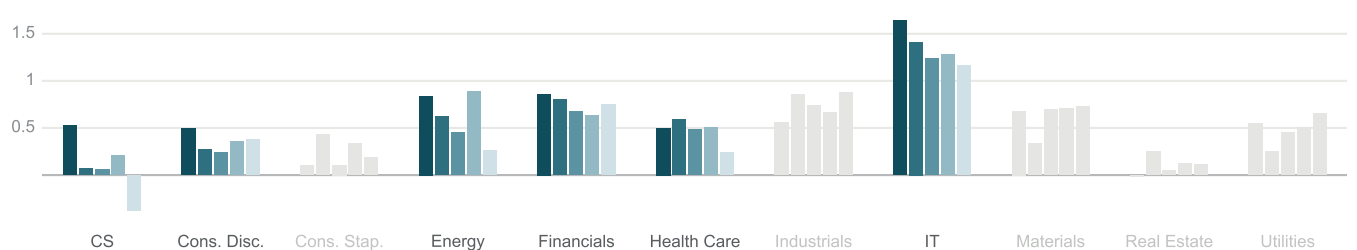
148 ● Change EPS to Sales

Earnings Quality L-S CAGR +5.2%



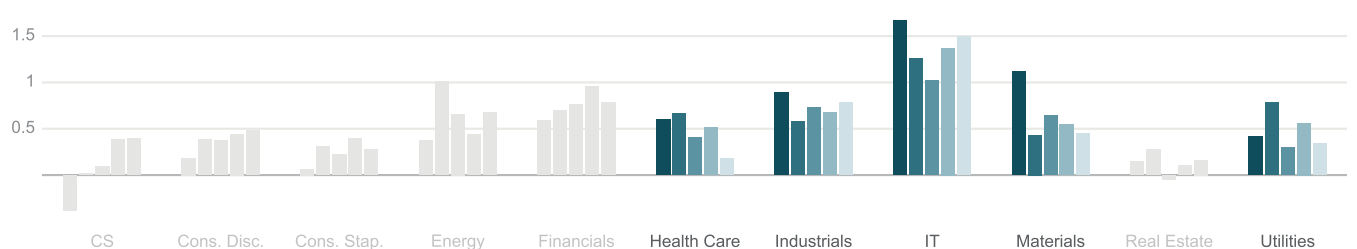
149 ● Cash to Total Assets

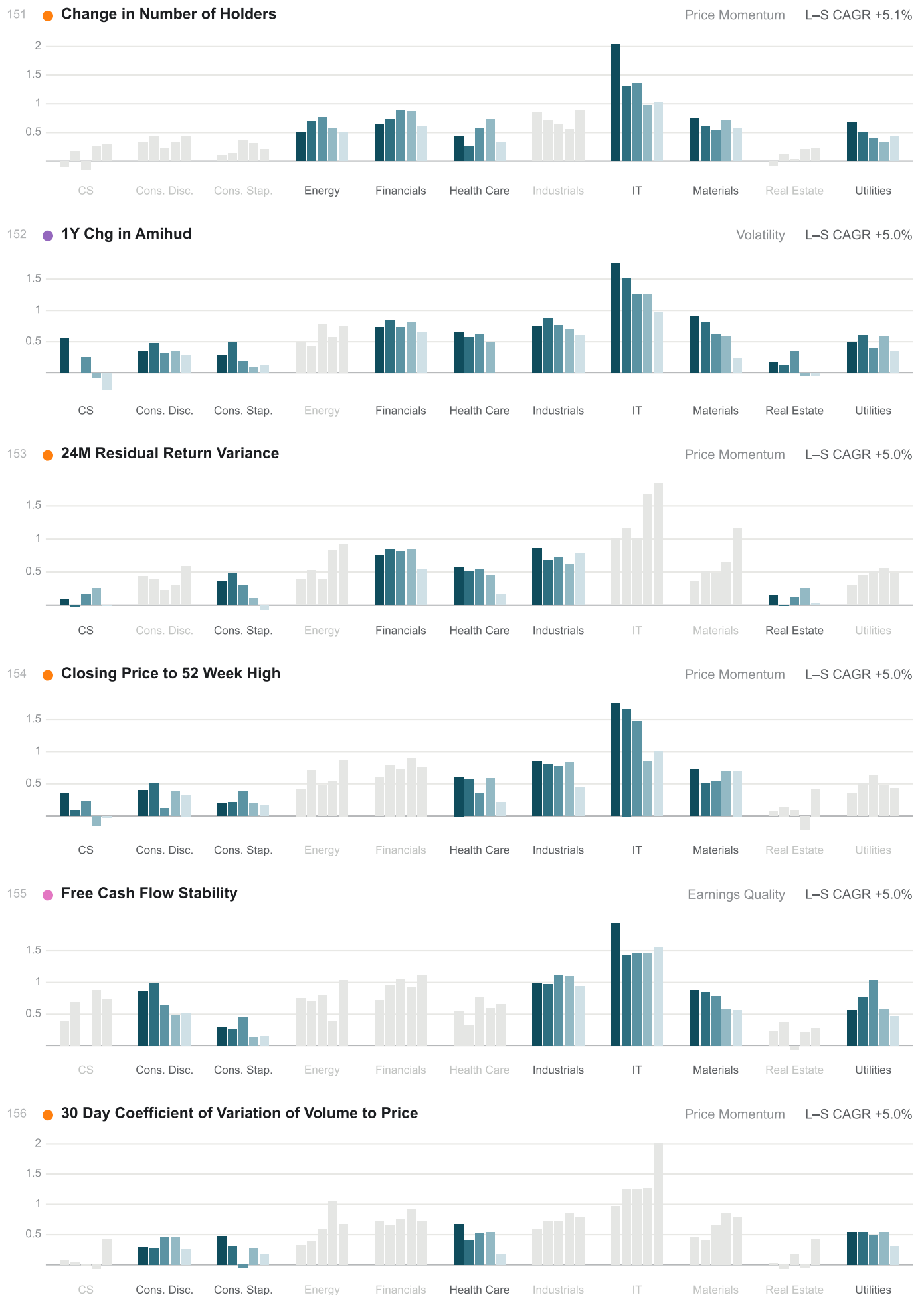
Valuation L-S CAGR +5.1%

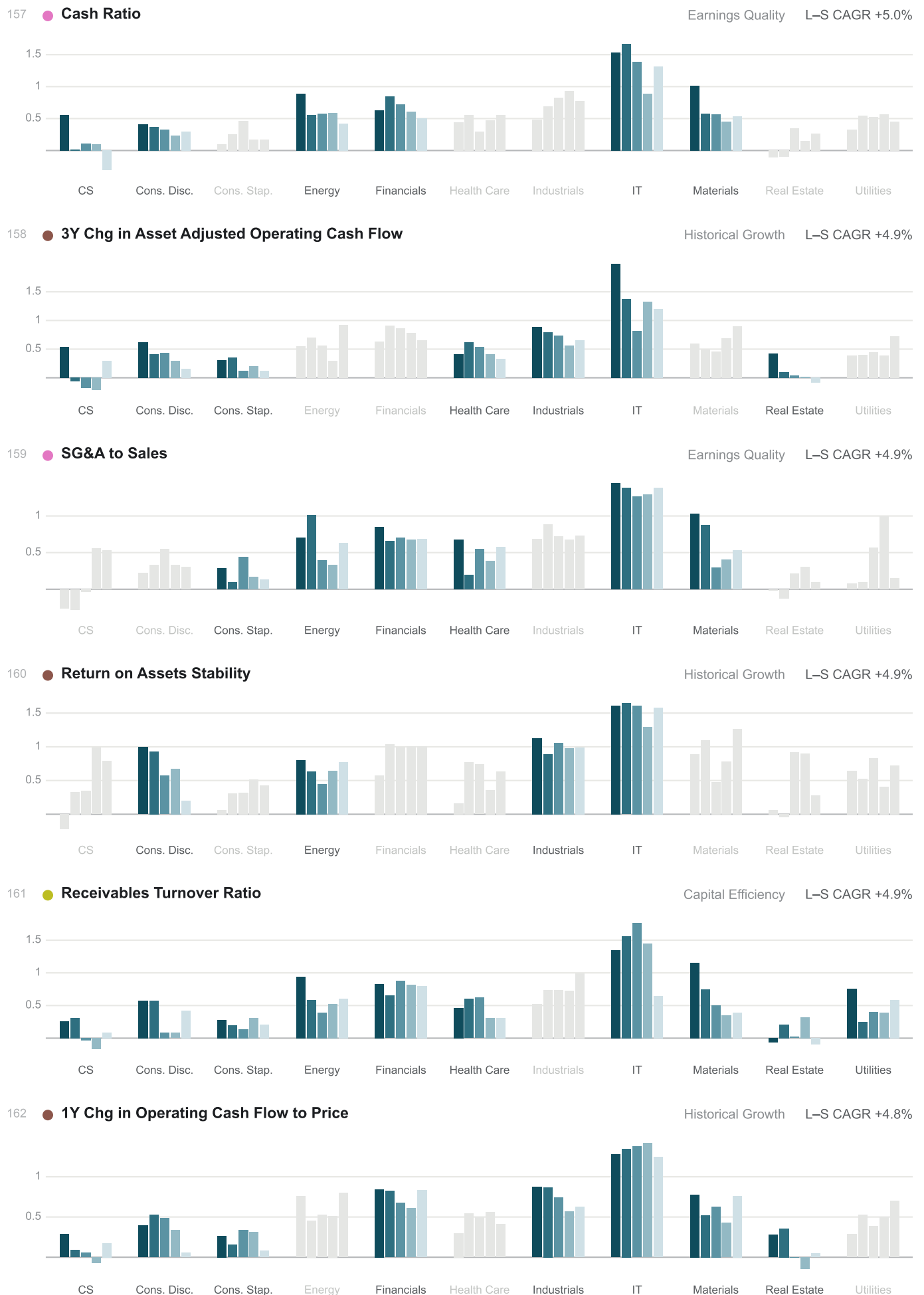


150 ● 6M Momentum in Trailing 12M Sales

Historical Growth L-S CAGR +5.1%

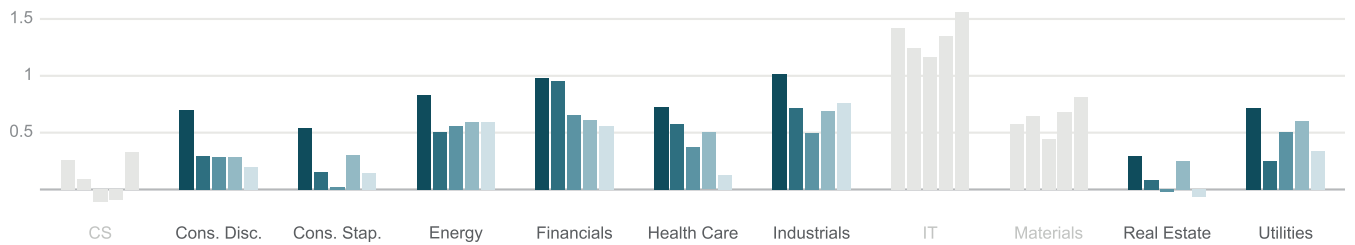






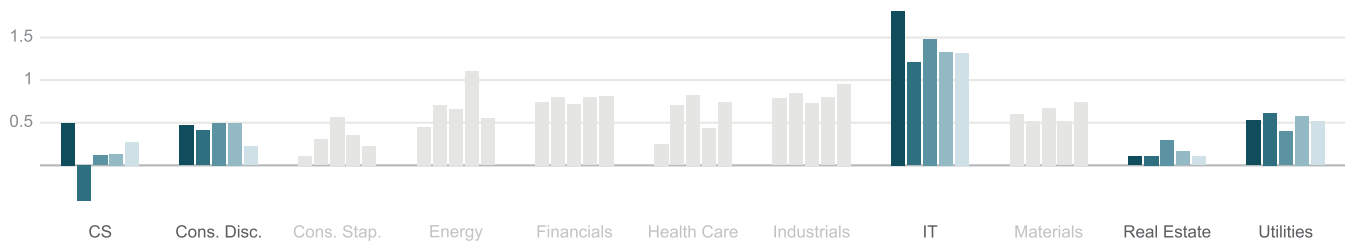
163 ● 1Y Chg in Shares Outstanding

Capital Efficiency L-S CAGR +4.8%



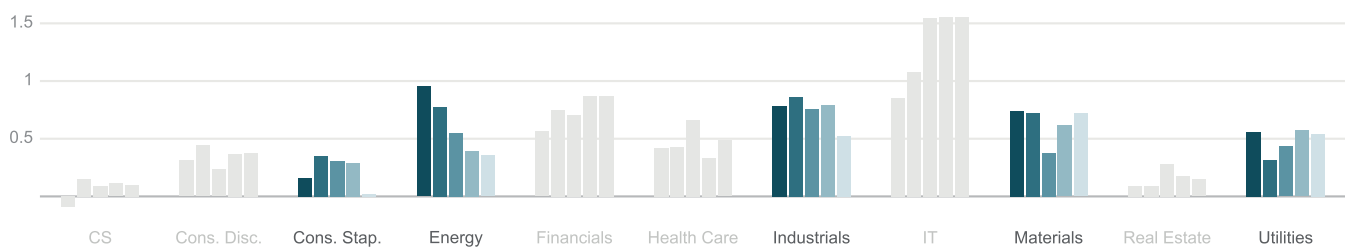
164 ● 1Y Chg in Stock Buybacks

Capital Efficiency L-S CAGR +4.8%



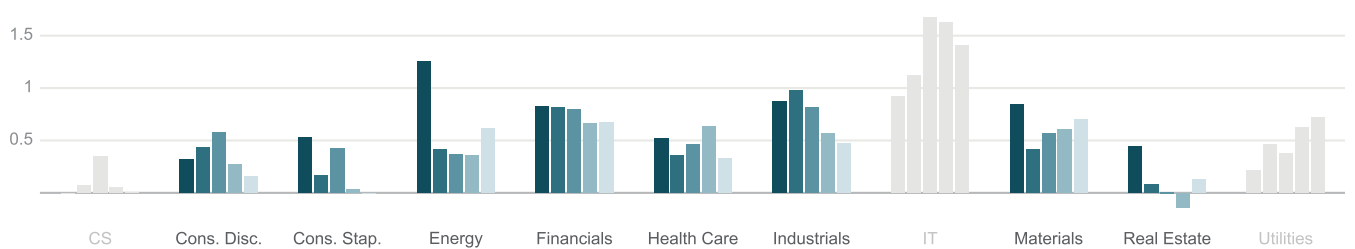
165 ● Depreciation to Capex Ratio

Earnings Quality L-S CAGR +4.8%



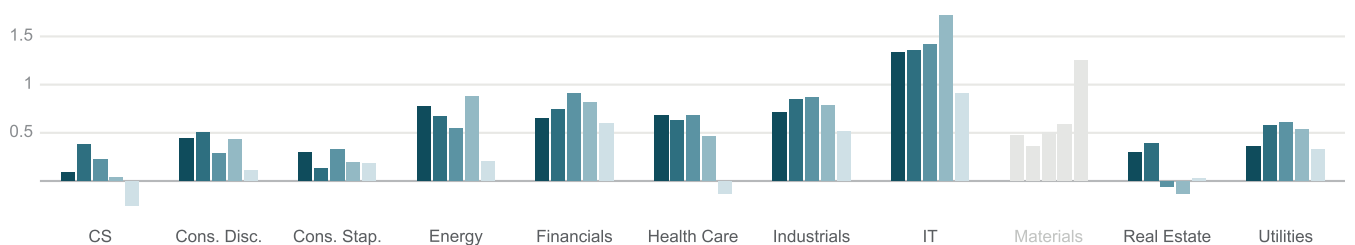
166 ● Free Cash Flow to Price

Valuation L-S CAGR +4.7%



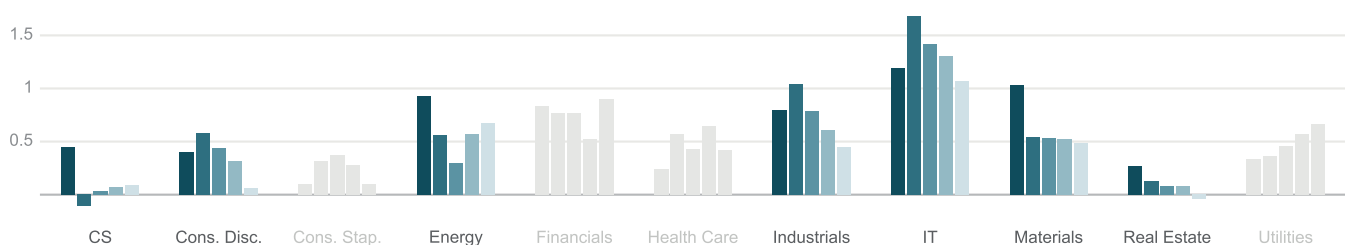
167 ● Asset Turnover

Earnings Quality L-S CAGR +4.7%



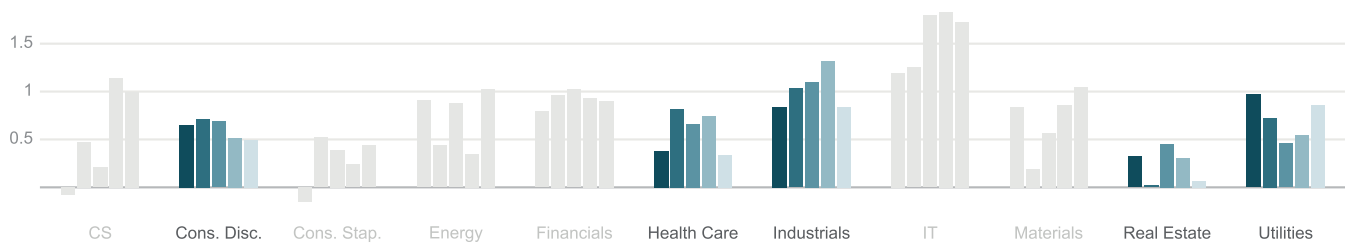
168 ● 1Y Chg in Free Cash Flow to Price

Historical Growth L-S CAGR +4.6%

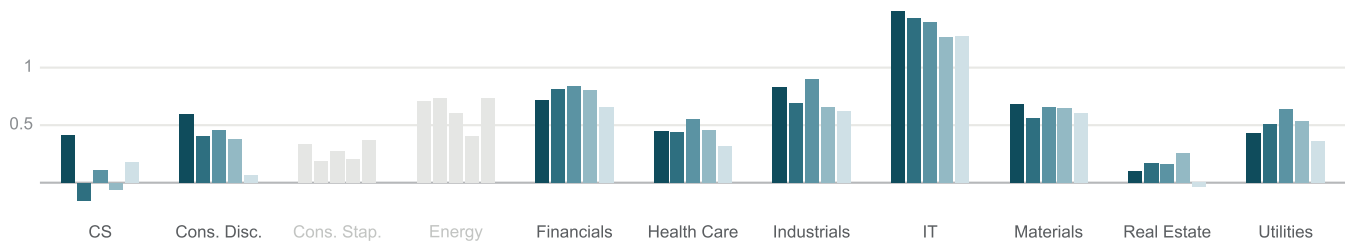


169 ● **Adj Forward Free Cash Flow to Price Ratio**

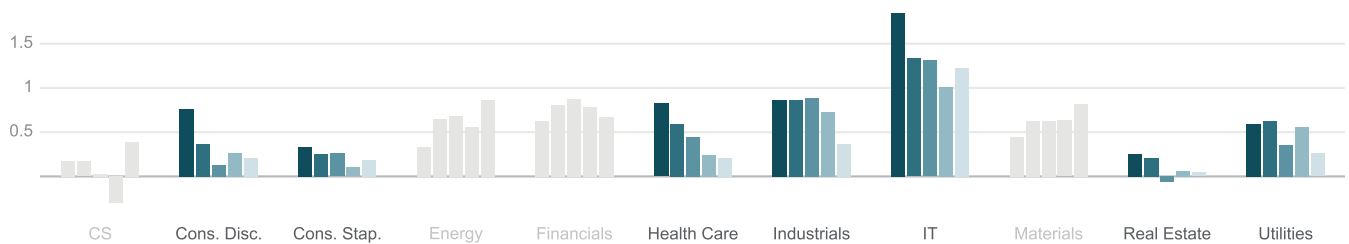
Valuation L-S CAGR +4.6%

170 ● **EPS Stability**

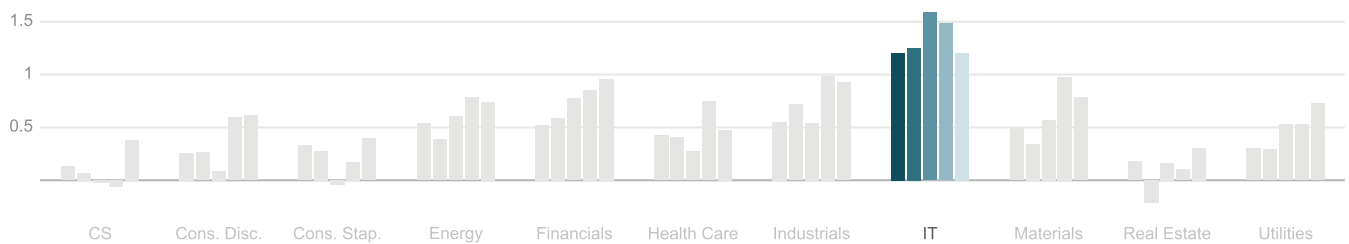
Historical Growth L-S CAGR +4.6%

171 ● **Cash Flow to Invested Capital**

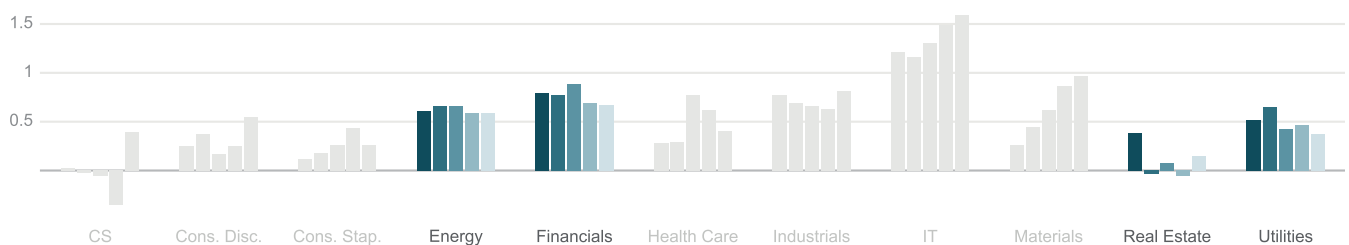
Earnings Quality L-S CAGR +4.6%

172 ● **1Y Chg in Sales Turnover**

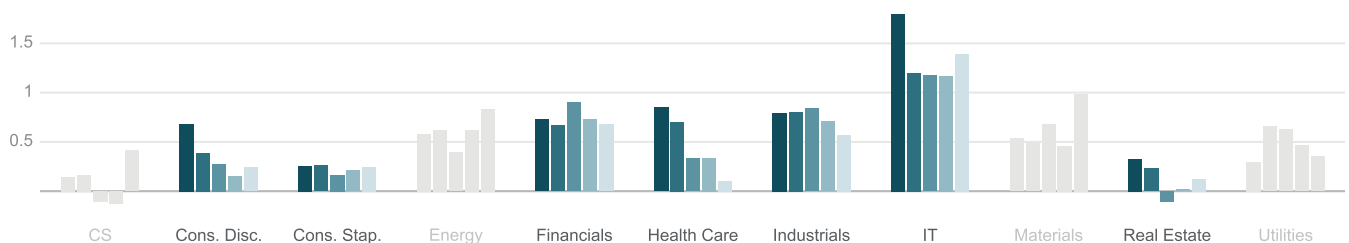
Historical Growth L-S CAGR +4.6%

173 ● **Dividend Growth**

Historical Growth L-S CAGR +4.5%

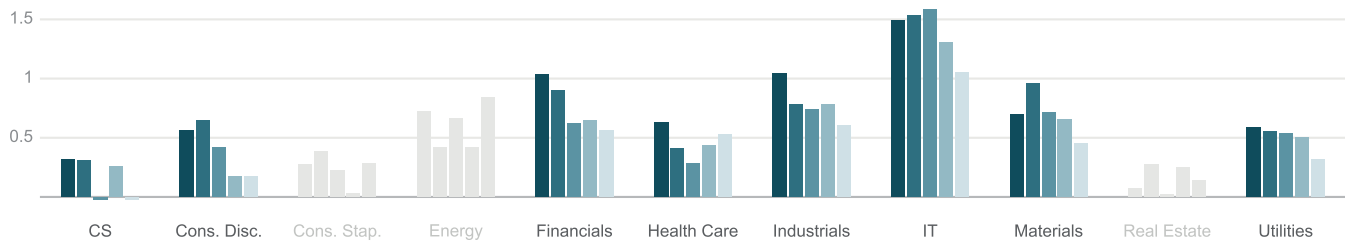
174 ● **Return on Invested Capital**

Capital Efficiency L-S CAGR +4.5%

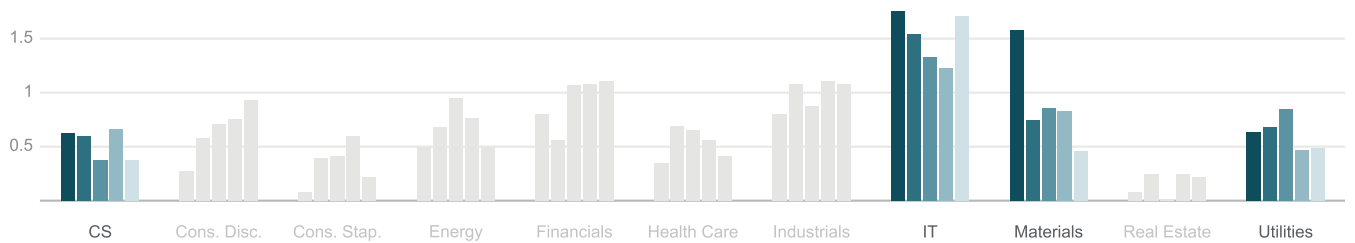


175 ● **Standardized Unexpected Earnings**

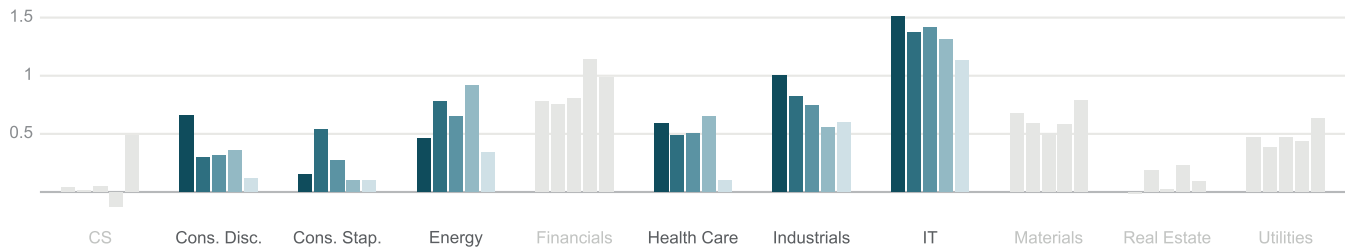
Analyst Expectations L-S CAGR +4.4%

176 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**

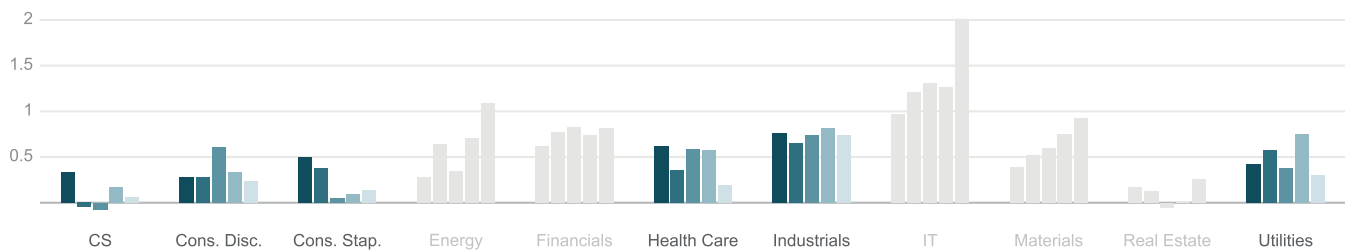
Historical Growth L-S CAGR +4.3%

177 ● **Cash Burn Rate**

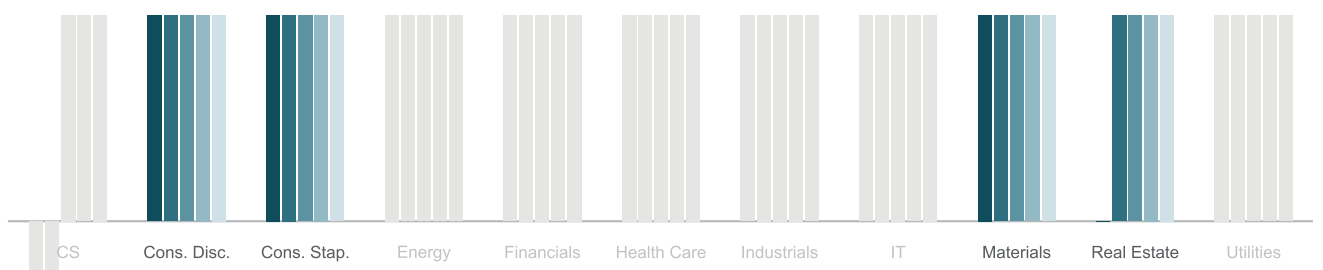
Earnings Quality L-S CAGR +4.2%

178 ● **20 Day Coefficient of Variation of Volume to Price**

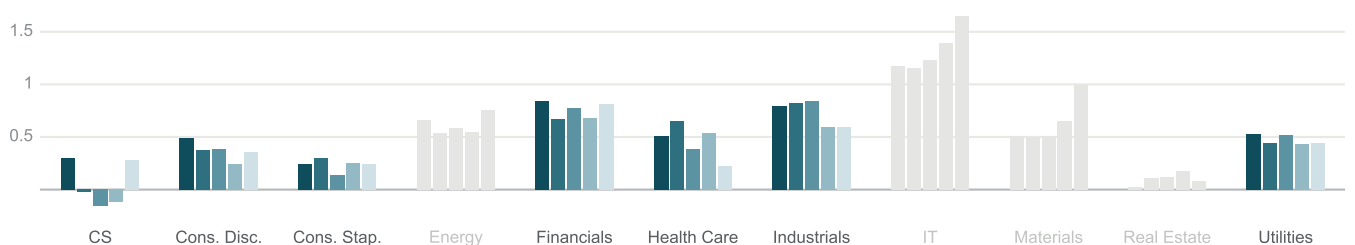
Price Momentum L-S CAGR +4.2%

179 ● **Inventory to Assets Ratio**

Earnings Quality L-S CAGR +4.1%

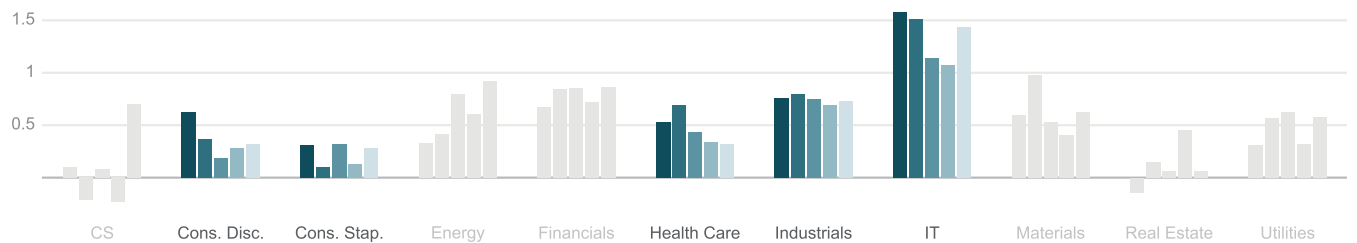
180 ● **3Y Chg in Cash Flow to Price**

Historical Growth L-S CAGR +4.1%



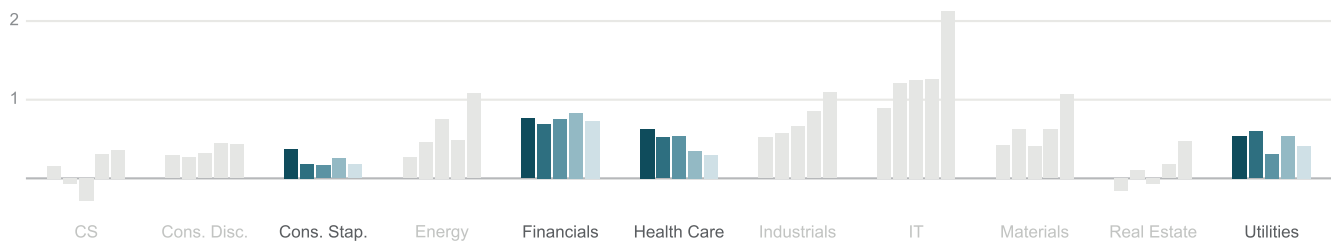
181 ● EBIT Margin

Earnings Quality L-S CAGR +4.1%



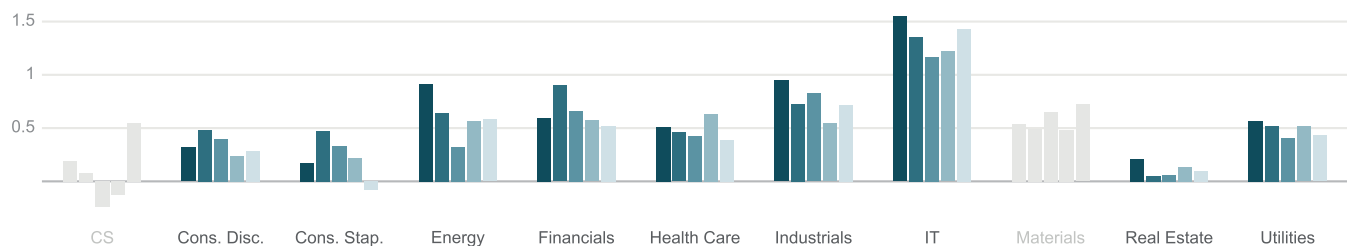
182 ● 60 Day Coefficient of Variation of Volume to Price

Price Momentum L-S CAGR +4.0%



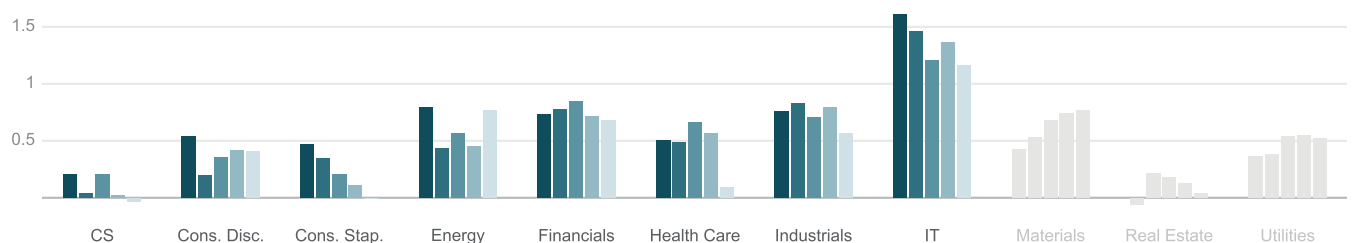
183 ● Degree of Financial Leverage

Capital Efficiency L-S CAGR +4.0%



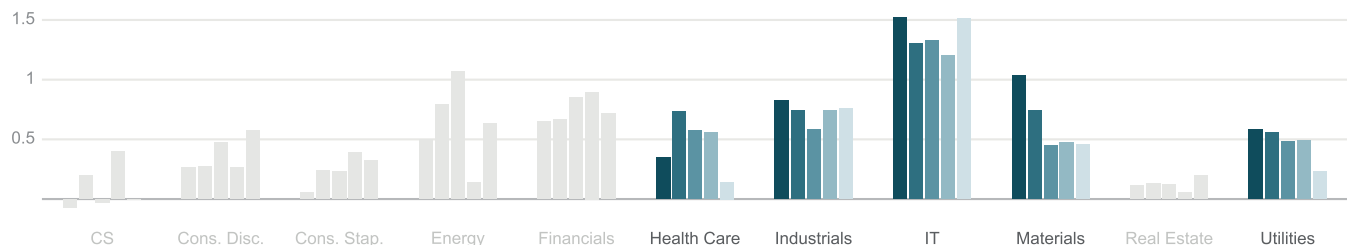
184 ● 3Y Chg in Sales Turnover

Historical Growth L-S CAGR +4.0%



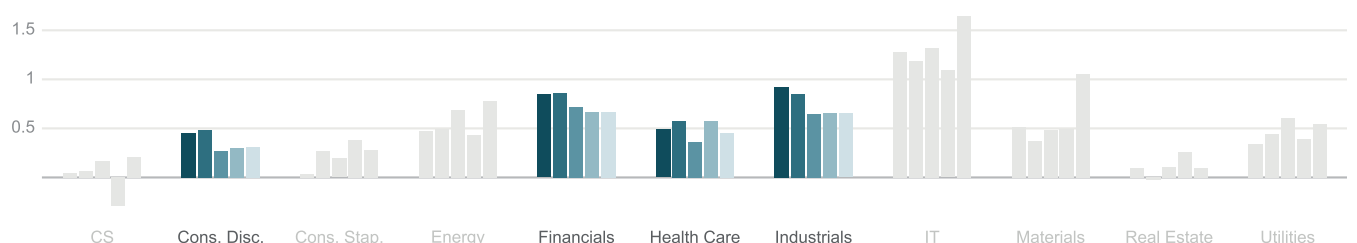
185 ● 3M Momentum in Trailing 12M Sales

Historical Growth L-S CAGR +4.0%



186 ● 3Y Average Annual Earnings Growth

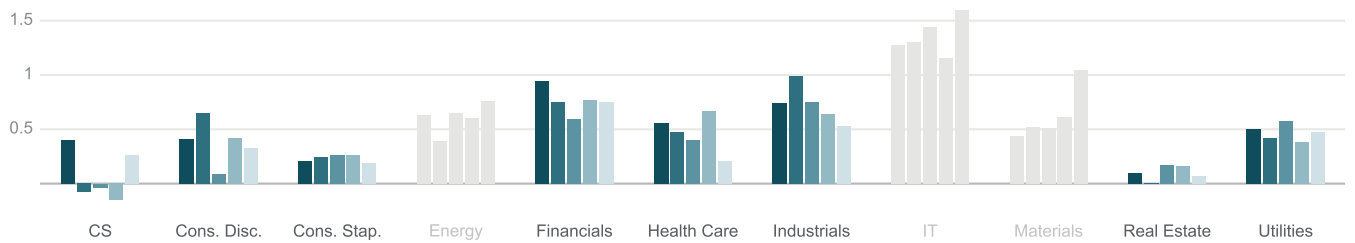
Historical Growth L-S CAGR +4.0%



187 ● **3Y Chg in Earnings to Price**

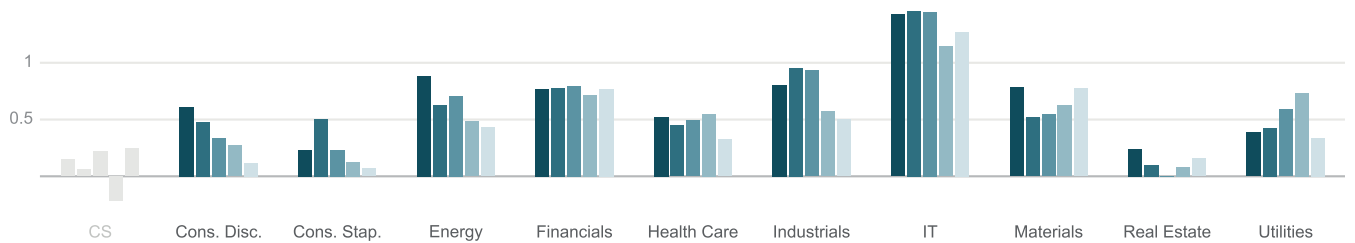
Historical Growth

L-S CAGR +3.9%

188 ● **Free Cash Flow to Sales**

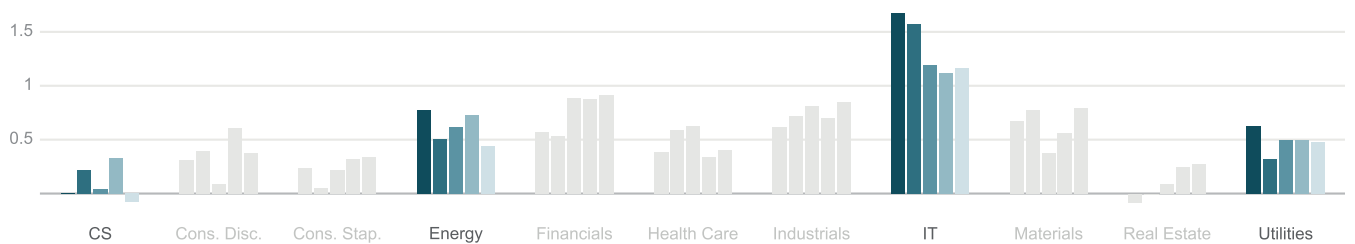
Earnings Quality

L-S CAGR +3.9%

189 ● **Change in Number of Buyers**

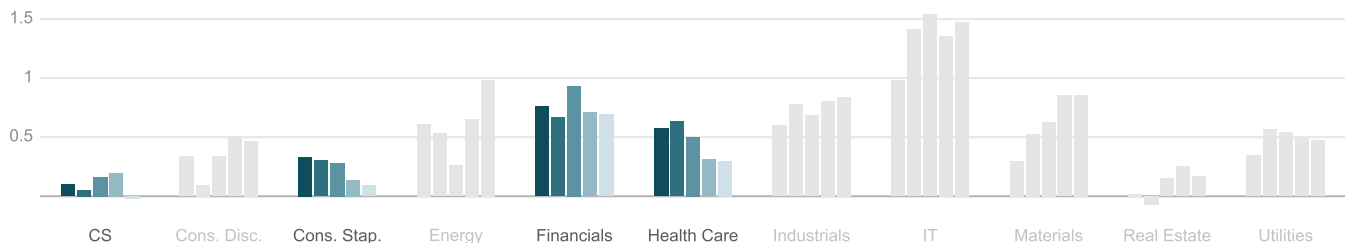
Price Momentum

L-S CAGR +3.9%

190 ● **130 Day Minimum Return**

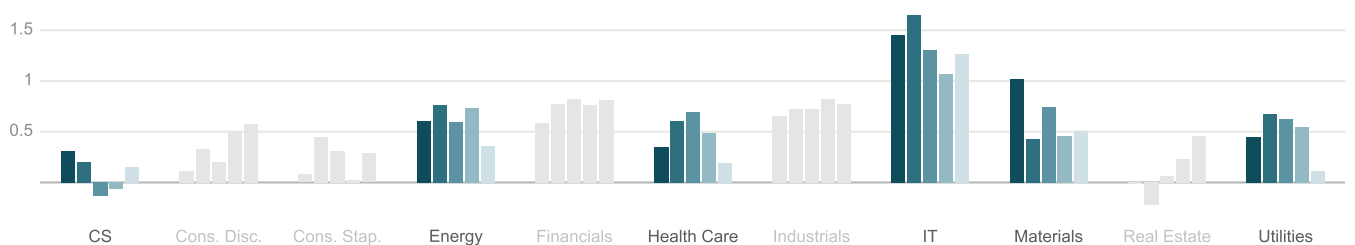
Volatility

L-S CAGR +3.8%

191 ● **10 Day MACD Trend**

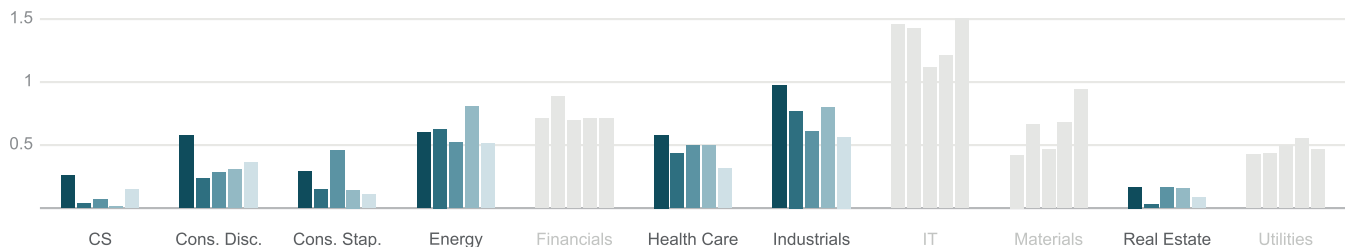
Price Momentum

L-S CAGR +3.8%

192 ● **ROA 60 Month Slope**

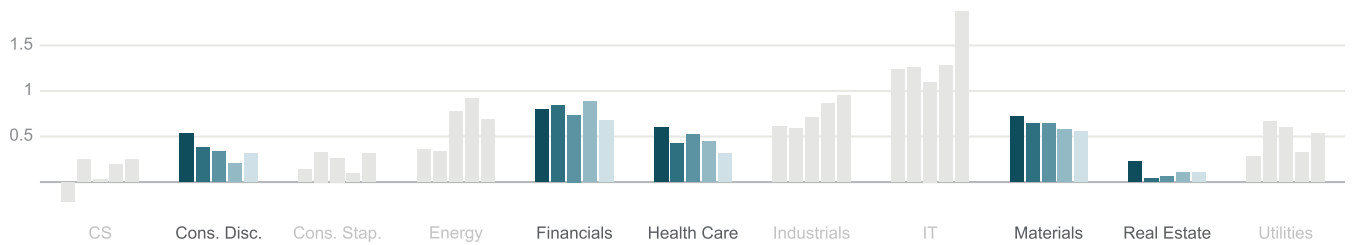
Earnings Quality

L-S CAGR +3.8%

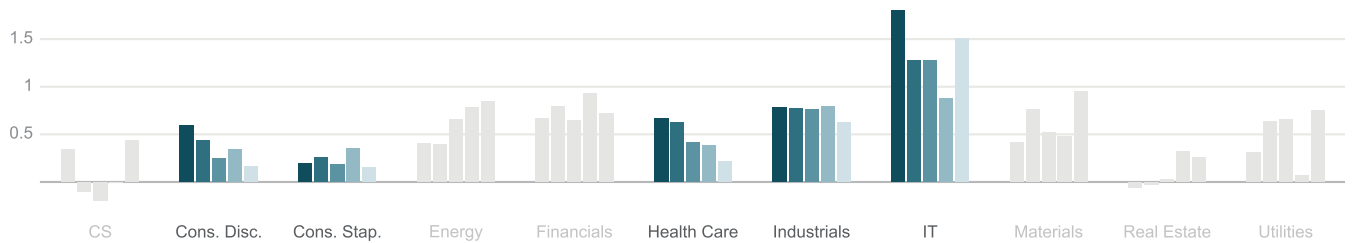


193 ● **Cash Conversion Cycle**

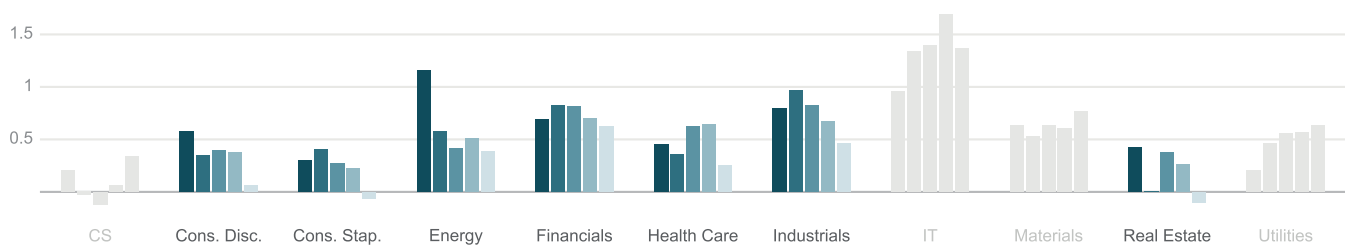
Earnings Quality L-S CAGR +3.8%

194 ● **Net Profit Margin**

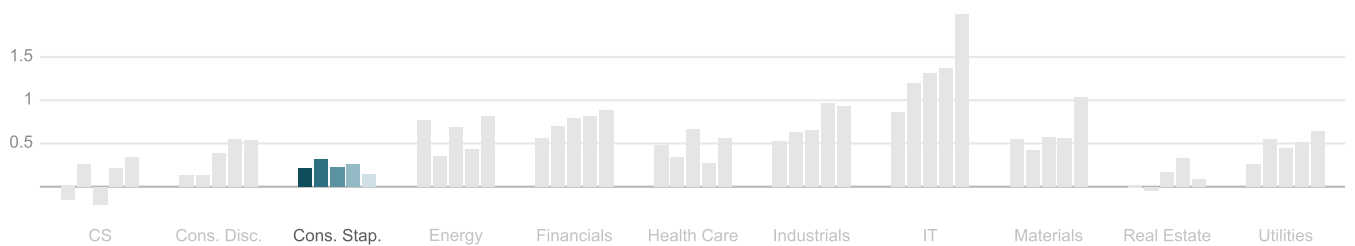
Earnings Quality L-S CAGR +3.8%

195 ● **Capital Acquisition Ratio**

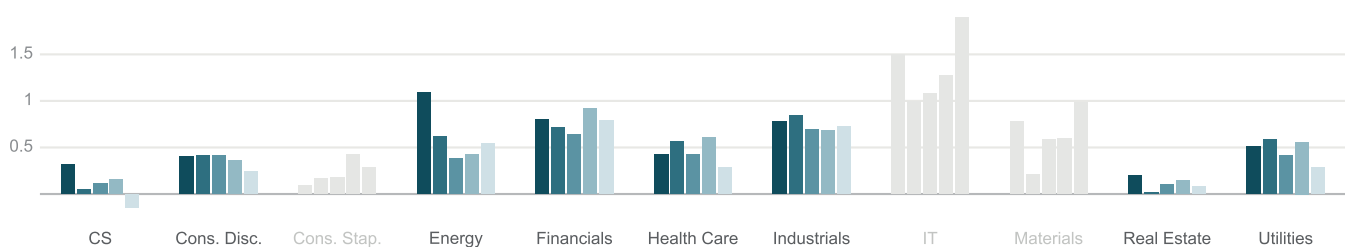
Capital Efficiency L-S CAGR +3.8%

196 ● **Share Turnover**

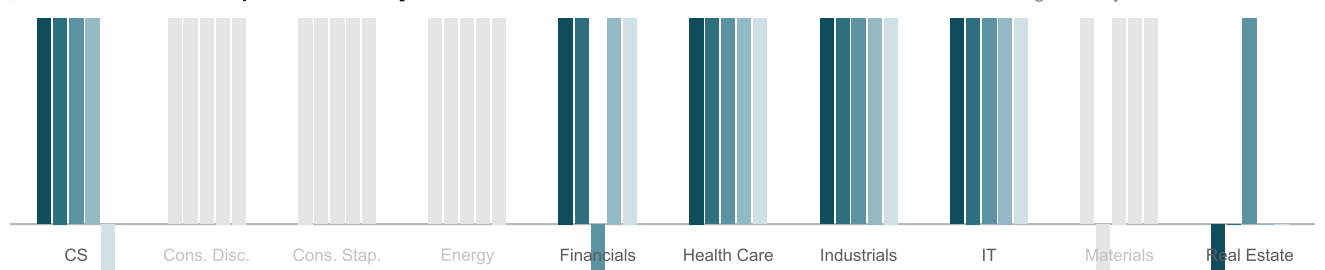
Price Momentum L-S CAGR +3.7%

197 ● **Sales to Gross Profit Margin Growth**

Earnings Quality L-S CAGR +3.7%

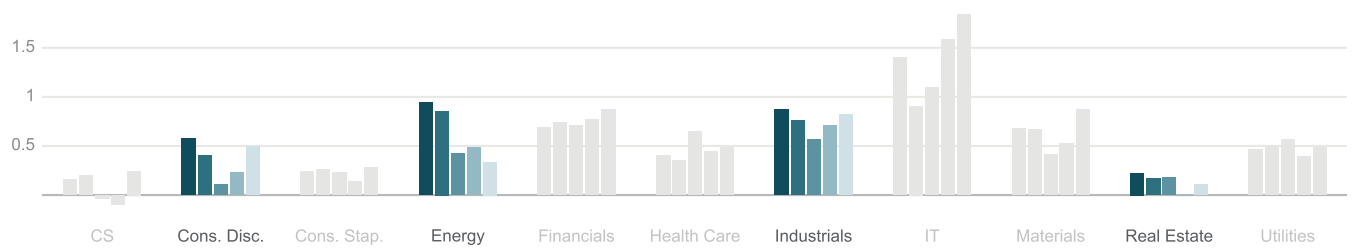
198 ● **Research & Development Intensity**

Earnings Quality L-S CAGR +3.7%

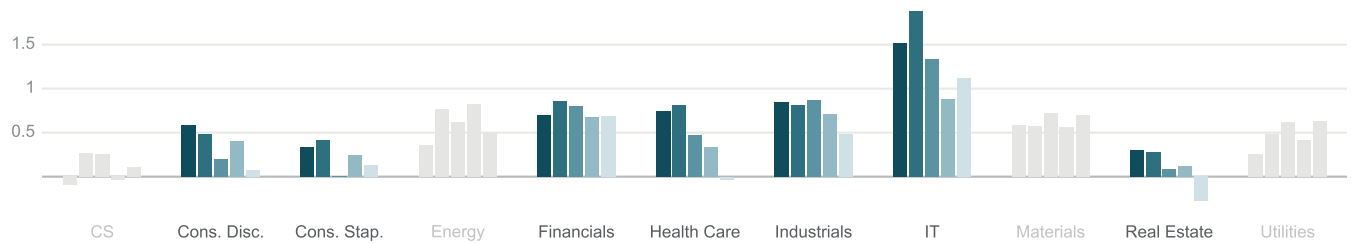


199 ● **Asset Adjusted Capital Investments**

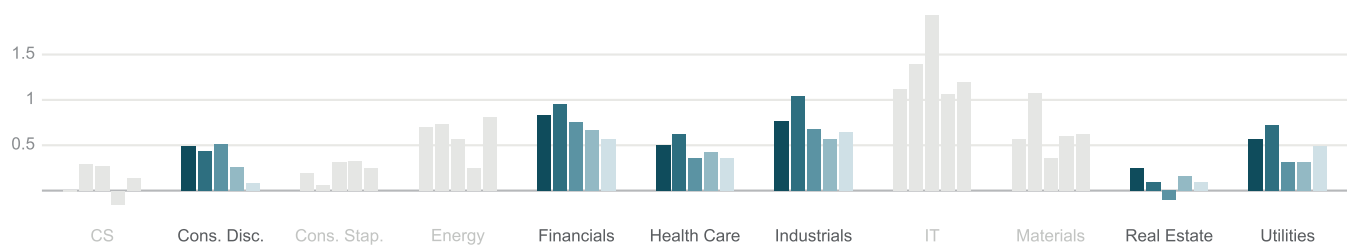
Earnings Quality L-S CAGR +3.7%

200 ● **Cash Flow Return on Invested Capital**

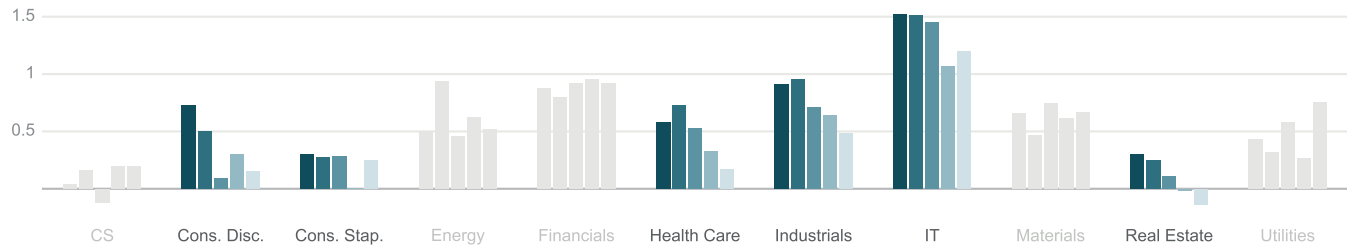
Capital Efficiency L-S CAGR +3.6%

201 ● **Institution Ownership Level**

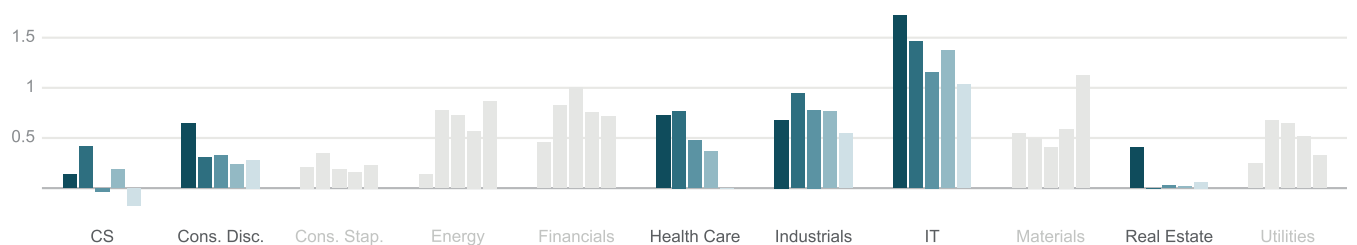
Price Momentum L-S CAGR +3.6%

202 ● **Operating Cash Flow to Equity**

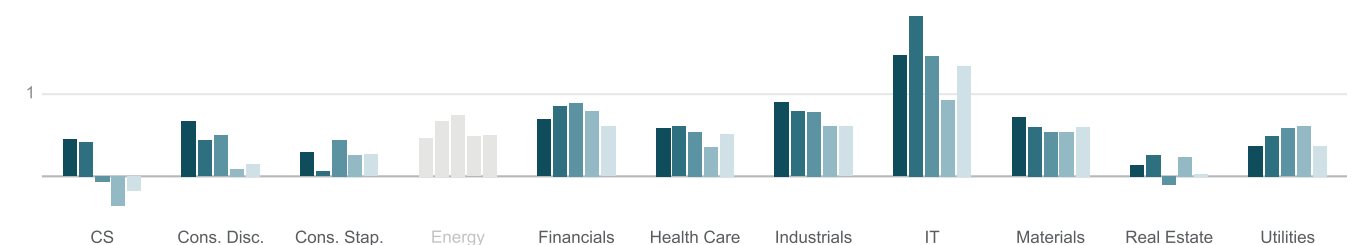
Capital Efficiency L-S CAGR +3.6%

203 ● **Gross Profit to Assets**

Capital Efficiency L-S CAGR +3.6%

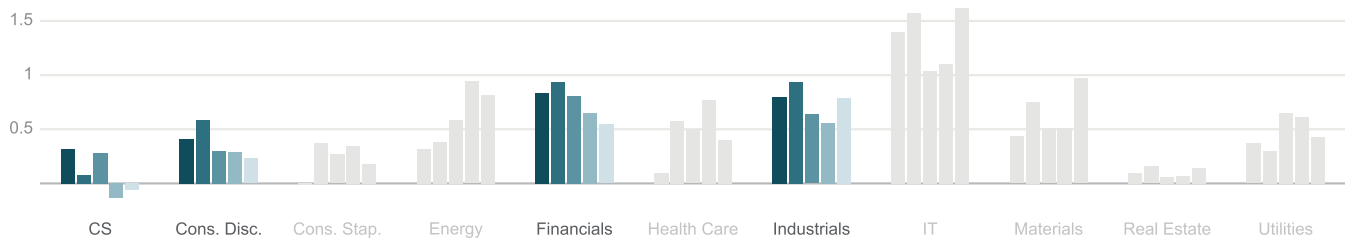
204 ● **Payout Ratio**

Capital Efficiency L-S CAGR +3.6%

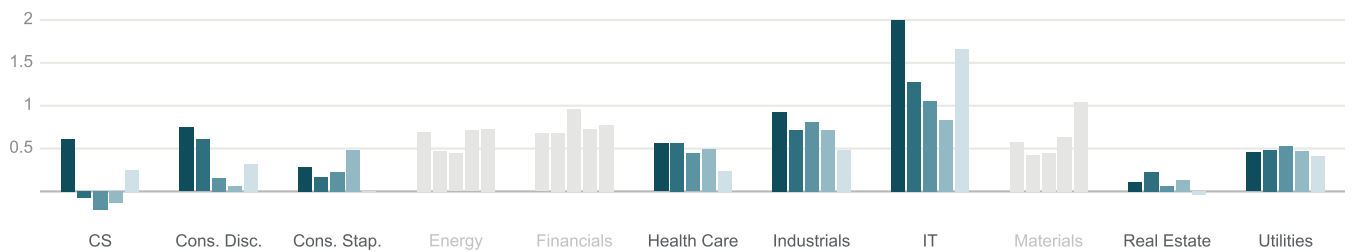


205 ● **3Y Average Annual Sales Growth**

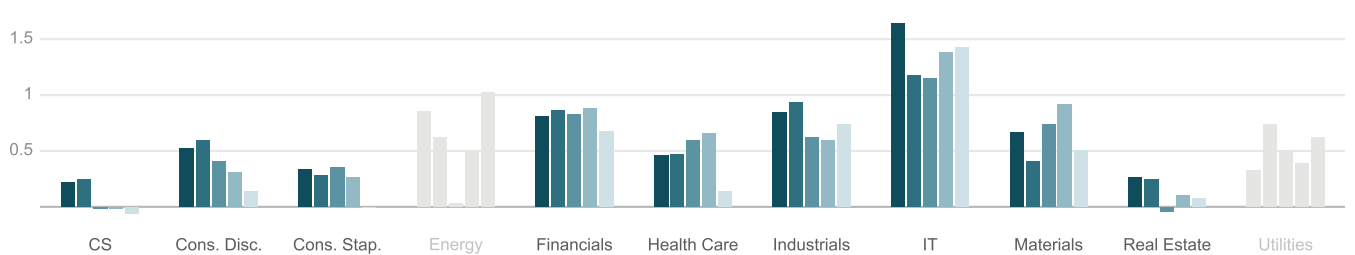
Historical Growth L-S CAGR +3.5%

206 ● **3Y Chg in Asset Adjusted EPS**

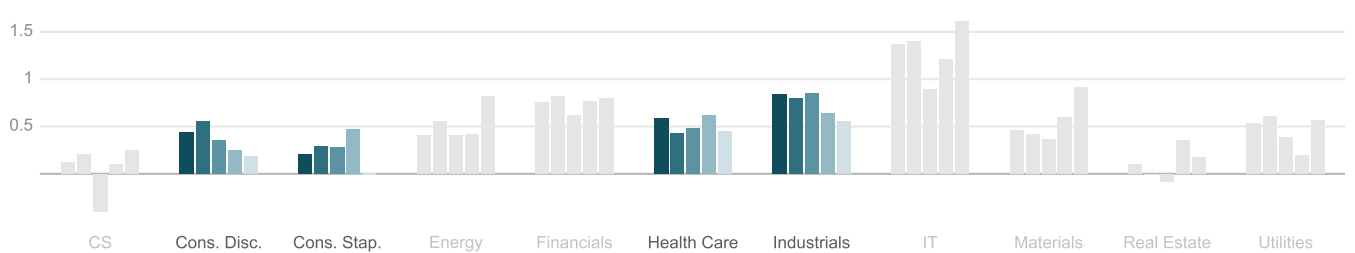
Historical Growth L-S CAGR +3.5%

207 ● **3Y Chg in Free Cash Flow per Share**

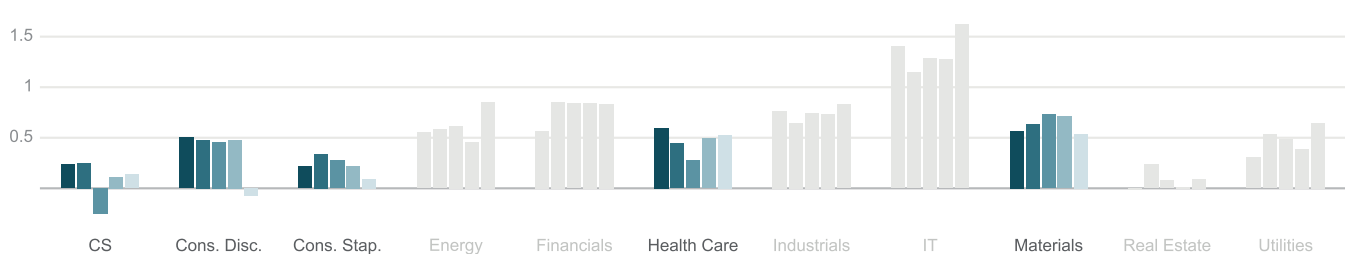
Historical Growth L-S CAGR +3.5%

208 ● **1Y Chg in Earnings per Share**

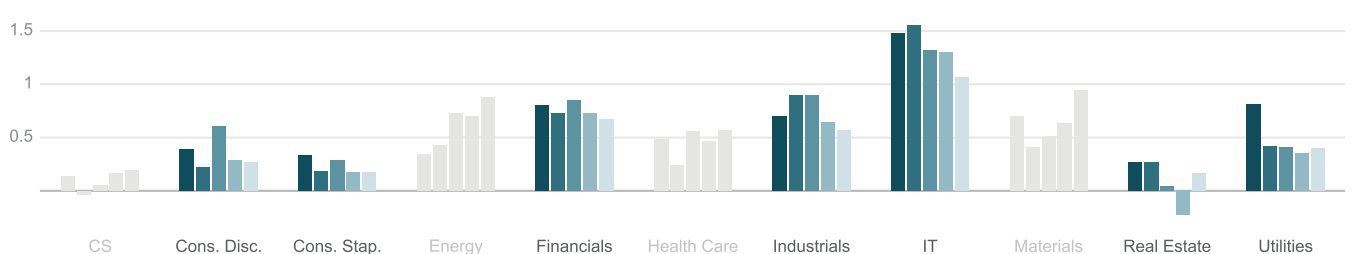
Historical Growth L-S CAGR +3.5%

209 ● **Unexpected Receivables Change**

Earnings Quality L-S CAGR +3.5%

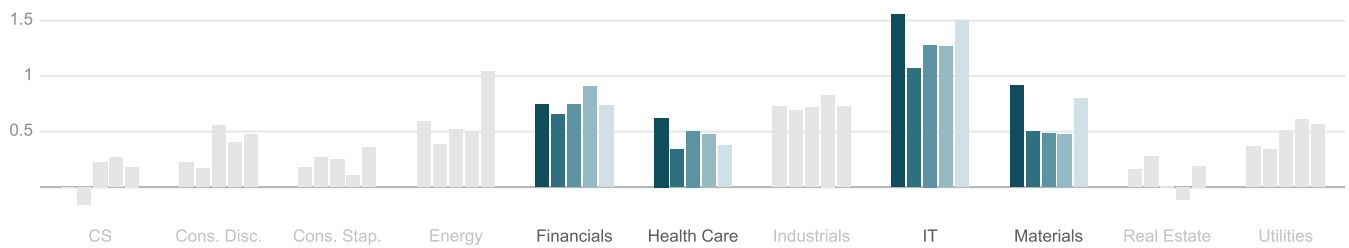
210 ● **6M Chg in 36M CAPM Alpha**

Price Momentum L-S CAGR +3.5%



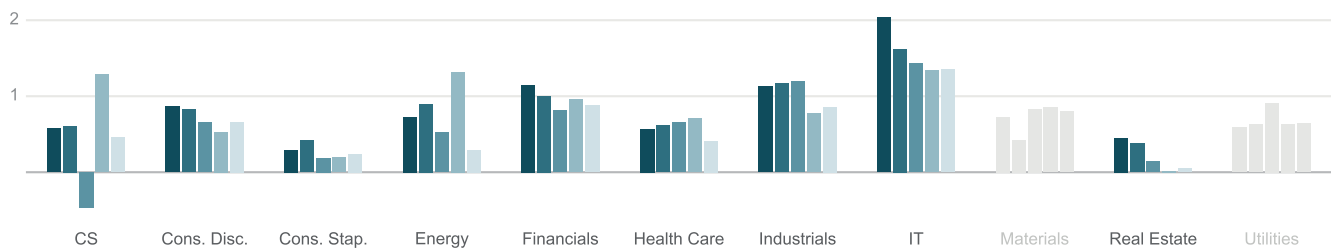
211 1Y Chg in Sales to Earnings

Earnings Quality L-S CAGR +3.5%



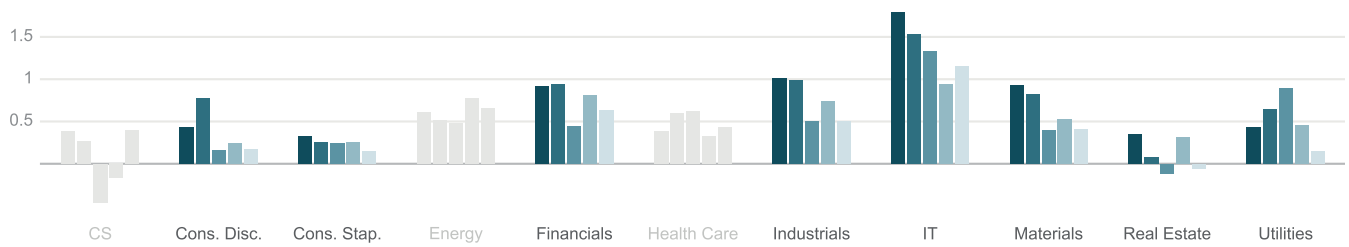
212 Operating Cash Flow Stability

Earnings Quality L-S CAGR +3.4%



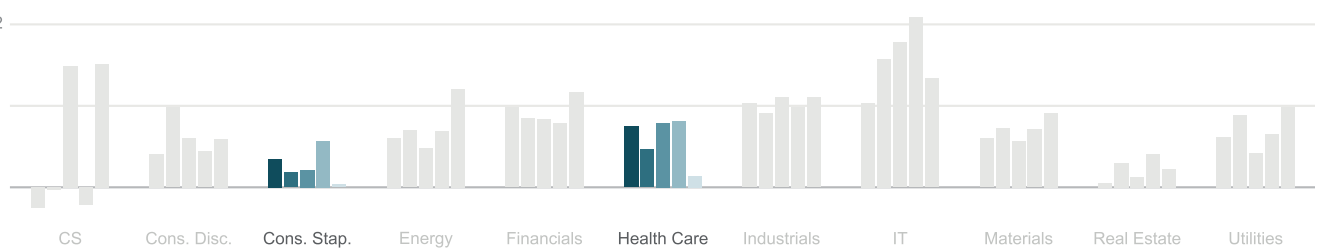
213 6 Month Average Share Turn Over

Price Momentum L-S CAGR +3.4%



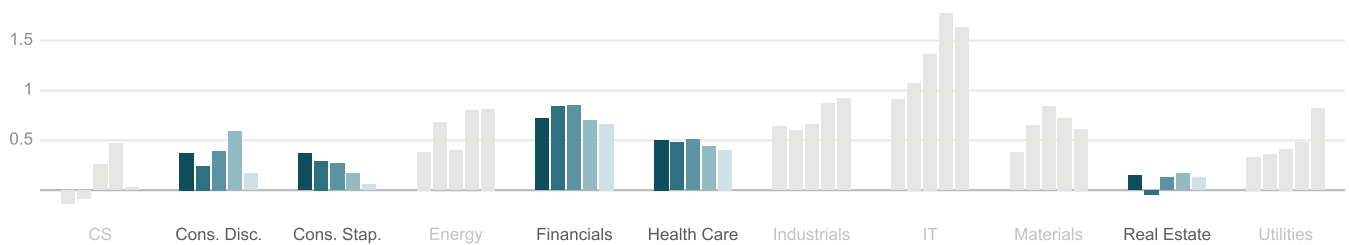
214 EBIT to Price

Valuation L-S CAGR +3.4%



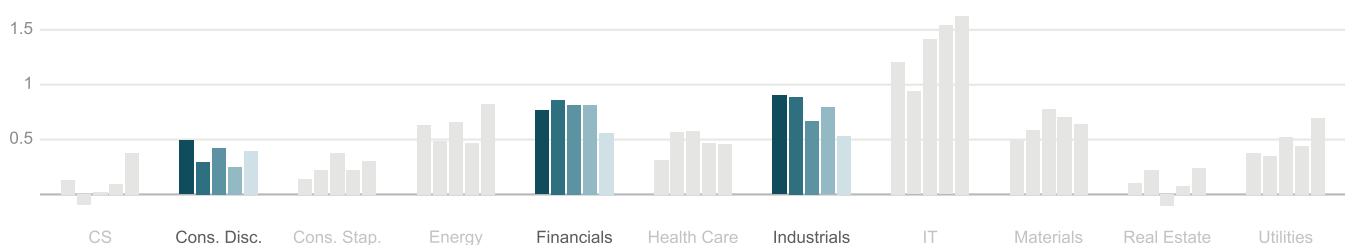
215 Ownership Breadth Stability of Institution Ownership by Holders

Price Momentum L-S CAGR +3.3%



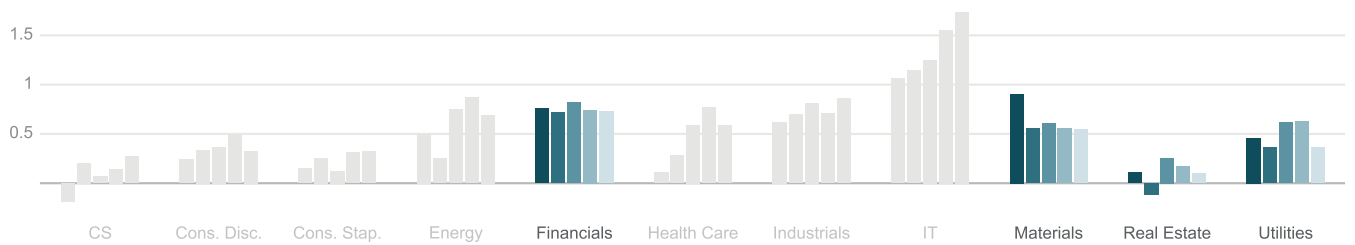
216 Year over Year Change of Total Debt

Capital Efficiency L-S CAGR +3.3%

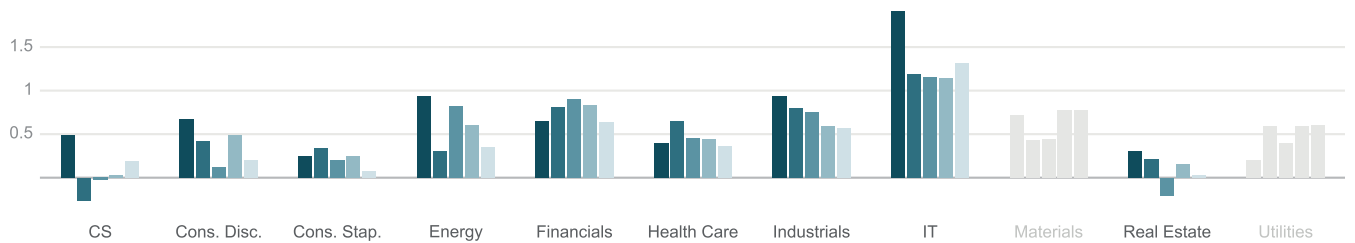


217 **Foreign Institution Ownership**

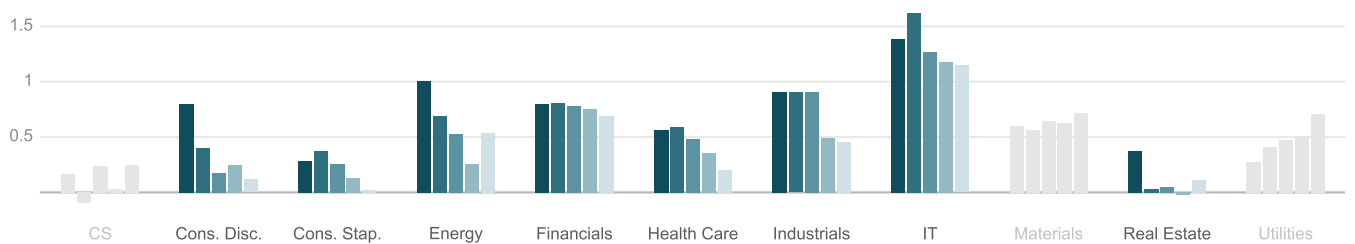
Price Momentum L-S CAGR +3.3%

218 **3Y Chg in Asset Adjusted Free Cash Flow**

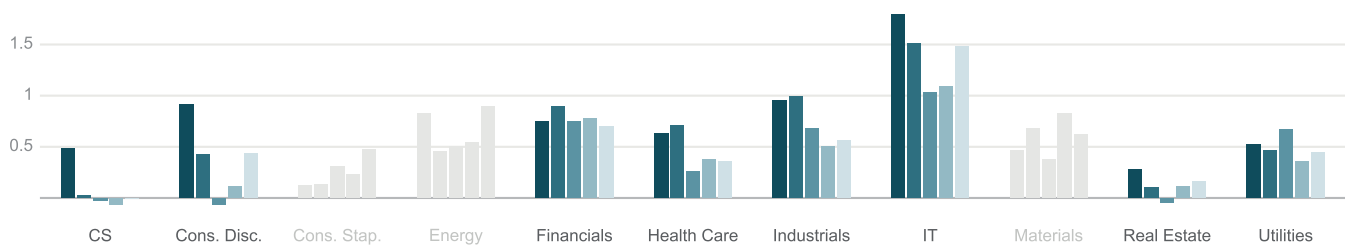
Historical Growth L-S CAGR +3.2%

219 **Free Cash Flow to Common Equity**

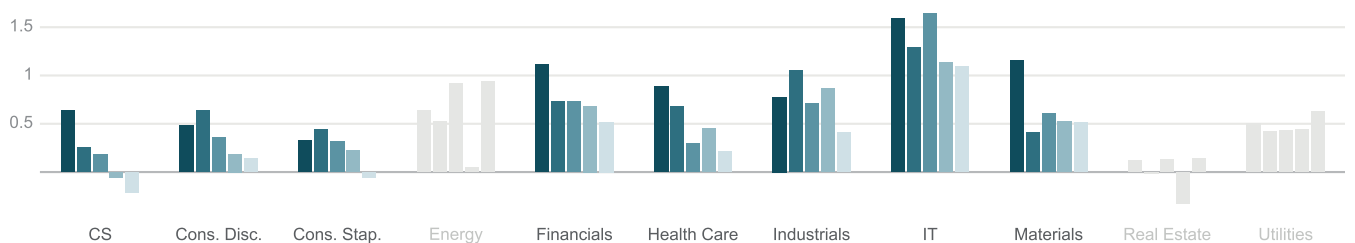
Valuation L-S CAGR +3.2%

220 **Sustainable Growth Rate**

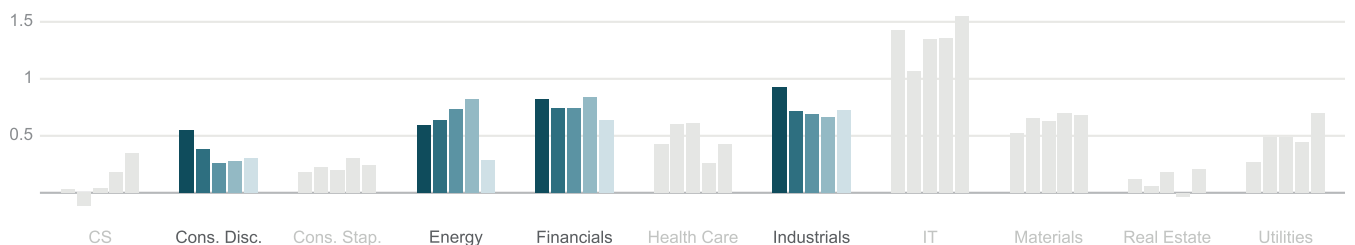
Historical Growth L-S CAGR +3.1%

221 **Adjusted Number of EPS FY1 Revisions**

Analyst Expectations L-S CAGR +3.1%

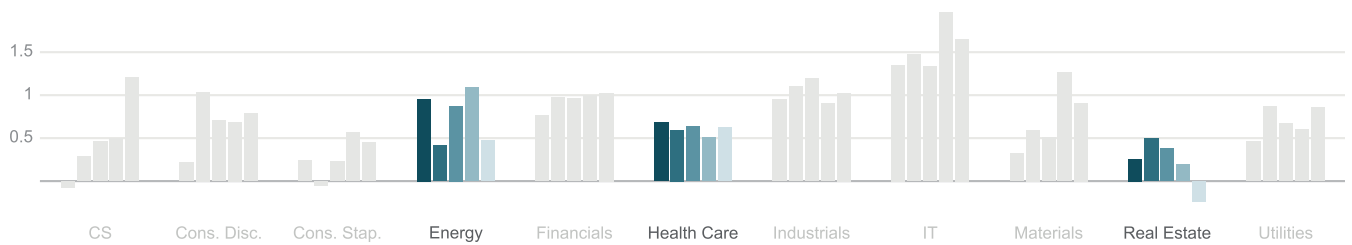
222 **1Y Chg in Debt to Assets**

Volatility L-S CAGR +3.0%

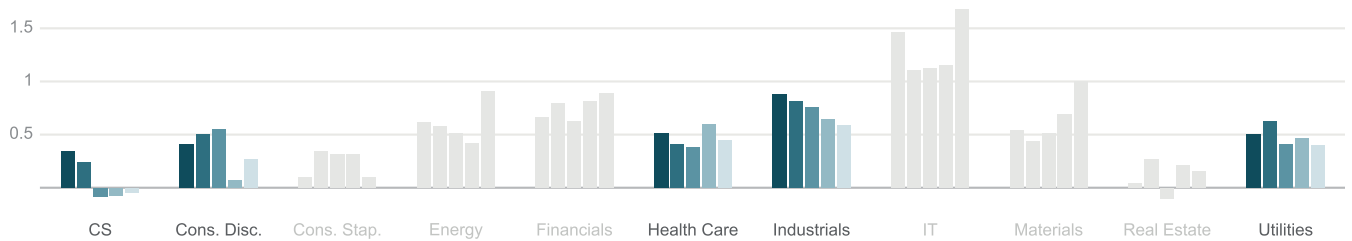


223 ● **Standardized 36M Residual**

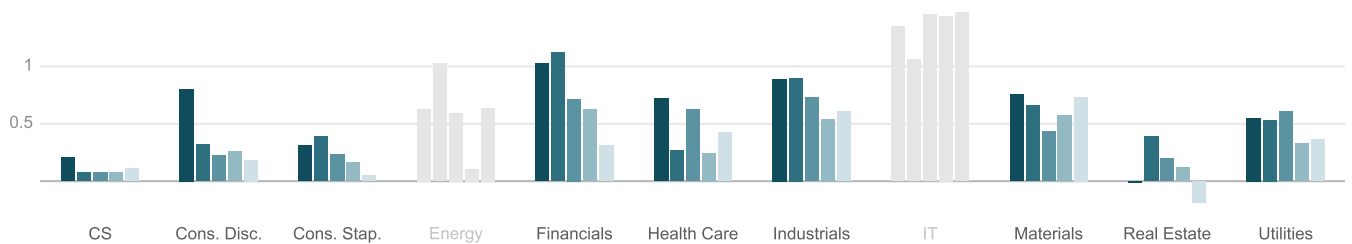
Price Momentum L-S CAGR +2.9%

224 ● **1Y Chg in Cash Flow per Share**

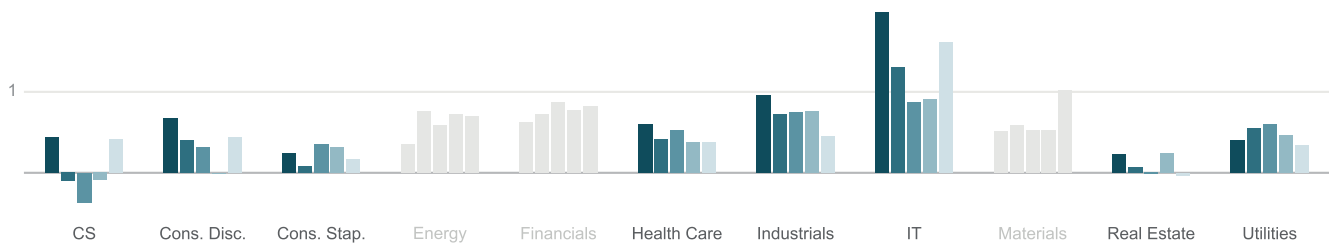
Historical Growth L-S CAGR +2.8%

225 ● **Target Price Gap to 6M EMA**

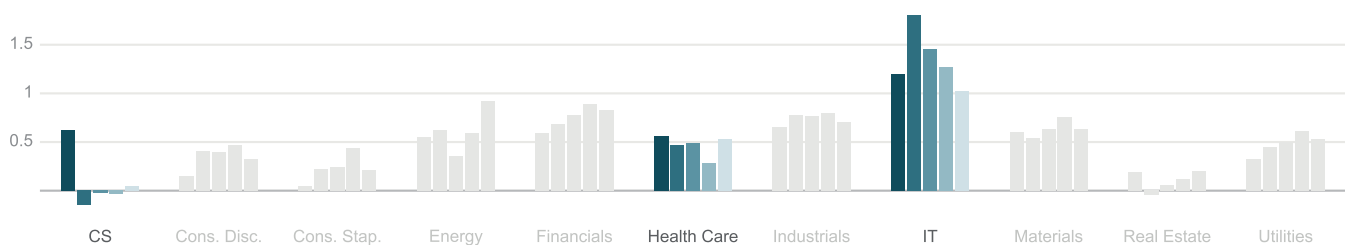
Analyst Expectations L-S CAGR +2.8%

226 ● **3Y Chg in Asset Adjusted Cash Flow**

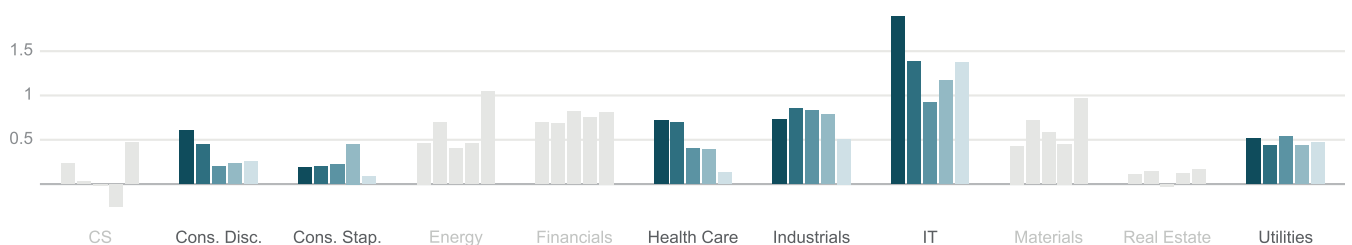
Historical Growth L-S CAGR +2.8%

227 ● **Net Current Assets to Price Ratio**

Valuation L-S CAGR +2.7%

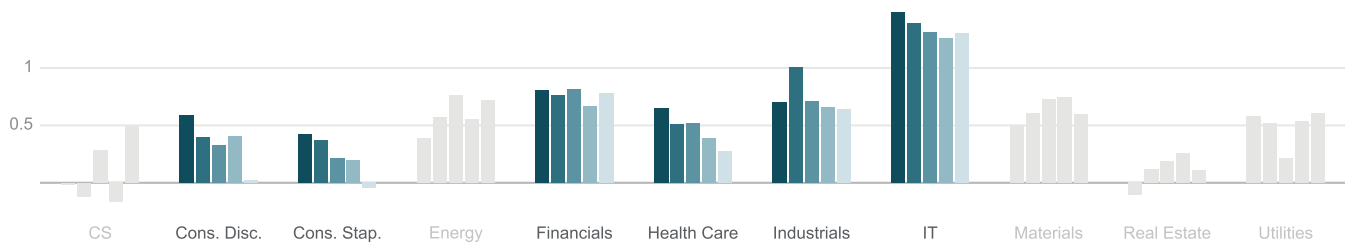
228 ● **Pretax Income to Net Operating Assets**

Earnings Quality L-S CAGR +2.7%

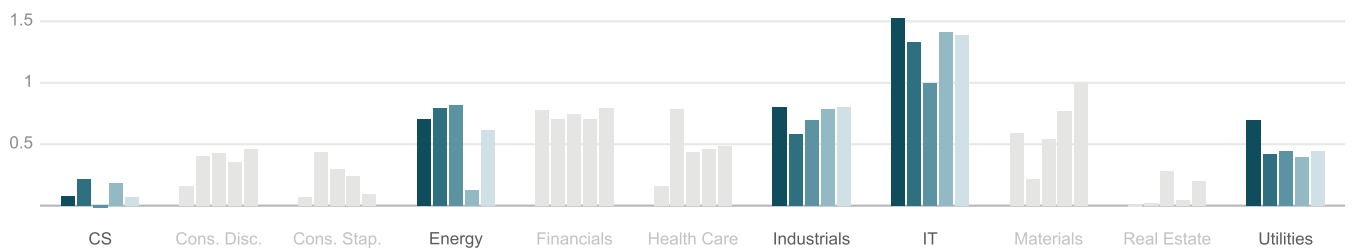


229 ● **EBIT Estimate Dispersion**

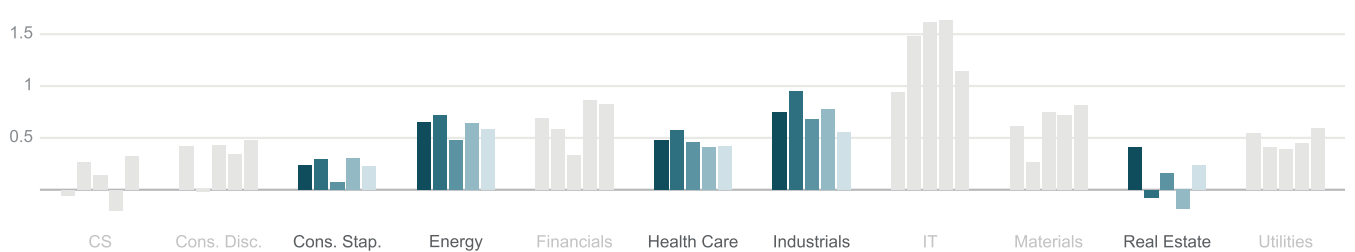
Analyst Expectations L-S CAGR +2.7%

230 ● **1Y Chg in EPS to Operating Cash Flow**

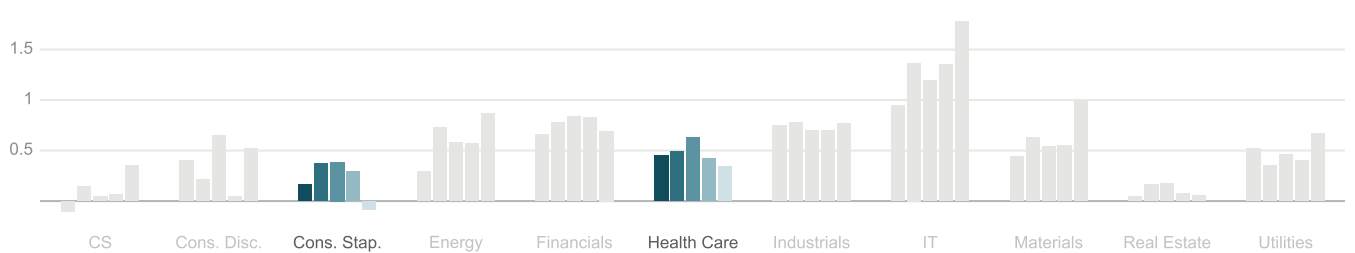
Earnings Quality L-S CAGR +2.6%

231 ● **Working Capital Turnover Ratio**

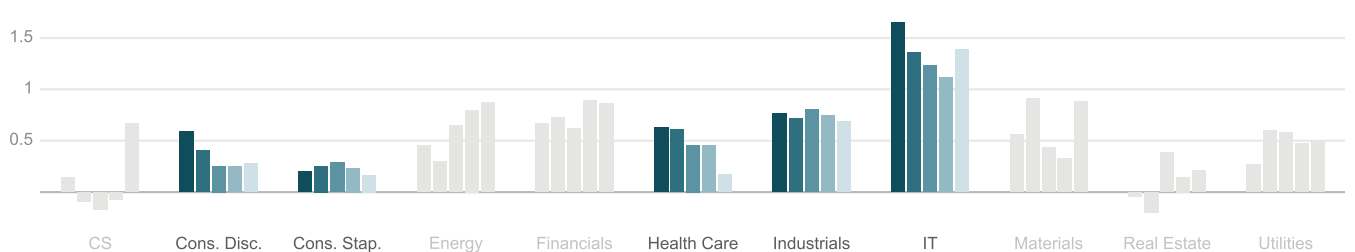
Capital Efficiency L-S CAGR +2.6%

232 ● **Change of Sales to Change of EPS**

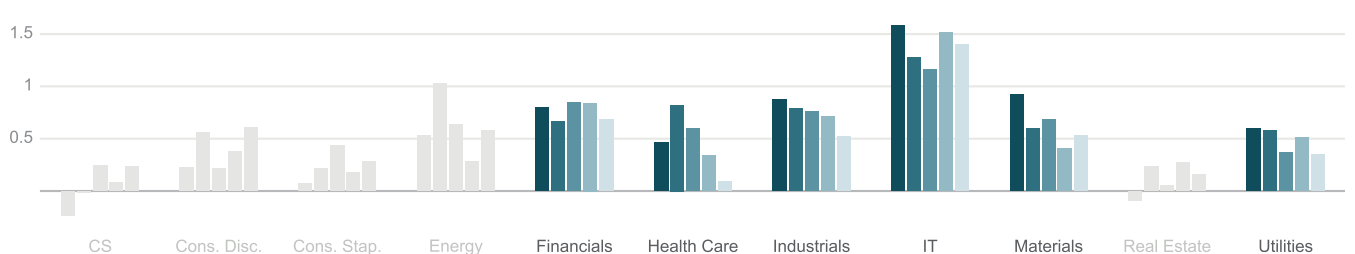
Earnings Quality L-S CAGR +2.5%

233 ● **Pretax Profit Margin**

Capital Efficiency L-S CAGR +2.5%

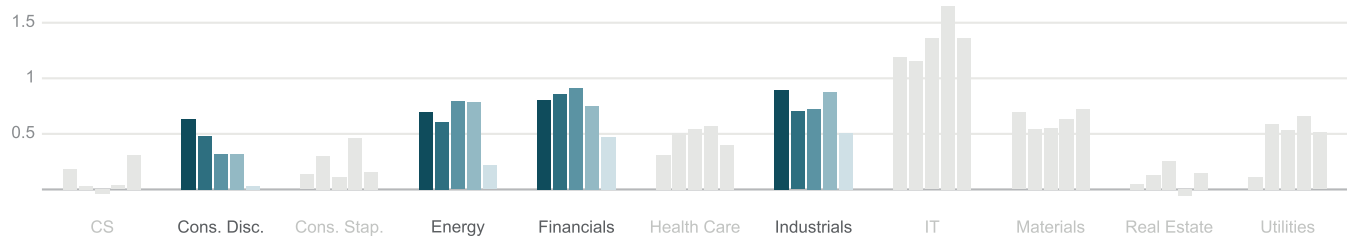
234 ● **Sales Acceleration**

Historical Growth L-S CAGR +2.4%



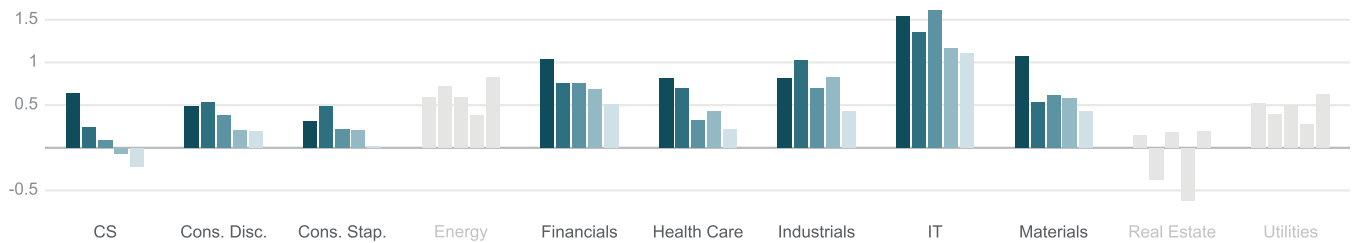
235 ● Risk Adjusted Relative Strength

Price Momentum L-S CAGR +2.4%



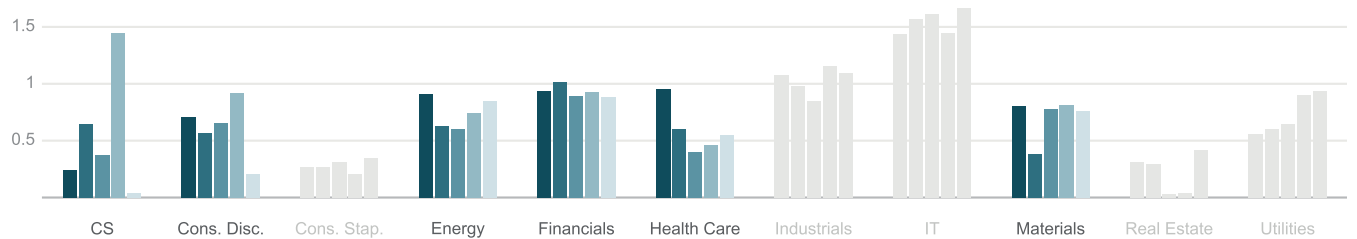
236 ● Analyst Earnings Estimate Diffusion

Analyst Expectations L-S CAGR +2.4%



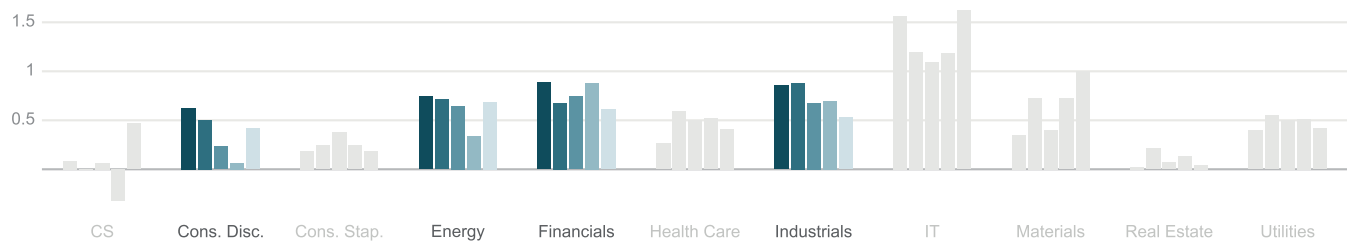
237 ● Change of Ownership Level - Hedge Fund

Price Momentum L-S CAGR +2.4%



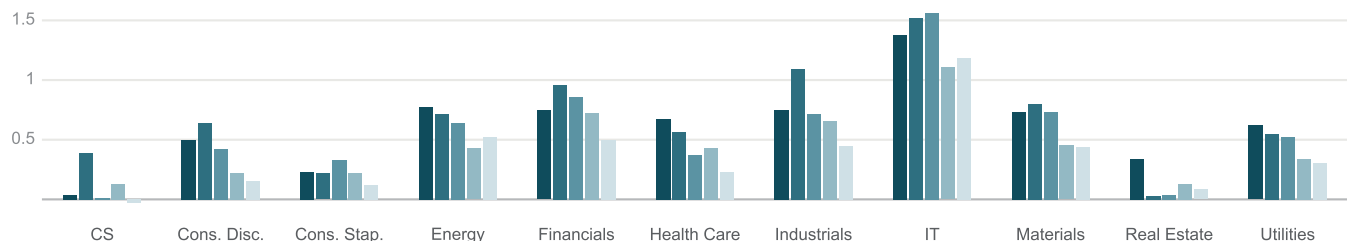
238 ● 3Y Chg in Cash Flow per Share

Historical Growth L-S CAGR +2.3%



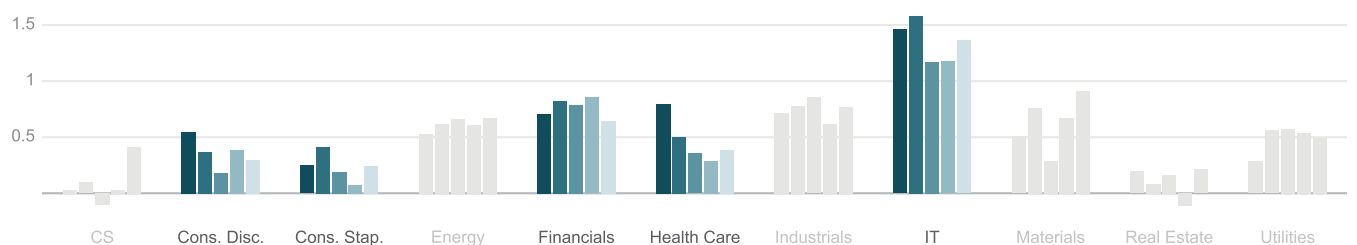
239 ● Investment Duration

Price Momentum L-S CAGR +2.2%



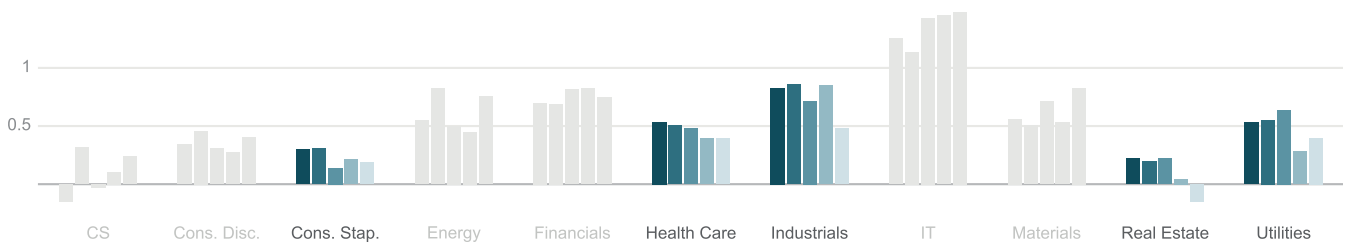
240 ● Normalized ROE

Capital Efficiency L-S CAGR +2.1%

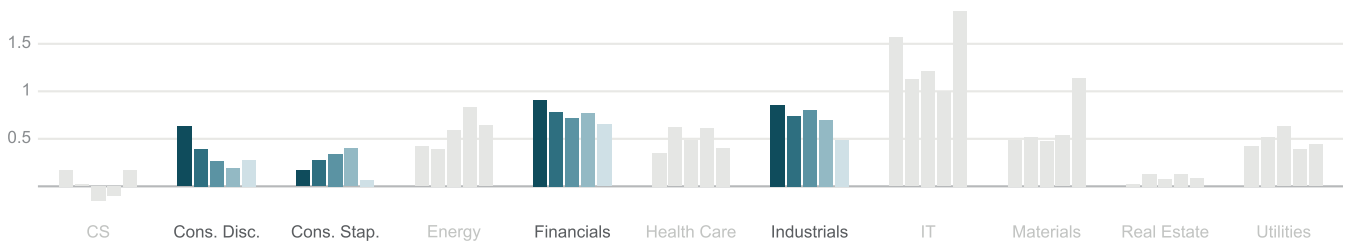


241 ● **Cash to Sales**

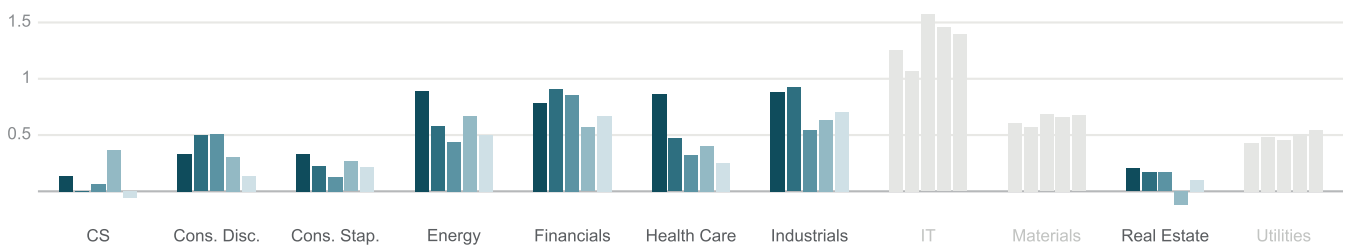
Earnings Quality L-S CAGR +2.1%

242 ● **3Y Chg in Earnings per Share**

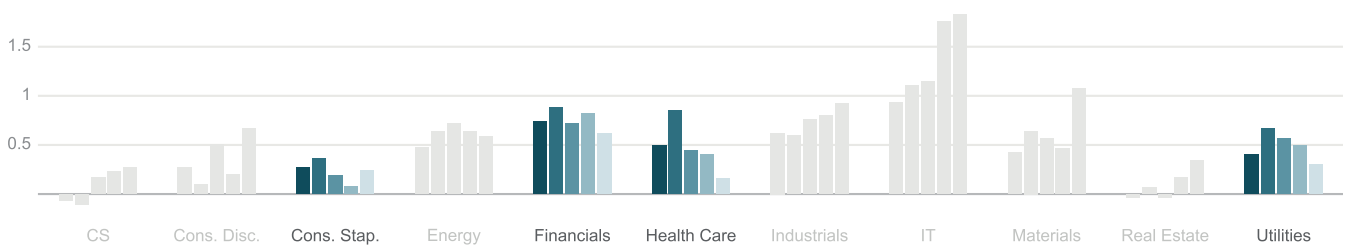
Historical Growth L-S CAGR +2.0%

243 ● **Institution Ownership Turnover**

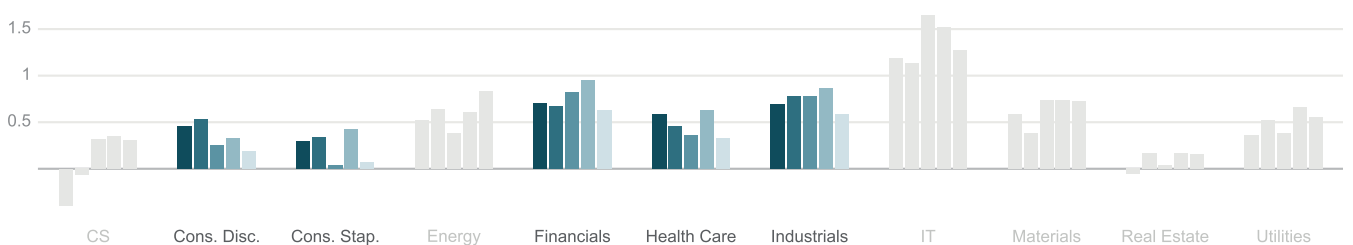
Price Momentum L-S CAGR +2.0%

244 ● **Max Daily Return in the Past 6 Months**

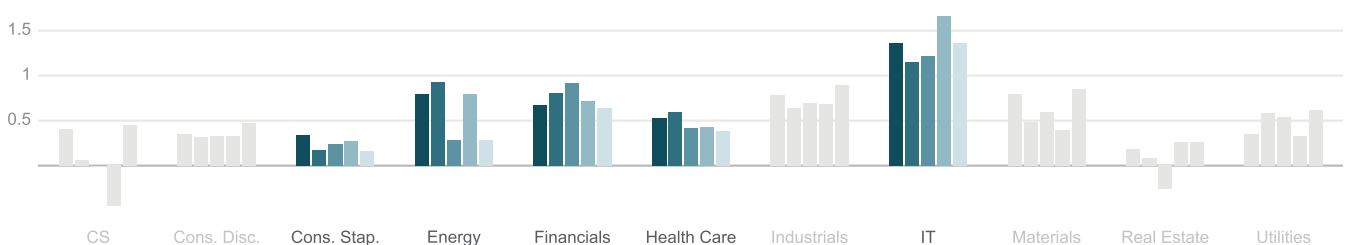
Price Momentum L-S CAGR +2.0%

245 ● **Revenue Estimate Dispersion**

Analyst Expectations L-S CAGR +1.9%

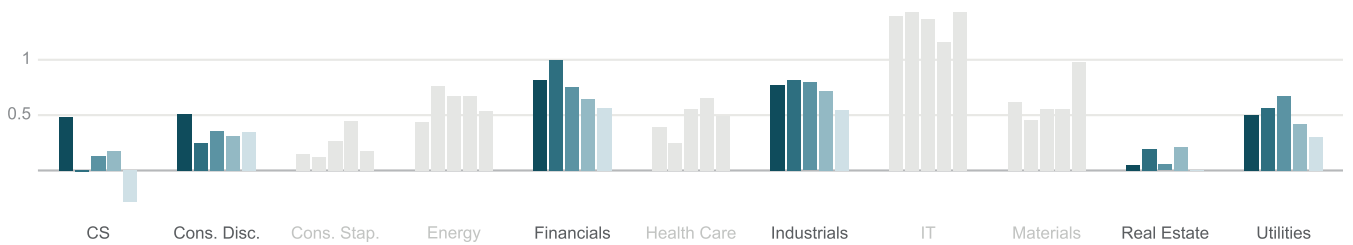
246 ● **Retained Earnings to Total Assets**

Earnings Quality L-S CAGR +1.9%



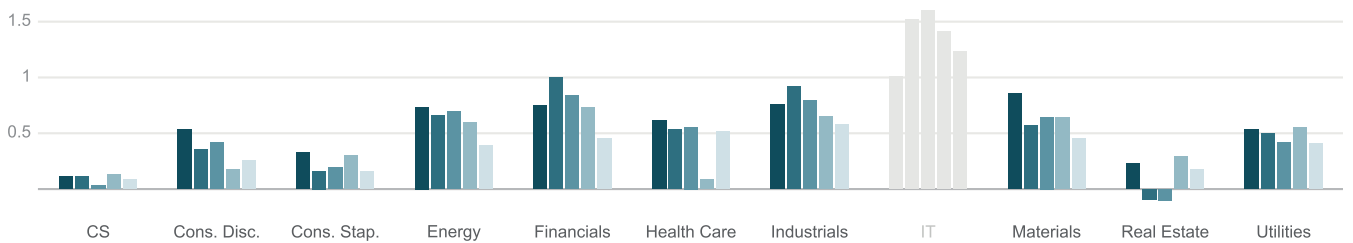
247 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

Analyst Expectations L-S CAGR +1.9%



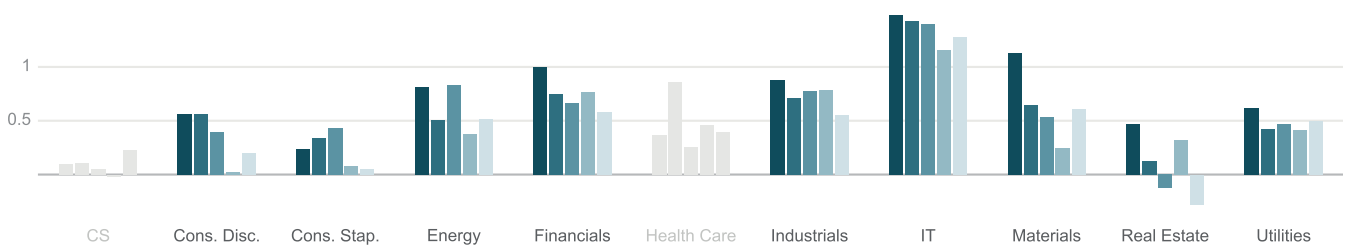
248 ● Institution Ownership Concentration

Price Momentum L-S CAGR +1.8%



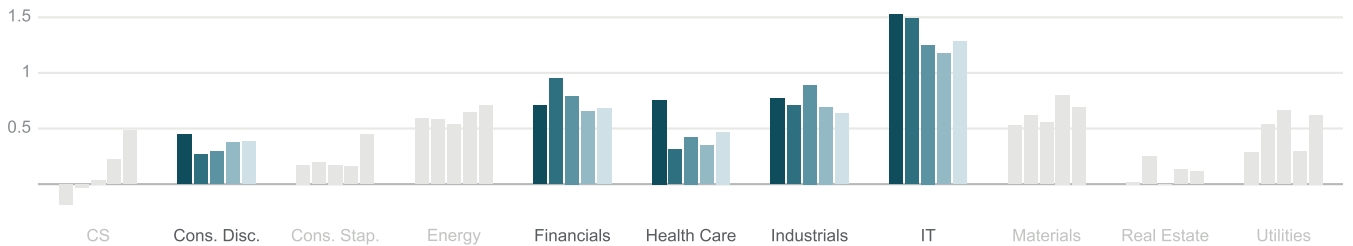
249 ● 5 Day Price Reversal

Price Momentum L-S CAGR +1.7%



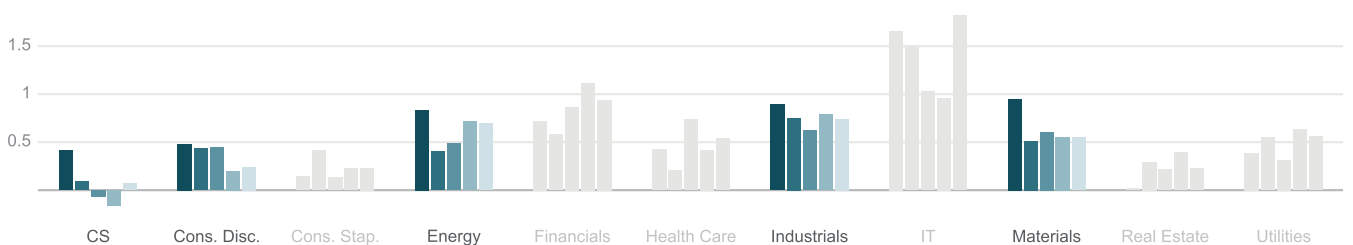
250 ● 6M Chg in 18M CAPM Alpha

Price Momentum L-S CAGR +1.6%



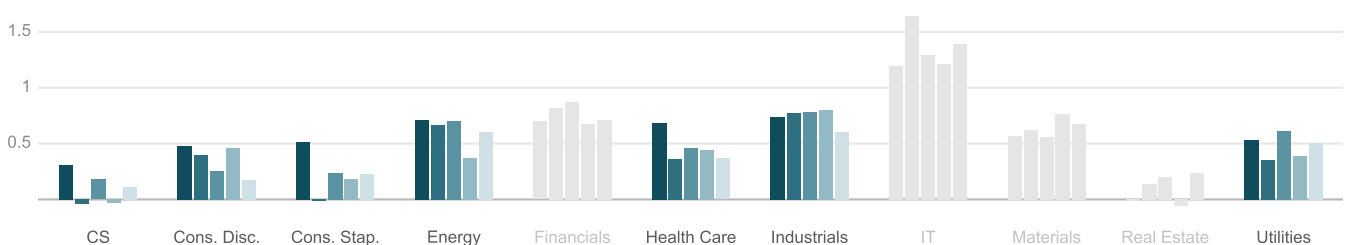
251 ● Year over Year Change of Inventory to Assets

Earnings Quality L-S CAGR +1.6%



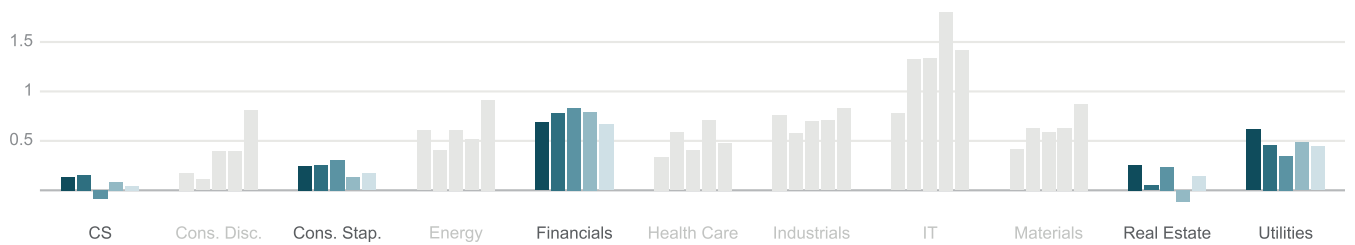
252 ● 6M Chg in 12M CAPM Alpha

Price Momentum L-S CAGR +1.4%



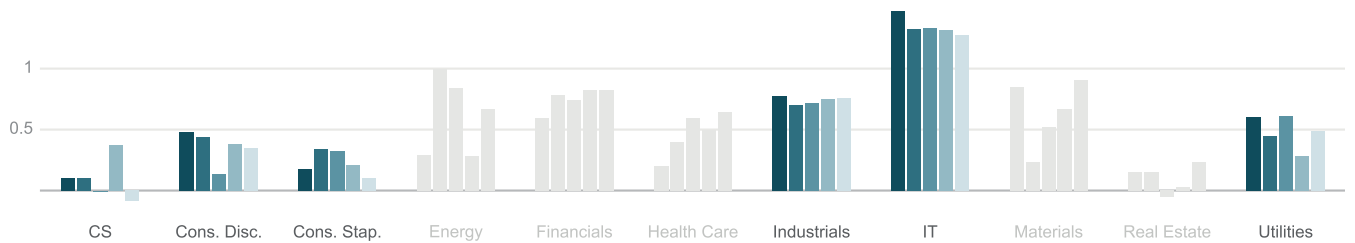
253 ● ROE 20 Qtr Standard Deviation

Earnings Quality L-S CAGR +1.4%



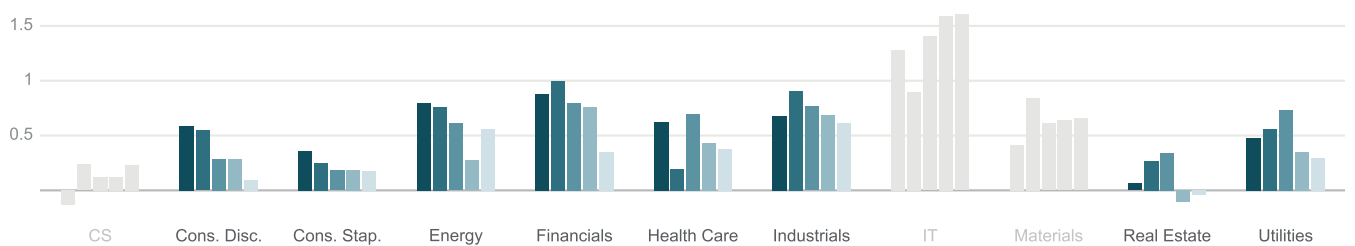
254 ● Change of Ownership Level - All Managers

Price Momentum L-S CAGR +1.3%



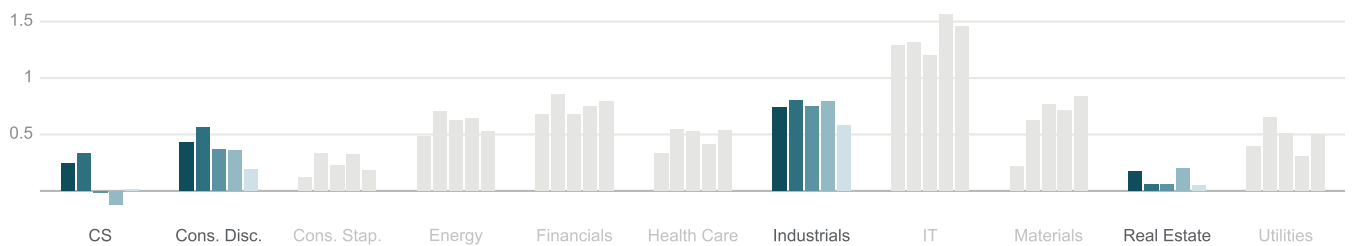
255 ● 6M Chg in Target Price Gap

Analyst Expectations L-S CAGR +1.2%



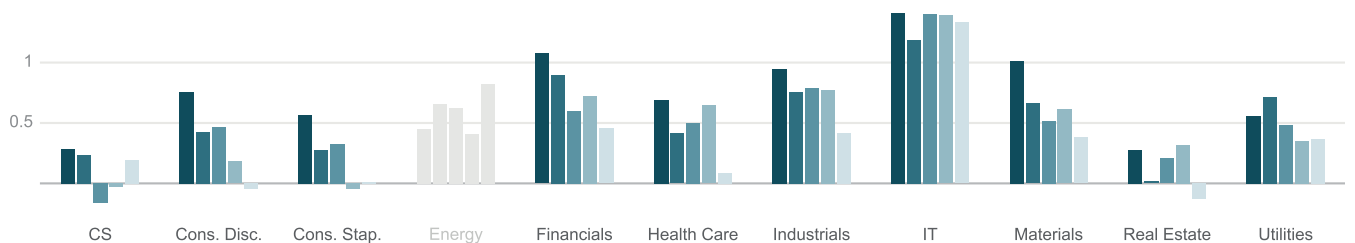
256 ● 3Y Chg in Operating Margin

Historical Growth L-S CAGR +1.1%



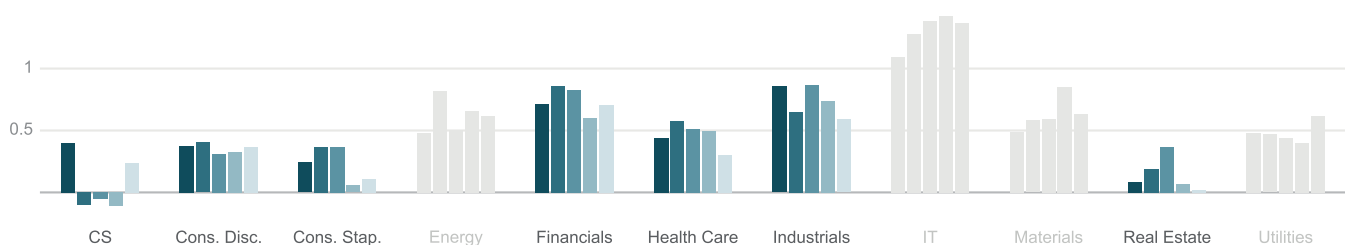
257 ● 8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation L-S CAGR +1.0%



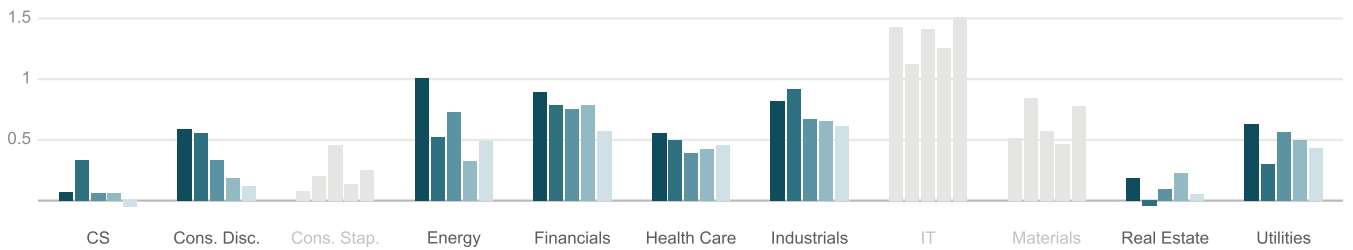
258 ● Change in TTM Depr. to Capex

Earnings Quality L-S CAGR +0.9%



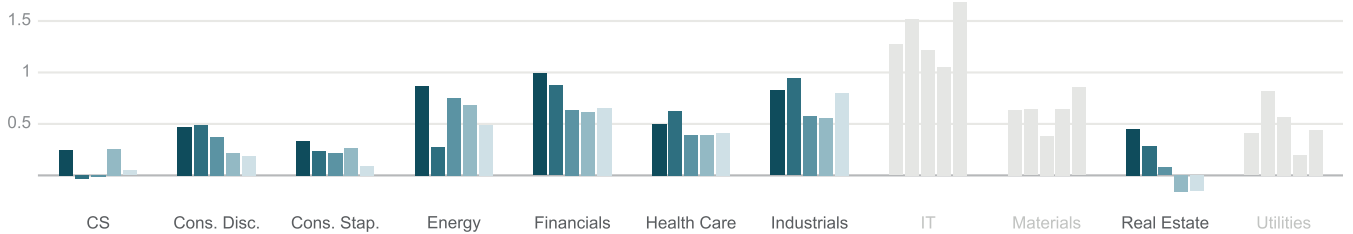
259 ● 6M Avg Chg 1M Recommendation

Analyst Expectations L-S CAGR +0.7%



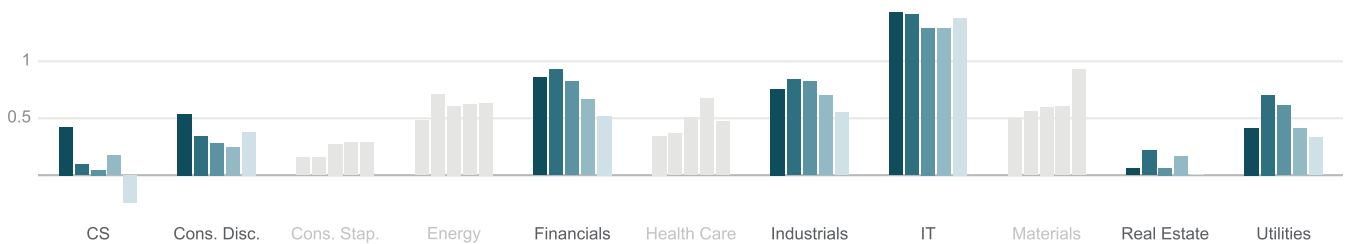
260 ● 14-Day Relative Strength Index

Price Momentum L-S CAGR +0.5%



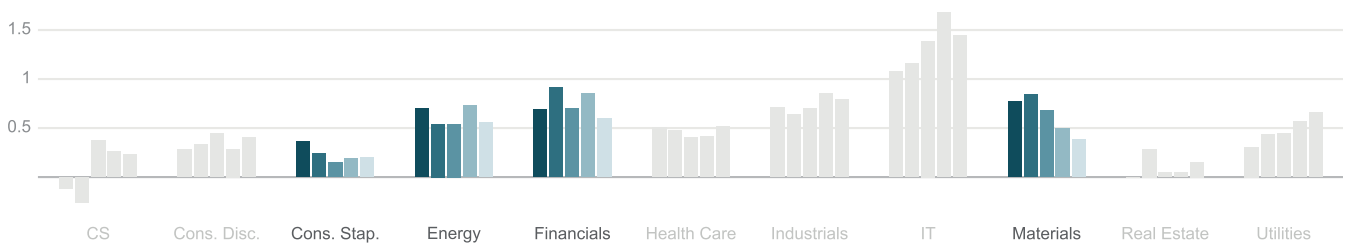
261 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations L-S CAGR +0.3%



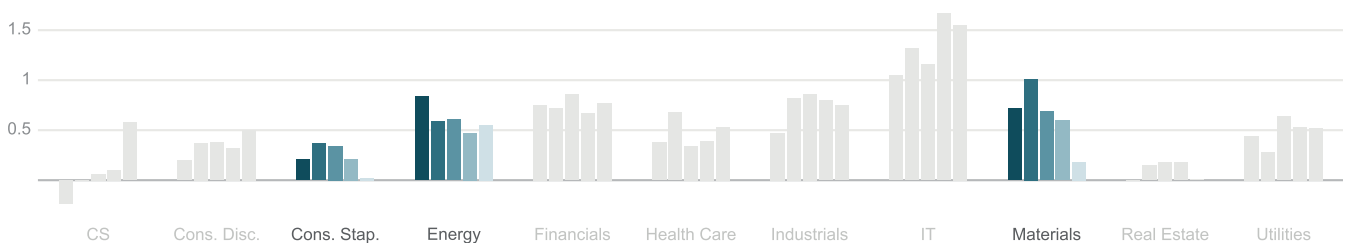
262 ● 1Y Chg in Share Turnover

Price Momentum L-S CAGR +0.2%



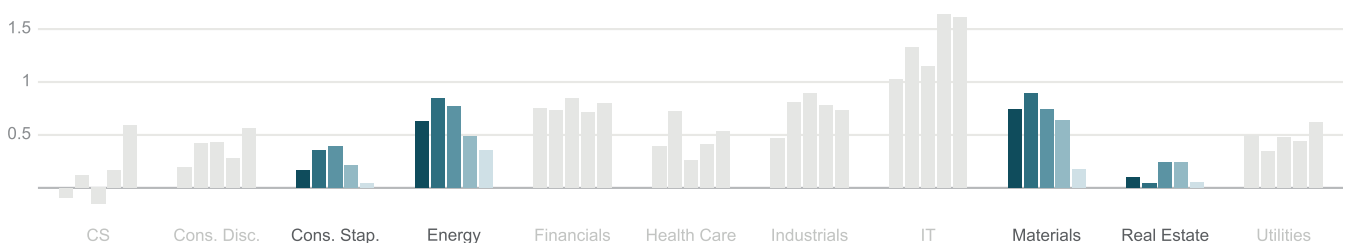
263 ● 50 Day Volume Signal

Price Momentum L-S CAGR -0.6%



264 ● Adj 50 Day Volume Signal

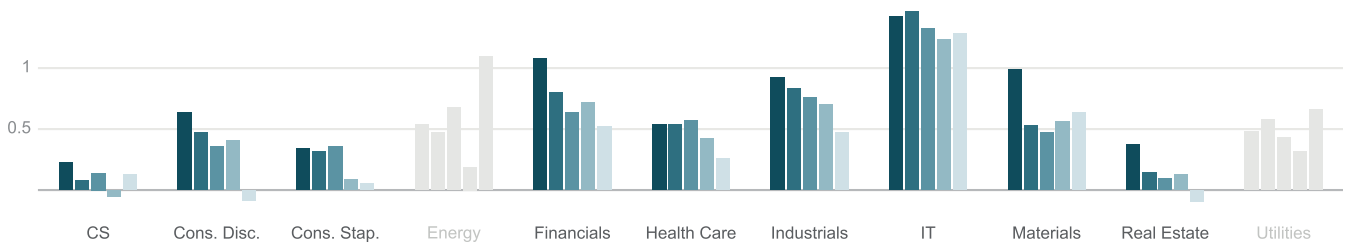
Volatility L-S CAGR -0.6%



265 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

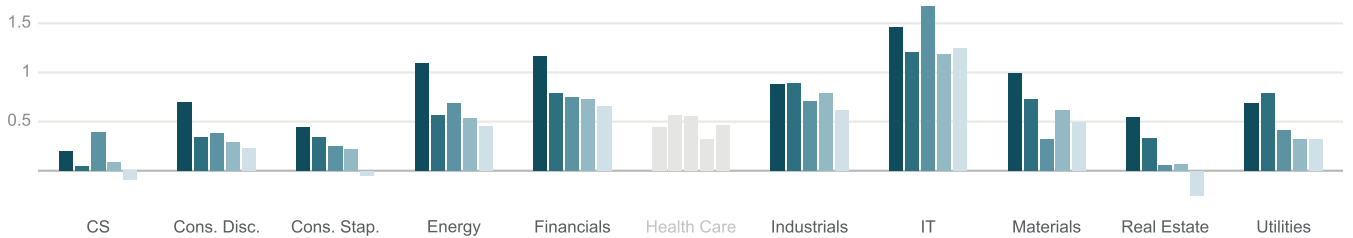
L-S CAGR -0.7%



266 ● 5-Day Money Flow/Volume

Price Momentum

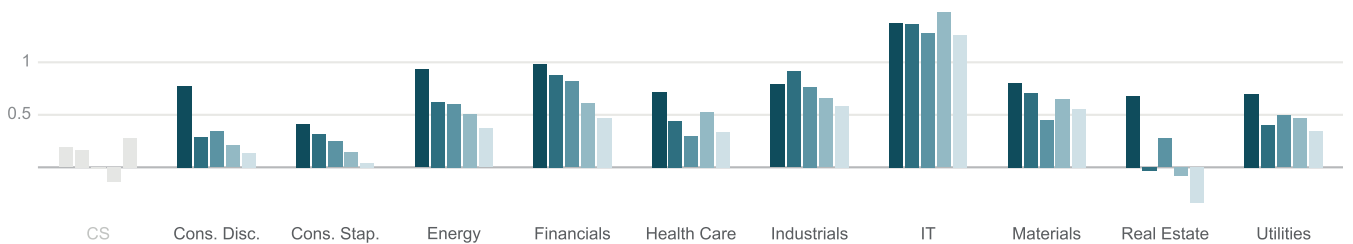
L-S CAGR -1.1%



267 ● 20-Day Lane's Stochastic Indicator

Price Momentum

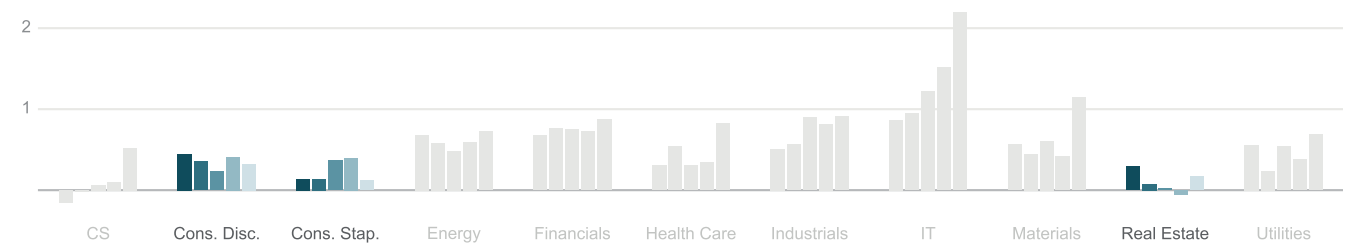
L-S CAGR -1.4%



268 ● 6 Months Price Reversal

Price Momentum

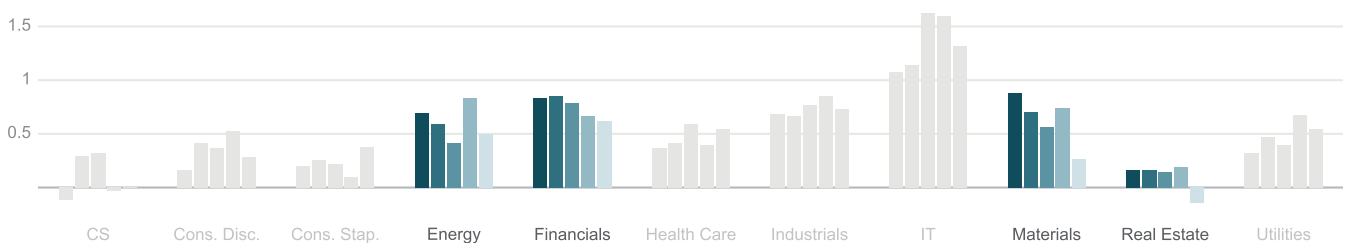
L-S CAGR -1.8%



269 ● 5 Day Volume Signal

Price Momentum

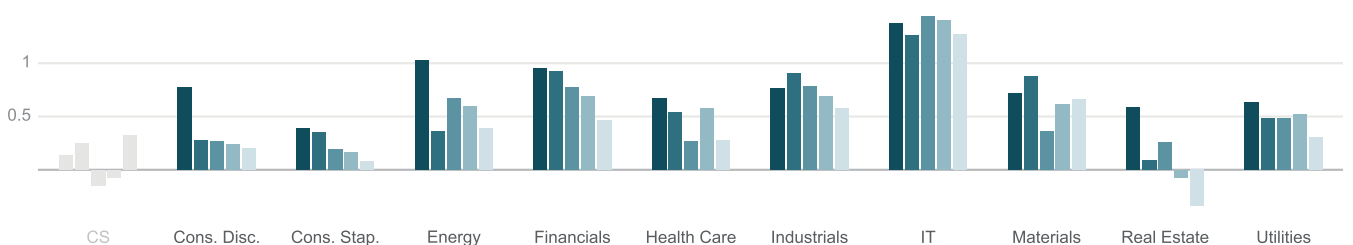
L-S CAGR -1.9%



270 ● 1M Price High - 1M Price Low

Price Momentum

L-S CAGR -1.9%



271 1M Price Reversal

Price Momentum L-S CAGR -4.0%

